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PRESIDENTIAL ADVISORY COMMITTEE

ON

GULF WAR VETERANS' ILLNESSES

PUBLIC MEETING

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Rutledge Room

Mills House Hotel

115 Meeting Street

Charleston, South Carolina

Dr. Joyce C. Lashof, Committee Chair

May 7, 1997

2

Advisory Committee Members Present:

JOYCE C. LASHOF, Chair

MAJ THOMAS P. CROSS

MAJ MARGUERITE KNOX

ROLANDO RIOS

ANDREA KIDD TAYLOR

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1 P R O C E E D I N G S

2 DR. LASHOF: We'll start this morning's meeting.

3 I'm Dr. Joyce Lashof, Chair of the Presidential Advisory

4 Committee. I'd like to welcome all of you here.

5 We have a fairly full agenda and we'll begin, as

6 usual, with public comment.

7 VOICE: We cannot hear.

8 DR. LASHOF: Are the mics not on? Is that better?

9 Okay.

10 Well, I said good morning. And we're ready to

11 begin our meeting.

12 The first person to appear is Mr. Kenneth and

13 Angela Queen. Are they here and ready to come forward? If

14 you would plan to limit your comments to about five minutes

15 and then we'll have about five minutes for questions to you.

16 And you can use the podium there, would be better. Thank

17 you very much.

18

19 STATEMENT OF KENNETH AND ANGELA QUEEN

20

21 MR. QUEEN: Thank you. First, let me thank

22 everyone for this opportunity. Hopefully what I say today

23 will help me and all over veterans who are suffering. If

24 not what I say, I've already sent a report in to the

25 Committee and maybe something in that report will help.

5

1 On January 6, 1991, we landed in Saudi Arabia.

2 The first few days we were waiting on our equipment. We

3 left on convoy on January 16, 1991, which was the day that

4 the ground war also started.

5 As we reached our destination, we set up our

6 perimeter and everything that goes with going into a new

7 site. There we were attached to the 7th Corps. My job was

8 a 62B20, which is a construction vehicle mechanic. For the

9 next few days, we spent our time building hospital sites,

10 chopper sites, et cetera.

11 After the ground war started, we started to go out

12 in contact teams to build and keep main supply routes open.

13 This would normally mean four pieces of equipment with

14 operators, assistant operators, squad leaders and two

15 mechanics with tools and some repair parts.

16 Like most of the soldiers that went to the Gulf, I

17 took all the shots, most of which I had no idea what they

18 were. Like most soldiers, you are not told what kind of

19 shot you're taking. I took several of the miracle drugs

20 which we were ordered to take. All we were told about them

21 was that they were to help nerve agent antidote work faster.

22 We were never told that they were not tested or anything

23 like that. I wish I could tell you exactly how many I took,

24 but I can't. My guess is between five and ten of them.

25 I did help paint the equipment. I've been asked

6

1 if I had burning skin while I was there. It was 110-120

2 degrees, it's nothing unusual to have burning skin when it's

3 that hot.

4 I saw several things blown up, which was vehicles,

5 what I believed to be ammo dumps -- I can't prove that, but

6 I'm going by the difference in the type of explosions that

7 we had. Chemical alarms would go off three to four times a

8 day, which was nothing unusual. We would go from MOPP 1 to

9 MOPP 4, whatever we was told to go to. I'd been in the

10 military almost eight years and I'd never seen chemical

11 alarms go off like that except in test situations when we

12 were testing the alarms and stuff like that.

13 When we would go out on contact teams to keep the

14 main supply routes open, we were given a compass and told

15 how to get there. Today, I have people asking me what I was

16 close to. I can't really tell you what I was close to, I

17 never saw a map. I had a top secret clearance, but yet all

18 I can tell you is we were in Iraq keeping main supply routes

19 open.

20 The job didn't stop once the ground war was over.

21 It stopped when most of the equipment and personnel were

22 back through. It required many long hours and very little

23 sleep.

24 Today, you have people saying that the illness is

25 in our head. I really wish that it was in my head. I would

7

1 love to go one day without hurting. I'd like to walk

2 through my house without bouncing off the walls and my legs

3 giving out and falling in the floor.

4 One time I was proud. I've never took charity

5 from anybody. Today, I have to worry about feeding my

6 family, keeping the power turned on. But yet when you go to

7 the VA Hospital, it's like you do not exist, you've got

8 leprosy or something.

9 On one trip to the local VA Hospital, which is not

10 in Charleston, it's in North Carolina, I was in more pain

11 that night than normal in my legs. I had already been

12 diagnosed with Desert Storm syndrome by Bowman-Gray

13 University Hospital. As we got there and got all the

14 paperwork filled out, the doctor called me back in. As I

15 got up to go, my wife picked up her book. She read one

16 paragraph and I came out and was ready to go. She later

17 counted the words, there was 153 words in that paragraph.

18 It was a real good checkup and nothing for pain.

19 In December of '96, I weighed 172 pounds. Today,

20 I weigh 173. I've been out of work since January 29, 1997.

21 Today, I'm looking at the real possibility of a wheelchair

22 because I keep falling as I go through the house, just

23 trying to get around. I'm in constant pain, I'm slowly

24 losing control of my kidneys. I don't know if it's coming

25 from the Gulf. I'm irritable. Thank God my wife loves me

1 or she would have already left, because she can't do nothing

2 right.

3 Recently, I've been diagnosed with diabetes. The

4 doctor tells me that this is a real possibility that it's

5 coming from some kind of virus or illness. And as my

6 medical records will show, I've had nothing wrong except my

7 legs and that's giving me a pain and I can't keep going.

8 Thank you for the opportunity to speak.

9 DR. LASHOF: Thank you very much. Are there

10 questions that anyone has?

11 MAJ KNOX: I don't have any questions, but I just

12 thank you for your service in the Gulf.

13 MR. QUEEN: Thank you.

14 MS. KIDD-TAYLOR: I have just one question.

15 DR. LASHOF: Sure, Andrea.

16 MS. KIDD-TAYLOR: I just wanted to clarify that I

17 understood. None of your symptoms started before you were

18 in the Gulf, all of this happened after?

19 MR. QUEEN: Yes, ma'am. I always had trouble with

20 my knees but never with my muscles in my legs.

21 MS. KIDD-TAYLOR: Just your knees.

22 MR. QUEEN: Just my knees. Now I have trouble

23 with the muscles.

24 MS. KIDD-TAYLOR: And they've just diagnosed it as

25 -- your physicians have diagnosed it as what now?

9

1 MR. QUEEN: They've run all kind of tests, ma'am,

2 they can't find anything. He wrote it down on my report as

3 Gulf War syndrome. He's still continuing to run tests.

4 MS. KIDD-TAYLOR: And you're receiving benefits?

5 MR. QUEEN: No, ma'am.

6 MAJ CROSS: Have you applied for benefits and

7 what's the outcome of that?

8 MR. QUEEN: Sir, it's still -- I go again next

9 Tuesday for another evaluation. It's still an open claim.

10 DR. LASHOF: Have you received any information

11 concerning the shots and the PB pills that you've taken?

12 Has the government sent you any information sheets to try to

13 explain about those shots and the PBs to you?

14 MR. QUEEN: No, ma'am. The only thing I've

15 received is what Mr. Longbrake sent from the last

16 Presidential Committee meeting.

17 Oh, she just informed me I said I weighed 172 and

18 173 in December. I weigh 153 now.

19 DR. LASHOF: Thank you very much, thank you for

20 coming.

21 MR. QUEEN: Thank you for your time.

22 DR. LASHOF: The next person is Laurie Hills.

23

24 STATEMENT OF LAURIE HILLS

25

10

1 MS. HILLS: Good morning, everyone.

2 I found out through the paper about a week ago --

3 I don't have a representative, I moved here in July from

4 Arizona where I discontinued my service.

5 For the Persian Gulf War, I was stationed in

6 Vilseck, Germany and I went with the 1st Armored Division,

7 123rd Support Battalion out of Nurnberg, Germany, and was

8 attached to that unit. I was in Saudi -- I went December,

9 '90, to -- no, I'm sorry -- December, '90 to May, '91.

10 I don't have a written speech but you'll get one

11 by the end of the month. I haven't had time to prepare.

12 I was at Log Base Alpha, Echo, Garcia, all the

13 rest of them in Saudi. Then I had volunteered to go to Iraq

14 before Desert Storm hit to push parts forward to the front

15 line. As far as Iraq, I'm not really exactly sure where I

16 was either. I knew I was near the Iraqi compound,

17 supposedly where they were taking the Iraqi soldiers that

18 were surrendering.

19 To back up a moment, when I was in Saudi in

20 January '91 -- it's a little grotesque, but basically what

21 happened, I thought I had diarrhea, I had to go to the

22 outhouse every 15 minutes and there was light discipline,

23 noise discipline, so I wasn't allowed to shine the

24 flashlight, et cetera, et cetera. Well, this lasted for

25 about five hours after guard duty. So I finally -- morning

11

1 time came and I looked in the outhouse and it was blood. So

2 needless to say, I went to sick call, I filled up about nine

3 bedpans with blood an hour, couldn't even make it to the

4 outhouse any more. They evac'd me to another compound where

5 I saw a Lieutenant Colonel. He didn't know what was wrong

6 with me, so then they evac'd me to 5th MASH where I saw a

7 full bird. He gave me some sort of medication, they never

8 explained to me what was wrong with me. They said it could

9 have been stress or they said something about breaking down

10 the inner lining of my eggs or ovaries or what-have-you, and

11 that was the end of it.

12 Needless to say, I don't really find it so much as

13 a coincidence that my medical records are missing. Everyone

14 I've spoken with that has had any medical problems in

15 Saudi/Iraq, their medical records are also missing. Now my

16 regular medical records are there, but my time period in

17 Saudi is missing.

18 I did go to the Persian Gulf Registry in Tucson,

19 Arizona. I did jam my left knee in Saudi in a pothole in

20 MOPP 4 running to a foxhole, so I got medically discharged

21 November '93 for arthritis and chondromalacia in both of my

22 knees. But I did go to Tucson to the Persian Gulf Registry,

23 they said I was one of the first legitimate cases, some sort

24 of diarrhea syndrome, I don't know what it is.

25 Also, as soon as I got back to Germany May '91

1 from Saudi and Iraq, I had what I thought was just a bug

2 bite right here on my hairline. Well, they wasn't sure what

3 it was, it was a cyst, so they removed it. Since I've been

4 out of Saudi -- before I went to Saudi, my cholesterol level

5 was 141, it's now 270. The same as the former speaker, I do

6 not know if the Gulf -- if this could have happened anyway.

7 All I'm saying is my health has basically went to hell.

8 Between cysts, yeast infections, bacterial infections -- I

9 catch every type of stomach virus, I have diarrhea at least

10 once a month. This past month, I think I caught a stomach

11 flu, the diarrhea lasted for two weeks and I was going to

12 the doctors and taking their medicine, et cetera. I've had

13 a lot of bowel problems.

14 I did have to burn feces and trash in Saudi. I

15 was told at the time over there that you could get seven

16 different diseases if you smoked cigarettes and five

17 different disease if you didn't. And that was from smelling

18 the smoke of burned feces -- rumor mill.

19 I did have to take malaria pills in Iraq. I only

20 took about four of them and we were supposed to take the NBC

21 tablets, I took one and then decided I wasn't going to be

22 the government's guinea pig for their drugs, so I just said

23 hey, if I'm going to die, I'm going to die anyway. So I

24 quit taking the NBC tablets because they were giving me

25 headaches.

13

1 I was in Kuwait for one day with the burning oil

2 wells.

3 I have filled out all the surveys sent to me for

4 the Persian Gulf, any related problems, what-have-you, I

5 send it in. I never get any response back. I wonder what

6 is going on. So I am very pleased to see that the President

7 has appointed this Advisory Committee for us.

8 That's about all I have to say. If you have any

9 questions.

10 DR. LASHOF: Thank you very much. Any questions?

11 MS. KIDD-TAYLOR: The only question I have, were

12 you anywhere near some of the incidents that occurred, such

13 as Khamisiyah, do you remember?

14 MS. HILLS: I have no idea. We were around a lot

15 of burned up Iraqi tanks, a lot of -- you know, of course, I

16 saw dead bodies, but I was pushing parts forward to the

17 front line, but I was nowhere near the front line.

18 MS. KIDD-TAYLOR: And you were in some MOPP gear,

19 occasionally you heard the alarms go off and were required

20 to get in MOPP gear?

21 MS. HILLS: Oh, that was a given.

22 DR. LASHOF: Are you on medication now? Medicines

23 that you take regularly now?

24 MS. HILLS: No, I just keep going back to the

25 doctors. I mean there's many different ones that I have

1 taken, but I'm 27 years old and I don't want to start taking

2 drugs now for the rest of my life. So -- same thing with

3 arthritis, I just do the hamstring stretches to ease the

4 tension off of it and, you know, they tell me in probably 15

5 years, I should get plastic kneecaps. But I think I'll wait

6 until I'm about 70 for that.

7 MR. RIOS: Do you have a disability rating now?

8 MS. HILLS: Pardon me?

9 MR. RIOS: Do you have a disability rating?

10 MS. HILLS: Yes, ten percent.

11 MR. RIOS: And then you have a pending claim on

12 your illness as well?

13 MS. HILLS: No. Well, like I said, the physician

14 in Tucson said that I was one of the first legitimate

15 claims, never heard anything again about it.

16 MR. RIOS: So that ten percent disability is for--

17 MS. HILLS: Is for my knees only, discharge from

18 the Army, medical discharge.

19 MR. RIOS: Not related to your service in --

20 MS. HILLS: No.

21 MAJ CROSS: Now when you say you go to the doctor,

22 is that a civilian practitioner or is it the VA?

23 MS. HILLS: It's -- well, the VA, I only go for my

24 knees. I've been to the Naval Hospital and the Air Force

25 Base because I'm currently married and he's in the Army, so

15

1 I've used those facilities. But after this month, I'll be

2 seeing a private physician. I work for the government now.

3 DR. LASHOF: Are you working full time, are you

4 able to work full time?

5 MS. HILLS: Yes.

6 DR. LASHOF: Okay.

7 MS. HILLS: I've taken a lot of sick days, but I

8 mean, to me, I'm perfectly fine, it's just the female

9 problems, the diarrhea that is a big inconvenience, calling

10 in sick because of that.

11 MS. KIDD-TAYLOR: And you were never given a

12 diagnosis?

13 MS. HILLS: No. I have no clue, and plus, my

14 medical records are missing, so I don't even know what the

15 Colonel physician had come up with. Or I don't even know if

16 they do know.

17 DR. LASHOF: Thank you very much, we appreciate

18 your coming.

19 MS. HILLS: Thank you, I appreciate your time.

20 DR. LASHOF: Corinthea Stack.

21 (No response.)

22 DR. LASHOF: All right, she apparently is not here

23 at the moment. Jeffrey Woods.

24

25 STATEMENT OF JEFFREY WOODS

16

1 MR. WOODS: First, I would like to thank you for

2 giving me this opportunity. I am not really a speaker, so

3 forgive me. I'm a paratrooper with the 82nd Airborne

4 Division out of Fort Bragg, North Carolina.

5 My first deployment was during the Operation

6 Desert Shield, in which we left in August of 1990. At that

7 point, we were in a location, and unfortunately I cannot

8 give you the name of that location. There were three

9 primary locations we were located at, the first one being in

10 Saudi Arabia, and though I don't know the name of it, I was

11 able to obtain pictures that I took of most of my entire

12 trip over there and absolutely everything that occurred

13 while we were over there. I have the pictures of the front

14 of the location. At that point, we were housed in a

15 building facility that we were just utilized as a -- and

16 please forgive me, I know this is going to sound like I'm

17 not together. I've written everything down, but this is one

18 of the difficulties that I have obtained since I've returned

19 from the Gulf.

20 During Desert Shield at the first location, our

21 primary mission was basically training. From August of 1990

22 until December of 1990, I obtained no medical problems that

23 were seen upon me. And I'm specifically speaking about the

24 diarrhea. Right when the air war started, SCUD missiles

25 started coming in on us near our location, prior to moving

17

1 to our secondary location. At that point, we were given a

2 drug and shots. The drug that we were given in pill form

3 was the PB pills. The shots, I could not tell you what they

4 are. None of those bits of information are in my medical

5 records, nor are they obtainable.

6 The first day that I took the pill, I noticed that

7 I had become ill with abdominal cramps and diarrhea by that

8 evening. The next day, I went over to the medics and they

9 explained to me that at that point in my life -- and I

10 assumed they were correct -- it was probably a bug that I

11 had caught. To this date, my bowel movements are still

12 nothing but fluid. More than likely now because I no longer

13 have a large intestine, it was removed in 1993.

14 As we moved on to our second location, which was

15 the location that our chemical alarms went off, we were

16 airlifted to that point, my medical problems had already

17 started. We were at that second location for approximately

18 two weeks. Again, that's where our chemical alarms went

19 off. We went into full MOPP gear, were later told that it

20 was a false alarm and told to remove our gear.

21 Our third location was when we breached the border

22 of Iraq, now into Operation Desert Storm.

23 I researched all the information that I could in

24 regards to other individuals that are still on active duty,

25 about exactly where we went. Most of the people on active

18

1 duty -- and I need to add that I am still active duty, have

2 been extremely reluctant to speak about these things, for

3 obvious reasons. Certain elements of the military turn

4 their back and if you speak about it or you're even somewhat

5 connected to it, you become an outcast.

6 Because I had become so sick and went from 220 to

7 160 in body weight, and at that time thought that I was

8 dying, I became extremely outspoken in April of '91 when I

9 came back to the physicians, telling them if you'd just

10 listen to me, maybe we could stop what's happening. Only

11 one physician listened to me, Dr. Burnett. She noted in her

12 medical records and told me that I was probably having a

13 reaction to the medical pyridostigmine bromide pills. That

14 physician was put out of the Army. For what reasons, I do

15 not know, but she is private practice in Atlanta, Georgia.

16 The Army sent me to several medical facilities --

17 Moncrief Hospital in Fort Jackson, Eisenhower at Augusta,

18 Georgia, where they removed my large intestine, and Walter

19 Reid, where I am presently assigned. I have also been seen

20 at the Naval Hospital here in Charleston, where I reside as

21 of today -- well, I've been living for quite some time. And

22 I've been seen at the VA Hospital.

23 My medical problems are my intestine being

24 removed, as a diagnosis at that time was put as ulcerative

25 colitis. I have extremely high uric acid intake, which

1 causes uric acid stones, which develop in my kidneys on a

2 rapid rate. I presently take medication for both those

3 problems. Even though some of the other symptoms that I

4 hear people talk about when they say joint pains, I feel

5 that if I sat down and explained to everybody what my body

6 feels like, you would think I was a hypochondriac. People

7 kind of look at you. Most people tell me, well, Jeff, you

8 look pretty healthy.

9 I have some pictures that I could show you, that

10 would show you the day I was on the plane to Saudi Arabia

11 and when I returned from Saudi Arabia.

12 Again, I've tried to explain to everybody, my

13 entire life has been a very physical life. I have always

14 maintained good health, a non-smoker, and I still try, to

15 this day, to try to maintain that, but the fatigue that I go

16 through has kept me on the couch probably five days out of

17 seven.

18 I don't want to give the impression in any fashion

19 that I am somebody that is sitting up here that gives a damn

20 about their disability at this point, because my health is

21 gone. But what the Committee has to realize is there is no

22 possible way that all the soldiers, not just myself, could

23 experience the medical problems, which one of your primary

24 problems I'm sure you've discovered are intestinal problems,

25 from something that happened over there. Being a combat

20

1 soldier, it doesn't affect me that it happened because of my

2 job -- you expect that probably defending this country, you

3 are going to get hurt.

4 What has taken the biggest toll on me is to watch

5 this country turn its back and to watch the military

6 personnel that I served with treat me like I'm a piece of

7 garbage because I became ill.

8 I could probably go on, stay up here and tell you

9 things that would devastate every person sitting at this

10 board, but legally I've been advised that I can't do that.

11 And because of issues pending, I have no choice.

12 Some of the diagnosed medical problems that I have

13 at this point are ulcerative colitis, the removal of my

14 large intestine, chronic abdominal pain, short-term memory

15 loss and at times confusion, headaches, cough, constant

16 spitting up of sputum, phlegm and shortness of breath. They

17 have diagnosed me with reactive airway disease. I've always

18 questioned that. Chronic stones due to ulcerative colitis,

19 hyperthyroidism, rashes on my arms and chest. And I

20 fractured, for some reason, my right foot three times on

21 just minor bumps. They are now in the process of testing at

22 the Medical University to find out whether or not the bone -

23 - what the bone density is in my body. I'm not quite sure

24 of the medical, what they're looking at, but for some reason

25 my bones are becoming brittle.

21

1 I want to thank you very much for allowing me to

2 come here and speak to you. I, myself, have written many,

3 many letters and have complained strongly. I, myself, -- I

4 assure you that if you tried to get my medical records,

5 you're going to have a difficult time. I fortunately

6 obtained quite a few in the very beginning on the advice of

7 Dr. Burnett and in my case, I think that that's probably one

8 of the reasons why I'm still on active duty, though I

9 haven't worked on an active duty job in my permanent MOS

10 since 1991, and since 1994, from '91 until '94, I only

11 worked one day a week and from '94 on, I haven't worked at

12 all. Not because -- I never said I didn't want to, the

13 military said it was mandatory retirement for me and I had

14 to get out, but let me make it perfectly clear, I never

15 wanted out.

16 Thank you very much.

17 DR. LASHOF: Thank you very much. Are there any

18 questions? We have time for just a couple.

19 (No response.)

20 DR. LASHOF: If not, thank you very much for

21 coming, we appreciate it.

22 Corinthea Stack here yet?

23 (No response.)

24 DR. LASHOF: We'll move on to Alvin Green Younce -

25 - Glenn Younce, I'm sorry.

22

1 STATEMENT OF ALVIN GLENN YOUNCE

2

3 MR. YOUNCE: Thank you for inviting me here today.

4 Like my brother, I've had a long journey.

5 I was stationed at Al Karsh, right outside of

6 Riyadh. And as you watched the news, you could see that

7 weapons were raining down all around us, the ground would

8 shake. During the nights, I could see the Patriots taking

9 out the SCUDs in the air.

10 I was an F16 crew chief and also a volunteer

11 chaplain. We didn't have enough chaplains and I'm a

12 minister and served as a volunteer chaplain. So I was all

13 over, trying to help where I could.

14 As time went on, my plane was flying four times a

15 day and I would scoop the intake four times a day. And they

16 asked me, they said, why did you do that. Well, my pilot

17 happened to be my best friend and he deserved a safe

18 airplane. So I would go down the chute of the F16 four

19 times a day on a wet t-shirt, many times exhausted to the

20 point that I couldn't even go take a bath, but go straight

21 to bed.

22 Later on, they started washing our clothes in bags

23 together, hundreds of us that had been exposed to this and

24 that. They were just increasing the amount of chemical that

25 we were exposed to through the laundry. Everything in my

23

1 tent had an odor, but it's just like if you came to my

2 house, it would have a specific odor and if I came to yours,

3 it would have a specific odor. So we became sort of immuned

4 to it, we didn't smell it any longer, until I came home and

5 started opening boxes that I brought back, which I disposed

6 of.

7 I became ill on the flight line with an ear

8 infection after the cease fire. And I have had chronic ear

9 infection ever since. I was put into the bed, not into the

10 hospital. The only way I was able to receive food and water

11 was from a dear friend that went and would bring it to me.

12 Until this very day, I still have chronic ear infections and

13 vertigo.

14 In the first year after returning home, I entered

15 into deep depression and night sweats and some of the most

16 horrible nightmares that I could possibly imagine. I served

17 as a medic in the very beginning of eight years between '78

18 and '76. So I've seen the worst of the worst, but this was

19 much more of a horrible experience, the things that I

20 witnessed as a chaplain and as a crew chief.

21 I've been hospitalized three times, the last time

22 -- and that was with urinary tract infections which

23 eventually led to prostatitis. My immune system is so low

24 that I can only take certain drugs for three to four days or

25 they will start counter-reacting with whatever pain

24

1 medicines I'm taking and will cause me to start having

2 seizures. I was in the hospital just over five weeks ago

3 and they called my wife in two times to tell her that I

4 would probably not live to the morning. They transferred me

5 by ambulance from -- this was in a private hospital -- to

6 the VA in Columbia, South Carolina, where they did further

7 tests and C&P exams.

8 During my C&P exams, my medical records were

9 completely ignored from an environmental specialist that I

10 became a part of his team. I did a lot of research from him

11 because he was receiving no research from the VA about Gulf

12 War syndrome and he was an environmentalist at this

13 hospital. So all the research I could pull from my

14 Congressman, I gave him -- I gave Lindsey Graham my research

15 and then he assigned an aide to help me get through this,

16 because as I got so sick, I couldn't do it for myself.

17 I've been promised by Social Security, by the VA,

18 by my own denomination that I had insurance with, that I

19 would be receiving benefits by now. My church has released

20 me because I spent more time at the VA than I did in my own

21 church.

22 I started having blackouts. I would park beside

23 the road and sometimes sleep for two to three hours,

24 sometimes go to sleep at the wheel and wind up beside the

25 road.

1 During the C&P exams, they brought outside doctors

2 in that would only have a form, a medical form -- not my

3 records. He had not reviewed my records, didn't talk to my

4 primary care physician in Augusta and when I was entering

5 into the Columbia VA, they couldn't even find my records.

6 Now, it's hard for me to believe, ladies and

7 gentlemen, that we are in the computer age and why can we

8 not communicate between two hospitals that are only an hour

9 away? We can put a man on the moon, but we can't find a man

10 that is dying's medical records.

11 The VA has treated me with respect. In the very

12 beginning when I went in, I had vertigo so bad that I fell

13 off the scales. They put me in triage where I stayed for

14 three hours and a urologist -- a neurologist was the serving

15 physician at this time. He put a shunt in my hand and sent

16 me to the psychological hospital, which they were going to,

17 when I got there, they had an admission tag on the desk and

18 was going to admit me and put me away until they got ready

19 to set me free.

20 There's nothing wrong except I have short-term

21 memory, chronic fatigue, ear infections, lower bowel

22 syndrome. I can't even eat food without being near a

23 restroom. I have close to 27 of the known diagnoses.

24 The VA has sent my records to the last regional

25 office, wherever that is, Tennessee or wherever, sent a note

26

1 back saying that the only thing that we see that is wrong

2 with you is that you have a sinus infection, hernia and

3 depression. I have chronic depression, I have chronic

4 fatigue, I can't sleep, I can't eat.

5 You are looking at a man that came back from Saudi

6 Arabia that was 185 pounds, in the best of condition because

7 I was in the gym every morning prior to flight, I ran every

8 day mile after mile and came home and began to have urinary

9 trouble, blockage of bladder. And my weight went from 185

10 to 210 and I can't get it down. I do what I can to

11 exercise, I watch my diet, but I can't get the weight down.

12 I just would like to read a statement and then --

13 I know we have very limited time, but I have it just right

14 here, I would like to just briefly -- these are all the

15 appointments that I've been to and some of these sheets have

16 been multiple appointments. I don't know how anyone in this

17 room behind me could keep a job and make all these

18 appointments. I was in the top 10 percent of Christian

19 educators in my denomination and I have lost my job, my

20 identity, but I have not lost my family. But I get sicker

21 every day and my doctor tells me he does not have any idea

22 how much longer I'm going to live because I can't even go in

23 the hospital now and visit because I pick up everything that

24 comes along.

25 I would like to share one thing. This is what the

27

1 senior pastor wrote as he practically forced me into

2 resignation. "To whom it may concern: The Reverend Glenn

3 Younce has been a full time employee at Northside Baptist

4 Church since October 1994. Prior to this, he was a part

5 time employee with our special education ministries." I

6 worked with disabled people.

7 "During this time, I have witnessed a change in

8 Glenn's ability to fulfill his duties. His desire and

9 commitment never changed, but his ability did. During this

10 time, he struggled with fatigue on a daily basis, his energy

11 was depleted quickly and it would be days before he felt

12 good again. Also, I watched with great concern as he

13 struggled with dizziness and blackout spells. He made it

14 through these periods by resting when he felt these periods

15 coming. This meant that at times he had to stop on the side

16 of the road, rest and sleep.

17 "During this period, I watched with interest the

18 television documentaries on the Gulf War syndrome, and I saw

19 in Glenn these symptoms. As I watched him, it was like

20 seeing what was being described in the news. It makes me

21 angry to hear officials saying that there was nothing to

22 these problems our soldiers are having. I saw these

23 problems in one of our soldiers, Glenn Younce.

24 "The other of my concern was Glenn's power of

25 concentration. He simply was and is unable to focus for any

28

1 length of time on any subject or event. I cannot tell what

2 is wrong with him, but I can tell you that he is struggling

3 with many different health problems that hinders his ability

4 to work."

5 I will be glad to answer any questions that any of

6 you have.

7 DR. LASHOF: Thank you very much. Are there any

8 questions?

9 MS. KIDD-TAYLOR: You never mentioned, are you

10 receiving any disability now or --

11 MR. YOUNCE: None. I have been put off and put

12 off since October and the only thing that they say that I'm

13 eligible for is an ear infection, a hernia and sinus

14 infection. That is not my main problems. I'm being seen by

15 a psychiatrist and a psychologist and the only reason I am

16 not in the hospital this morning is I wanted to be here, I

17 wanted to be heard. Because we have been shoved from one

18 place to another and just like one of the gentlemen said

19 before, we have been treated like trash.

20 I have been treated with great respect in the

21 hospitals, but our regional offices, we cannot get any

22 answers, not even our Congressmen can get answers from

23 regional offices. I have every month records where they

24 send me that they are reviewing my case. They skipped

25 Christmas, they took Christmas off and didn't send one.

29

1 DR. LASHOF: John.

2 MR. LONGBRAKE: As a crew chief, can you tell us

3 the exposure -- the flight crew chief, can you tell us what

4 exposure you had?

5 MR. YOUNCE: Yes. I'd go down a chute that

6 there's barely room for one person, and it's a long way

7 because in the back of an F16 is where the engine is

8 located. I was in a wet t-shirt and wet fatigues because of

9 the high heat, plus the high heat of the engine bay itself

10 after landing.

11 I would check it before every flight because I

12 felt like Charlie had to have a safe airplane. I checked

13 the blades, the turbine blades, to make sure that there was

14 no fractures or no debris because he went through many

15 ground-to-air attacks.

16 I watched the films as he would come back because

17 I was in charge of six airplanes. I would watch the films

18 to make sure that my bomb loaders were loading the bombs to

19 be on target, to make sure we were hitting our designated

20 targets. Come to find out, in those films, we say them

21 bombing chemical factories. We found canisters on

22 unexploded ordnances, SCUDs, that were sitting within five

23 miles of our base with canisters of biological weapons made

24 by the United States, that they were using against us, that

25 was sold to them during the Iraq/Iran war.

30

1 And ladies and gentlemen, that's a fact. I was

2 there. I saw the films and to this very day, I am so sick

3 that I can't work and I am so sick and I am fixing to lose

4 everything that I worked for for 19 years, because the VA

5 won't realize that we're sick.

6 DR. LASHOF: Thank you very much. There are no

7 further questions, we do appreciate your coming today.

8 Todd Temple.

9

10 STATEMENT OF TODD TEMPLE

11

12 MR. TEMPLE: Hi. I, unlike the rest of these

13 people, know where I was in Iraq. I was fortunate enough to

14 have maps. I worked on military intelligence track for 4th

15 Battalion, 64th Armored of the 24th Infantry Division. We

16 were the ones that spearheaded the attack into Iraq. We

17 never went into Kuwait, we went to what they called the

18 supply depot or the supply point, which turned out to be

19 Khamisiyah.

20 We advanced, kept advancing. We would blow up

21 ammunition along the way, get into fire fights, continue

22 marching on and on. We went straight for three days, we

23 never stopped except when we would outrun our support, we

24 would have to stop.

25 Before we went into Khamisiyah, the tactical

31

1 operations center for 4th Battalion, 64th Armored, came

2 under artillery attack. There was a big cease fire. We

3 didn't know if it was our own artillery or if it was

4 somebody else's. All we knew is we were near a bunch of 55-

5 gallon drums that were commonly used as markers of pre-

6 plotted points for artillery. We advanced right to the back

7 end of the tanks, got in the middle of the fire fight, which

8 we weren't supposed to be in. We continued to advance, went

9 through Jabar air field, we also went to Ramoyah (ph) oil

10 field, where we stopped. That's where we had our cease fire

11 finally. I got an award for this because I drove for three

12 days straight, refused relief, always volunteered for

13 security. I shouldn't have been given an award for not

14 sleeping -- I couldn't.

15 If any of you guys would ever take the anti-nerve

16 agent pill that I was taking, you wouldn't be able to sleep

17 either. My heart would race, I had diarrhea, I was sick,

18 nauseous, nervous, chemical alarms going off. Especially in

19 that, you've got so much anxiety, you don't want to go to

20 sleep because you're afraid you won't wake up.

21 Let me back up for a moment. On August 25th of

22 1990, I left work early from the 24th Infantry Division,

23 went and bought flowers and took them home to my wife. I

24 had tears in my eyes when she opened the door because I had

25 just had a two month old baby. She knew where I was going,

32

1 we had suspected it. The next day I went in, I got so many

2 shots that I could barely move my arms -- typhoid, malaria,

3 you name it. I got more shots than I've got -- I've got the

4 records for what shots I had.

5 I left and landed in Saudi Arabia on August 27th

6 of 1990, which was my anniversary -- kind of a happy

7 anniversary. And stayed there, we were deployed out into

8 the desert for Desert Shield. We didn't know if we were

9 going to go over the border yet or not, we didn't know what

10 we were going to do. I stayed in the desert, we kind of

11 made our own little porta-potties out of materials that we

12 could get and I also was put on duty for burning feces and

13 burying feces.

14 I didn't get to take a shower for the first four

15 months that I was there. We didn't have tents for the first

16 three months that I was there. We were in the middle of the

17 desert. We stayed there until January of 1991 and then

18 started to slowly move around until we finally went to the

19 main point where we were going to leave and go over the

20 border.

21 When we finally went over the border, like I said,

22 we kept advancing and attacking and attacking and then we

23 got the cease fire, we had met our objective. We were all

24 happy, but during that cease fire, a day after, you probably

25 heard of a minor incident of the 24th Infantry Division

33

1 attacking an armored convoy trying to go back into Iraq.

2 That was us, it wasn't minor, it was 400 armored vehicles.

3 We did that and we stopped. After that, we had to blow up

4 ammunition, we destroyed things, confiscated weapons, a

5 whole multitude of things. And fortunately, since we were

6 the first ones there, I got to be one of the first ones

7 back. So we left.

8 When I came back, I had diarrhea, was skinny.

9 That's probably because -- a lot to do with the heat,

10 started suffering rashes. I had a rash in my groin area for

11 over two years that I couldn't get rid of and finally got

12 rid of through a private physician. I've had skin lesions

13 since I got back, I've been diagnosed with PTSD. I'm always

14 sick. My immune system is so low, it's not even funny. I

15 was working with six people last summer at my current job

16 with ticks climbing all over every one of us and I was the

17 only one that got Rocky Mountain spotted fever. It was

18 originally diagnosed as lymphoma. I have bad reactions to

19 medication. I've been rushed to the hospital here in town

20 for an emergency CAT scan because I was having blackouts and

21 seizures. It was from taking Lorazepam, one milligram of

22 Lorazepam.

23 I've recently been prescribed and told that I'll

24 be on it the rest of my life, Paxal. I don't know if any of

25 you know what that is, it's an anti-depressant. I've had

1 indigestion problems so bad, and they've called it reflux

2 disease now, that I take 20 milligrams of Pepcid on a

3 regular basis. It's nothing uncommon for me to have severe

4 stomach problems. I have diarrhea two, three times a month.

5 My legs will give out on me. I have sore joints, I have

6 aches and pains in my spine, headaches that won't go away.

7 I keep going to the VA here and they say it's all

8 in your head. Well, the headache is, but that's it, that's

9 the only one that's only in my head. I get turned down,

10 shunned away, don't get much support from this local VA

11 Hospital.

12 I opted to -- I'm fortunate, I work full time on a

13 full time job that allows me a little bit of flexibility

14 because they care about me. I had 108 hours of sick leave

15 last year and burned up more vacation being sick on top of

16 that. And my company paid me. I get private insurance.

17 It's to the point where last week, I had such a hard time

18 getting my prescription filled and getting an appointment to

19 be seen for Paxal, that I'm not supposed to stop taking, I

20 went through my private physician and got it prescribed.

21 I hear all these stories, I read articles. I've

22 got friends looking out for me. But the only way I hear of

23 anything -- the only way I heard of this meeting today was

24 through my mother. If it wasn't for my mother knowing about

25 this meeting, I would have never known. There's a total

1 lack of communication for people trying to help or give us

2 information. I'm on every registry known to man for Persian

3 Gulf illness and unfortunately all I do is send them stuff

4 and they've never sent me anything. Nobody communicates

5 with us.

6 I think there's a real serious lack of

7 understanding or medical monitoring. They say there's

8 medical monitoring, I don't see -- if I go to the VA, I

9 don't see one team of doctors, one team of doctors that

10 would be monitoring my illnesses and being able to compare

11 them with other people's illnesses around the United States

12 from the Gulf War.

13 Like I said, I work full time and when I'm not

14 sick, I need to work to feed my family. I don't have time

15 for all these appointments. I try to, but I don't have

16 time.

17 I was playing softball one summer and my legs gave

18 out on me. Well, I ended up with big pains in my leg, my

19 left leg had such pain I was having a hard time walking and

20 I couldn't even go to work for three days it hurt so bad.

21 But my whole body just gave out on me.

22 I went to the doctor because it kept bothering me

23 two weeks later, and I saw a nurse practitioner instead of a

24 doctor and I kept asking her, I said I want an answer, what

25 is it, I don't understand. I said something is wrong, I

1 want an answer. So she got frustrated with me because I

2 wasn't going away that easy. My mom was with me, they go

3 and get a doctor to come in and the answer was do you want

4 more medication? That's not the kind of treatment or answer

5 I'm looking for. I'm looking for somebody to help me.

6 I left my country and kissed my two month old

7 daughter goodbye on August 25th of 1990 and didn't think I

8 was ever coming back. Well, I did come back and now I

9 deserve it and every man here that was in the Persian Gulf

10 war deserves it.

11 MS. HILLS: And woman.

12 MR. TEMPLE: This country owes it to us. Every

13 man and woman, yes.

14 DR. LASHOF: Thank you very much. Are there

15 questions that anyone needs to ask?

16 Let me just stop a minute and ask if Corinthea

17 Stack has arrived.

18 (No response.)

19 DR. LASHOF: If not, I understand your mother

20 wanted to make -- take a few minutes and speak, and so we'll

21 recognize her at this point. I will have to ask you to be

22 brief because we are running late. Thank you very much.

23 Thank you very much for coming, Mr. Temple.

24

25 STATEMENT OF JAN TEMPLE

37

1 MS. TEMPLE: Thank you. My name is Jan Temple,

2 and I'd like to speak from two perspectives -- one of a

3 parent and then one of an observer. We had two sons in the

4 war, so the price of being a patriot to this country, we

5 know.

6 When our son was there and when he came back, the

7 form of help Todd had from the VA was very loose and very

8 weak until I went with him. I would bet that the majority

9 of the military personnel were running through the same

10 experience. Our son had problems psychologically,

11 physically, emotionally, mentally, neurologically and there

12 were a lot of weaknesses in each of those that we could

13 address.

14 I now am concerned for our grandchildren. Our

15 three year old couldn't move the other day, he was in sweats

16 and we couldn't figure out what was wrong even when we took

17 him to the hospital. During Todd's time while he was here,

18 numerous calls have been made, we've taken him to the

19 hospital, we had EMS, EMS has indicated to us on a number of

20 occasions the symptoms that our son had for the blackouts

21 were very common in Gulf War veterans.

22 One of the concerns I have as an observer and

23 somebody that deals with the areas of risk and risk

24 communication, as I look around this room, I'd like to ask

25 how many active personnel that have been in the war are in

38

1 this office -- in this meeting. The communication

2 oftentimes is one way and not two way, nor is it at the

3 level of where it needs to be to hear from the veterans who

4 served. We may need to look at optional approaches and I

5 would like to share some of those with you at a later time.

6 But I think we have got to do that. We are only hearing

7 from a few, and the majority that are here are hitting it on

8 a hit and miss basis.

9 I tracked this meeting and it was in the newspaper

10 approximately ten days ago and I've been watching the paper

11 every day since and I have seen very little. And I've been

12 looking at all the papers in the state, I've been listening

13 to the radio, I've been listening to the talk shows and

14 watching the TV, and this room should be packed based on the

15 number of veterans that have served in that war that are in

16 the state of South Carolina.

17 Other things I am concerned about are looking at

18 the issues associated with biomarkers. This basically

19 reflects a biological event in the system, when there's

20 either an exposure, an effective exposure or the

21 susceptibility of the diverse individuals and I think we're

22 at a cornerstone here with this committee to take that

23 message back, that we've got to be looking at issues

24 associated with biomarkers with the various veterans that

25 have served.

39

1 The denial of exposure -- and that has been very

2 much the issue here because the issue of trust and

3 credibility is very optimum -- denial of exposure is

4 stopping us from addressing health concerns based on

5 symptoms that we're hearing now. There are health problems,

6 from the very few people that are here, there are definite

7 health problems.

8 I also want to address the concern of active duty

9 versus inactive duty and looking at those as potentially two

10 control measures and consideration of looking at physicians

11 who are practicing physicians. The majority of physicians,

12 studies show, have no background in environmental

13 implications, chemical, biological warfare issues. Do they

14 have a direct line of communication other than through the

15 VA, that we have heard is not meeting the needs?

16 There will be more issues in the future. I love

17 my son and I want to help him. And I at least have the

18 contacts to do that. What about everybody else?

19 Thank you.

20 DR. LASHOF: Thank you very much.

21 That concludes our public testimony and we'll now

22 proceed on to the report on the Marine breaching and Fox and

23 256 detections, LTC Art Nalls from the Chemical and

24 Biological Warfare Agents Division.

25

40

1 Marine breaching and other Fox and M256 detections: DOD

2 investigations

3

4 MR. VESSER: Good morning, Dr. Lashof. My name is

5 Dale Vesser and I am Dr. Rosker's deputy, appearing in his

6 stead for the Department of Defense today. With me are MS

7 Anne Rathwell Davis, Chief of our Investigations and

8 Analysis Division and her Deputy for Chemical Weapons, LTC

9 Art Nalls, U.S. Marine Corps.

10 Dr. Rosker regrets that he cannot be present

11 today. As agreed with your staff, he is continuing

12 previously scheduled town hall meetings with veterans on the

13 west coast. In these meetings, he is collecting information

14 on their concerns about treatment and care for Gulf War

15 illnesses, and sharing information about our investigations

16 and efforts to find out what is making them sick.

17 You asked us to testify about a number of chemical

18 incidents today. Before I introduce our witnesses, I want

19 to tell you a bit about myself. I have served over 42 years

20 in the Defense Department, mostly in uniform in the Army.

21 During the Gulf War, I served in a civilian capacity as an

22 Assistant Deputy Undersecretary of Defense, whose duties

23 included organizing, planning within the office of the

24 Secretary, to support the courses in the Gulf. I went to

25 the Kuwaiti theater of operations during Desert Shield and

1 after the fighting ended, oversaw preparation of the

2 Department's official Gulf War history.

3 Given this experience, I find Gulf War illness

4 issues frustrating because we lack so many answers for basic

5 questions. Therefore, I am dedicated to finding more

6 information to give answers to veterans felled by what we

7 have yet to understand.

8 Today, you have asked that LTC Art Nalls tell you

9 what we know about our investigation of the Marine Corps

10 breaching incident and Fox vehicle and M256 kit detections.

11 His background qualifies him to speak in depth about our

12 approach to chemical warfare incident investigations and to

13 the relevance of ongoing military operations on the

14 incident.

15 LTC Art Nalls will now address these issues.

16 DR. LASHOF: Thank you very much. I appreciate

17 you coming, Mr. Vesser.

18 LTC NALLS: Good morning, Dr. Lashof and members

19 of the Committee, ladies and gentlemen. I'm LTC Art Nalls

20 and I've been asked here to address a range of issues

21 concerning chemical and biological warfare agents. Since

22 February, I've been the Chief of the Chem/Bio Warfare Agents

23 Division of the Investigation and Analysis Directorate,

24 working for Ms. Davis.

25 Chem/bio warfare agents are obviously of great

42

1 concern to many of the veterans and many of whom that I have

2 spoken with, and I am pleased to be here today to address

3 how we have refocused our organization, how we have

4 reprioritized and what assets we've put toward it, and how

5 we're going about business.

6 This is an overview of what I'll talk to today.

7 I'll cover our chemical incident process and how we're going

8 about answering the questions, who, what, where, when, why

9 and how. You've asked for an assessment of -- our

10 assessment of the Fox vehicle, the M256 kit and the training

11 aids during Desert Shield and Desert Storm and that is key

12 to understanding what happened during some of these events.

13 Then I will update on two cases under investigation for the

14 Marine breaching operations of the 24th of February and the

15 Fox vehicle tapes that we've received from Edgewood. And

16 this afternoon, LTC Morris will pick up with this same slide

17 and she'll cover the other four events that are listed

18 there.

19 VOICE: Excuse me. Is there a reason why the

20 audience can't see this?

21 DR. LASHOF: No.

22 VOICE: Is there a way it could be rearranged so

23 the audience could at least see what is being presented?

24 DR. LASHOF: Is that possible? We'll see what we

25 can do. Pretty much, they're saying what's on the slide, I

43

1 think you'll find, but we'll look into it. It certainly was

2 not a deliberate effort. Try tilting it, please.

3 And LTC Nalls, I would appreciate it if you would

4 concentrate in on the 256 and the Fox and the breaching and

5 short-circuit the discussion of how you're set up and what

6 investigations are going on since we've had that.

7 (Pause.)

8 DR. LASHOF: Can the committee sort of move this

9 way? Well, Andrea, I think you can see from there.

10 MS. KIDD-TAYLOR: Yes.

11 DR. LASHOF: But have a seat at the other end. We

12 do have slides in our book, so you can follow the slides in

13 your book.

14 LTC NALLS: Yes, ma'am, you have all these slides

15 and I can zip through this pretty quickly. I understand the

16 time constraints.

17 DR. LASHOF: We've got a time crunch.

18 LTC NALLS: Yes, ma'am.

19 To understand why people are sick, there are two

20 key pieces to the problem. The first part is what medical

21 science says in drawing a link between what we know and the

22 second part is what actually happened during the Gulf War.

23 And that is my mission as the Chief of the Chemical Weapons.

24 And both parts must be interactive and feed information back

25 and forth to people.

44

1 Next slide please. Last year when I testified, we

2 had two people on the Persian Gulf investigation team

3 studying chemical weapons and I wanted to emphasize that we

4 have increased that significantly. We've got over 26 people

5 and we're prepared to add more people as the need be, but we

6 have military people at the top, myself as the head, and a

7 chemical officer as my deputy, and I have five teams

8 organized to study the highest priority incidents under

9 investigation.

10 Next slide please. The recommendation made to us

11 in your Final Report was that all reports of positive 256

12 kits and Fox detections be thoroughly investigated. We have

13 a plan to do that.

14 Next slide please. We took the original matrix,

15 which was not intended to be a product, but a tool, and that

16 was a starting point and we aggregated that by time, date

17 and location, because there were multiple reports of what

18 turned out to be the same incident. And then we

19 prioritized, being that we do have finite resources, and we

20 had to pick the most important things to investigate. So we

21 prioritized them and that's where we started.

22 Next slide please. The types of things that we

23 are investigating. This is the general guidance that I gave

24 to my investigators, that we would interview the appropriate

25 people who were directly involved. That includes the

1 operational commanders, the medical people, NBC personnel

2 and substantive experts on background for how the equipment

3 works. We would also research the medical health of

4 individuals, research weather and environmental factors and

5 we would document as thoroughly as we can going back to the

6 original source document. And then once we have that

7 picture, we would coordinate externally to the other

8 agencies within the U.S. government. And we would prepare a

9 narrative and that is our product to the American people --

10 a narrative of what we know.

11 Next slide please. This slide is in your packet,

12 this is a more detailed methodology of how we're going to do

13 investigations and it is based in part upon the United

14 Nations protocol for chemical armaments verification.

15 I'd like to point out that the Fox vehicle is a

16 very, very key part of this and it's in the upper line

17 there, detections and secondary detections for corroborating

18 the evidence under Fox, CAM, 256. It is a very, very

19 important piece of our investigation, but it is not the only

20 piece of information that we're searching for and it must be

21 taken into greater context of everything else that was going

22 on around it.

23 Next slide. Now I'll focus on the Fox

24 reconnaissance vehicle and our assessment. The Fox

25 reconnaissance vehicle was operationally fielded by the

46

1 German army prior to the war, but it was designed for the

2 mission of warfare in Europe. It was considered by the U.S.

3 Army for a non-development procurement with some

4 modifications to Americanize it and because of the urgency

5 of war, the German government donated 60 vehicles for Desert

6 Shield/Desert Storm. We had to take something off the shelf

7 and we had to get it quick.

8 It is designed -- the primary mission this thing

9 was designed for was as a reconnaissance vehicle for heavy,

10 persistent ground contamination. And that is a very

11 important point. For the warfare as the Germans saw it in

12 Europe was heavy, persistent ground contamination. It is

13 designed that it will give an alert when a chemical agent is

14 present and because we want to be absolutely sure that that

15 vehicle will alert if something is there and because it is a

16 two-step process -- it gets an alert and requires the

17 operator to go back and confirm it through a detailed

18 spectral analysis -- because of those design considerations

19 and because of the mission, we have accepted that there

20 could be false alarms generated. You get the initial alert

21 and the operator goes back and performs a secondary follow-

22 up step.

23 And because it was designed for heavy ground

24 persistent contamination, it is not a suitable airborne

25 detector. And that is also a key point. The air flow that

1 goes in through the sampling tube is on an order of

2 magnitude of 300 to 400 times less than the M8A1 alarm --

3 very, very low air flow rate and it is not a suitable

4 airborne detector. It can be used in that capacity, but the

5 operator has to understand what he's getting out of the

6 machine. We have a lot better equipment designed

7 specifically to sample airborne vapors.

8 And also, as a result of Desert Shield/Desert

9 Storm, the operator is not able to perform an M256 kit test

10 from inside the vehicle. This has implications for

11 determining when it is safe to unmask and the 256 kit will

12 cover that a little bit later. But there were several

13 reports where someone did not follow up with a 256 kit, and

14 that's important, because he cannot do it from inside the

15 vehicle. He can't open up the vehicle and go outside to do

16 that.

17 Next slide please. For operational use in Desert

18 Shield/Desert Storm, it was designed for ground

19 contamination, but found it, because of the tactics and the

20 scenario, being used many times as an airborne detector.

21 And during combat operations -- this is a key point that I

22 would like to make, especially as it pertains to the mine

23 field breaching operations -- troops were ordered to

24 expeditiously maintain the momentum going through the mine

25 field. A mine field is a very dangerous place for troops to

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1 be because they're subject to direct and indirect artillery

2 fires and they don't want to be there very long. And so

3 they were directed to maintain the tactical momentum and go

4 through the mine field.

5 That prevents them from stopping and doing a more

6 detailed spectral analysis as we would like to do if we had

7 the luxury of time. They were not permitted to do that

8 during Desert Shield and Desert Storm, specifically in the

9 breaching operations.

10 Because the equipment was so new, the tactics were

11 not well defined or standardized between the units. On the

12 issues of training, crews were expeditiously trained at two

13 schools, both the German school and at the Army chemical

14 school in Alabama. And they had a time line to meet. We

15 were on the way to war and we had to do the best we could

16 with what we had at the time.

17 What is turning out to be a significant factor in

18 our investigation though is that there were no tactics in

19 place or no doctrine in place that required people to

20 archive the tapes. The tapes that we have, someone thought

21 that they were important enough that they kept them. And

22 that's the way we found them, somebody kept them as a

23 souvenir and some way they've made their way forward, but

24 there was no procedure for them to keep the tapes and

25 archive the tapes. And that's unfortunate.

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1 Next slide. We have a slide of several comments

2 from various documents that are posted on the Internet that

3 show that the Fox vehicle was not a perfect system, as it

4 was fielded for Operation Desert Shield and Desert Storm.

5 We've collated those on a handout and I'll be glad to

6 provide a handout and it has Internet addresses where people

7 can go and search some of these other documents themselves.

8 Next slide please. Now for the 256 chemical agent

9 detection kit. This is a Ministry chemical set and its

10 primary purpose is not as a confirmation tool, but it is to

11 determine when it is safe to unmask. The key factor here is

12 that it takes about 15 to 20 minutes for proper results and

13 it requires some familiarity and some training in order to

14 do it properly.

15 Next slide please. Accordingly, there are several

16 things that can cause misleading results. It's color -- you

17 read the results by color and if a person has color

18 blindness or uses them at night, that's an obvious

19 possibility of getting misleading results, but you can also

20 get misleading results if you use outdated samplers, -- and

21 some were in the theater -- if the samplers become

22 contaminated or if they're wet or exposed to heavy rain.

23 But there are numerous things that can lead to a

24 misleading result and the directions to the troops are if

25 you get a positive result, take out another kit and do the

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1 test again -- another 15 to 20 minutes.

2 Next slide please. We also have some comments

3 from the 256 detection kit that showed that it had some

4 problems in the desert environment and it had some problems

5 that require the operator to be familiar with it and

6 thoroughly proficient with it. And again, we've taken some

7 comments from the Internet and we've collated those and we

8 have a handout concerning that. The point being is that was

9 not a perfect system either, as it was fielded for Operation

10 Desert Shield and Desert Storm.

11 Now if we take both -- if we take that bit of

12 knowledge and we look at the cases that are under

13 investigation, I think that sheds a little bit of light on

14 some of the things of where we are and where we're going.

15 Next slide please. First, I'll address the Marine

16 breaching operations. There are many reports of Marines

17 encountering possible chemical agents and the Marine

18 breaching operations are focused only on the 24th of

19 February, through the two mine field belts in southern

20 Kuwait. Those other incidents that have been reported in

21 testimony are covered under other investigations. But our

22 purpose is to determine what happened in the Kuwait mine

23 fields as the Marine divisions attacked north into Kuwait.

24 There were two reports of chemical detections by

25 Fox vehicle crewmen, both of whom are here today and will

1 speak later on and the similarity between the two, one on

2 the western side of the battlefield and one on the eastern

3 side of the battlefield, caused us to combine them into one

4 investigation. There are also reports of possible chemical

5 injuries, so that made this a very high priority for us.

6 Next slide please. To date, we have interviewed

7 many of the Fox crewmen, the divisional, regimental,

8 battalion, NBC officers and NCOs. We found one of a

9 possible three individuals who may have been injured. We

10 have interviewed many of the eye-witnesses, personal eye-

11 witnesses, to this young Marine. We've interviewed the

12 operational commanders of the Marine units, the medical

13 officers and personnel.

14 There are several reports of this. One was

15 reported in what has been called the Marine Corps history.

16 It is the Marine Corps -- the Marines in the Persian Gulf

17 1990 to 1991, and the 2nd Marine Division, specifically

18 mentions this incident. It is also repeated in several

19 operational logs and records and reflected up the chain of

20 command and we have the original yellow canary NBC report.

21 We also were fortunate enough to locate the

22 original Fox vehicle tape, which we have with us here today

23 and we're prepared to lay out on the table and discuss that

24 with any of the Marines who were involved in that, if they

25 would like further information on that. We have an

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1 assessment of that tape which has been provided to Dr.

2 Brown, and so we were fortunate to find that original tape.

3 Next slide please. We've had some unproductive

4 leads though. There still doesn't appear to be any

5 substantive evidence to confirm or deny the agent was

6 present. The source was suspected to be a chemical mine;

7 however, there has not been one single chemical mine located

8 in Kuwait during the war or after the war during contractor

9 operations to clear 350,000 mines -- not one of them was

10 found to be a chemical mine. So that is unusual.

11 Even though we have the original Fox tape, the

12 tape shows that the operator ran a spectrum, the spectrum

13 did not print to the tape, so we do not have the evidence to

14 sit down and analyze that in the laboratory here.

15 We are still interviewing additional witnesses to

16 the potential injuries and attempting to locate additional

17 possible injured personnel -- the other two Marines.

18 Next slide please. Much of the evidence we have

19 is anecdotal. There is nothing wrong with anecdotal

20 evidence except that it is often contradictory, even to

21 things such as weather, which way the wind was blowing, it

22 was raining, the sun was shining, the oil well fires. There

23 were no secondary tests done for confirmation performed, no

24 subsequent 256 kit tests or anything. The only thing we

25 have is the Fox vehicle and the Fox vehicle tape and the

1 tape itself was inconclusive.

2 Also of note is that the mine fields were heavily

3 saturated at the time with oil from the oil well fires, oil

4 soot, smoke. It saturated the ground, it saturated the

5 equipment, it saturated the people. And they are known

6 interference with the MM1 mass spectrometer. The status of

7 this case is that we have an interim report that's in review

8 and we expect to publish very soon.

9 Next slide please. The next case you've asked me

10 to talk about is the Edgewood Fox reconnaissance vehicle

11 tapes. This is an investigation into the operational events

12 surrounding seven tapes that were submitted to

13 Chemical/Biological Defense Command by the Department of the

14 Army for analysis. The tapes were received by an unknown

15 individual who intentionally deleted identifying information

16 from the tape and it makes it difficult to locate that

17 individual for follow-up investigation.

18 It was initiated after the Persian Gulf

19 investigation team visited CBDCOM in 1995 and knew of the

20 existence of the tapes and initiated an investigation into

21 the operational events surrounding them.

22 Next slide please. We've interviewed personnel at

23 CBDCOM who analyzed the data. We've interviewed the program

24 manager and the results and analysis of that tape is -- has

25 been -- there was a written response to the Deputy Chief of

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1 Staff for Operation and Plans in this memorandum that's

2 noted here.

3 Next slide. We've had some unproductive leads so

4 far. We've been unable to positively identify the actual

5 locations. Even though the grid coordinates are printed out

6 on a Fox vehicle tape, the two identifying letters that

7 precede those coordinates are missing. They are never on

8 the tape, and so that leads us to -- we have to guess where

9 we think a probable location was or some other piece of

10 information that ties us to that. We've been unable to

11 identify the specific vehicles, the specific Fox vehicles,

12 that took these tapes. We have, however, just had a recent

13 string of luck in that in interviewing some of the

14 operations officers with S3 and G3 of the various units, we

15 have been able to find a unit that does match -- does appear

16 to match up with the location of these tapes, at least on

17 the surface. And that operations officers has provided us

18 some leads that we are following up right now as we speak.

19 We have people back in Washington making phone calls trying

20 to identify the specific people involved with these tapes.

21 So we are proceeding.

22 We are still seeking the original tapes, the unit

23 identification or any of the crew members. We are trying to

24 correlate these with Fox crewmen from an outreach. We have

25 an active plan to go out and talk to every single Fox

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1 vehicle crewman, to try to give us something else to follow

2 up on this.

3 Next slide. Even with that late breaking

4 information, the technical assessment does not change. The

5 technical assessment available from the tapes as they were

6 presented highlighted several operational errors, omitted

7 key steps, most notably they didn't perform a spectrum

8 analysis and it did not print out to the tape. None of the

9 detections matched against the spectrum library of 60

10 compounds and the detections were at very low levels, that

11 in the minds of the analysts who looked at the tapes said

12 this is indicative of a false alarm.

13 One event that did have a complete spectrum and

14 did print out the spectrum, under further analysis showed

15 that it was normal atmospheric air, there was no chemical

16 warfare agent there.

17 Next slide. This is a direct quote from that

18 memorandum that I mentioned earlier. There was a team of

19 experts that looked at this, military and civilian, and

20 based on the evaluation, they cannot confirm any of the

21 reported chemical warfare agent detections from the

22 information supplied. But equally important, nor can they

23 deny with 100 percent certainty that chemical warfare agents

24 was detected by the MM1.

25 That's their conclusion; that's not our

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1 conclusion. As I said, we are continuing to investigate the

2 events surrounding that and we're going to dig deeper.

3 That's all I have for this morning, I'll be --

4 DR. LASHOF: Thank you very much for the speed

5 with which you spoke. I know we put you under tough time

6 constraints. But I think there are questions that the staff

7 and committee may have.

8 Andrea.

9 MS. KIDD-TAYLOR: I have a few questions. It goes

10 back to one of your earlier cites about the Fox vehicle not

11 a suitable detector for airborne vapors.

12 LTC NALLS: Yes, ma'am.

13 MS. KIDD-TAYLOR: It's more for ground

14 contamination basically.

15 LTC NALLS: Yes, ma'am.

16 MS. KIDD-TAYLOR: Now the second question is

17 regarding the M256 kit. Is that like a universal air pump

18 where you actually can --

19 LTC NALLS: No, ma'am. It -- did you guys bring

20 one? I think that's going to be covered in more detail by

21 Mr. Vigus and Mr. Stern, who will follow us. But it is a

22 small kit, about this big (indicating), and it has numerous

23 little chemistry experiments, for lack of a better way to

24 describe this thing.

25 You crush ampules, you rub things on different

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1 parts of it and you wave it in the air for 15 to 20 minutes

2 and that is -- by waving it in the air, you get air flow

3 over these things and hope to get a reaction. I think they

4 have one that they can show you on a slide and we have some

5 actual kits back in Washington we would be glad to let you

6 see.

7 MS. KIDD-TAYLOR: I'm curious, when it changes

8 color, is it changing color for a specific chemical, which

9 is what a universal pump does.

10 LTC NALLS: Yes, ma'am.

11 MS. KIDD-TAYLOR: And whether it gives you one or

12 several --

13 LTC NALLS: I think that question is better suited

14 for Mr. Vigus and Mr. Stern.

15 MS. KIDD-TAYLOR: All right. So then I guess the

16 other question that I have, given that the Fox vehicle was

17 not suitable for airborne vapors, what else did you have

18 that could actually detect levels of chemicals.

19 LTC NALLS: That's a very good question. Every

20 single Fox vehicle that we had was outfitted, when we

21 Americanized them, with an M8A1 alarm. The alarm was

22 physically inside the vehicle, the sensor was outside the

23 vehicle, specifically for that purpose, to pick up airborne

24 vapors, recognizing that the Fox doesn't do a very good job

25 of that.

1 Also, the threshold -- another reason why it is

2 unsuitable, the threshold detection level for the Fox

3 vehicle, depending on the substance, can be as much as 1000

4 times higher than some of our other detectors. And I also

5 have a handout that has a little chart that speaks to that,

6 that I'll be glad to provide. It's an excellent article

7 that addresses some of the chemical agent detectors and some

8 of their capabilities.

9 MS. KIDD-TAYLOR: I'd be interested in that.

10 DR. LASHOF: Mark.

11 MR. BROWN: Art, under the -- in your description

12 of the memo that was written in 1993 that represented the

13 Department of Defense's own experts' analysis of these

14 tapes, the Edgewood -- so-called Edgewood tapes, that you

15 picked up --

16 LTC NALLS: Yes, sir.

17 MR. BROWN: -- that were donated anonymously.

18 That memo, as I recall, also mentions that there was an

19 anonymous -- someone who anonymously brought the tapes

20 forward. Is part of your efforts to try and track down that

21 individual since that individual may have more information

22 that might supplement -- in the memo that you cited, it said

23 that you can't be 100 percent if there were or were not real

24 detections there. It seems to me that trying to track down

25 the original source might be useful.

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1 LTC NALLS: Yes. That is still an active case

2 under investigation. We're trying to locate that

3 individual, realizing that they took positive steps not to

4 be located. We think the way we're going to be successful

5 in doing that is by calling one of the other members of that

6 particular crew. There were four people in that Fox vehicle

7 and maybe we can talk to one who says yes, I was in this

8 particular unit, I do remember that. And as I stated

9 earlier and would like to reiterate, we were fortunate

10 enough to talk with an operations officer and he showed on a

11 tactical map -- his tactical map -- where they were day by

12 day and the Fox coordinates, making the assumption that

13 there was -- if they had a Fox with them, which he said they

14 did, those coordinates match up very closely, which leads us

15 to believe that this tape may come from one of his Fox

16 vehicles. And that has given us some more --

17 MR. BROWN: But you have not been able to locate

18 this individual yet, but you are still trying to?

19 LTC NALLS: No, sir, not yet. We have people

20 making phone calls right now.

21 MR. BROWN: Similarly, this same memo that makes

22 this point that, you know, you can't be 100 percent certain

23 based on the evidence at hand, also mentions a named

24 civilian that has -- might have more tapes. Is there

25 efforts to try and locate that? I didn't notice that on

1 your list.

2 LTC NALLS: We were. That's the dead end we ran -

3 - one of the dead ends we ran into. When we interviewed the

4 people up at Chemical/Biological Defense Command who were

5 still there and that we could get ahold of who had some

6 familiarity with this information, they were unable to shed

7 any light on it. It apparently was just passed to them with

8 a tasking from Department of the Army, take a look at these

9 tapes, convene some experts and tell us what you think.

10 MR. BROWN: Well, I guess the reason it sounded

11 interesting is that the implication in the memo was that

12 this person might have additional information that could

13 help you come to some certainty about what these tapes say.

14 But you think now that that's a dead end, that you'll be

15 unable to --

16 LTC NALLS: It has been a dead end so far. We

17 have not been able to find that person.

18 MR. BROWN: Are you still investigating --

19 LTC NALLS: I'm sorry?

20 MR. BROWN: Are you still pursuing it?

21 LTC NALLS: Yes, sir.

22 MR. BROWN: Thanks.

23 MAJ CROSS: LTC Nalls, can you just review the

24 analysis of the tape again, what the experts found, what are

25 some of their comments?

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1 LTC NALLS: I have a slide that addresses that.

2 If we could go back to slide number 20. If you would put

3 that up please, Caroline.

4 The technical assessment was based on the

5 information they had at the time and there was something to

6 suggest that it was an incomplete tape. But the parts that

7 they did have showed that the operator had omitted some very

8 key steps.

9 MAJ CROSS: Operating error?

10 LTC NALLS: Operating error, yes. And that may be

11 a function of the training, it may be a function of the

12 operational scenario. Remember, there may be something else

13 here similar to the Marines going through the breaching

14 operation where they couldn't stop -- which is key.

15 Ideally, you would like to be able to stop the vehicle,

16 you'd like to be able to back it up to the spot where you

17 first got the contamination. You'd like to be able to cool

18 the probe down and take a more detailed -- a bigger sniff of

19 air, a better sample, and run through a complete analysis of

20 it. And you'd like to take some soil samples too if you

21 could. So we don't know what events were surrounding this.

22 But based on what the technical assessors had to look at,

23 they said well there were some key steps here that were

24 missing, but when they did run a spectrum, that nothing

25 matched up with the substances in the 60-substance library.

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1 That's key, because there are 60 substances that you're

2 asking the computer to make a detailed match of, that are

3 the suspected chemical warfare agents that Iraq had at the

4 time. So you're matching it up with something that you

5 suspect is out there and when it doesn't match, that's key

6 to the possibility of being a false alarm. And I think Mr.

7 Vigus and Mr. Stern can probably add a little more detail on

8 that.

9 MAJ CROSS: The last bullet there where it says

10 one event did include a complete spectrum, was that in the

11 breach area?

12 LTC NALLS: No, these tapes were not associated

13 with the Marine breaching -- this particular tape was not.

14 So --

15 MAJ CROSS: Do you know what event this was

16 associated with?

17 LTC NALLS: No, sir. I was getting back to the --

18 this is the tape that was provided anonymously to Edgewood,

19 that we don't have sufficient identifying information to be

20 able to pinpoint exactly where it was and who was in the

21 vehicle. And that's what we are actively trying to find out

22 now.

23 MR. VESSER: I think it's appropriate that we say

24 that we think that the unit that this Fox vehicle was

25 assigned to could have been near the depot of Khamisiyah on

1 the day that the 24th Infantry Division took it, but we

2 haven't been able to validate that yet. We only got this

3 lead on Monday this week and have spent a good deal of time

4 on it, but that's the extent of our knowledge on these

5 particular tapes.

6 MAJ CROSS: Now you also talk about the M8

7 detector, which also was used I believe along with the Fox

8 vehicle. But the drawback with that, that doesn't give you

9 a printout or any analysis, it's just an alarm.

10 LTC NALLS: No, that is similar to a smoke

11 detector in your house, type of thing.

12 MAJ CROSS: Right.

13 LTC NALLS: You have a remote agent and it's

14 connected by a com wire to an alarm box. The Marine Corps

15 had some allocated to them, but through testimony and

16 interviews, we've talked with people and because Marine

17 Corps operations were very dynamic, the only ones that were

18 used were the ones inside the Fox vehicle -- it's my

19 understanding. Someone else may have a different opinion on

20 that, but we haven't found any Marine units that actually

21 took out their M8 alarms and deployed them, other than some

22 headquarters units.

23 MAJ CROSS: One of the things I'm getting to is

24 that we heard testimony earlier today -- we've heard it

25 throughout the previous year -- that alarms were

1 continuously going off and obviously people ignoring them

2 because at some point they go off, they go off and it's a

3 regular occurrence. The drawback there is well if they do

4 go off, there's no record of it, there's nothing to

5 substantiate any of that. So you're adding to the confusion

6 here.

7 The Fox vehicle is the only thing that will give

8 you a printout, but --

9 LTC NALLS: That's correct.

10 MAJ CROSS: -- the tactics suggest that you want

11 to confirm that because of the false positives it could give

12 you. Sometimes you did it, sometimes you did not.

13 LTC NALLS: That's right. And we have an example

14 -- we actually have the original tape that was in MSG

15 Bradford's vehicle. We were fortunate enough to locate

16 that, someone kept it as a souvenir and when we interviewed

17 that person, we asked the question, oh, by the way, do you

18 know of anybody that has any Fox tapes, and he said well

19 yeah, I've got one. And we were obviously quite excited.

20 He went back and got the tape and we laid it out on the

21 table and analyzed it and it verified what MSG Bradford had

22 said, that he'd run a spectrum, however the printout was not

23 there. There's another step that requires the operator to

24 hit a print button that would actually print it to the tape.

25 That's unfortunate that wasn't done, because that would have

1 given us a piece of concrete, hard evidence to analyze

2 today. And you don't get any of that with the M8s or any of

3 the other detectors.

4 MS. KIDD-TAYLOR: I just have an additional

5 question regarding that. So what I'm hearing is that you

6 could not actually measure airborne concentrations with the

7 record, the follow-up record or tape. The question that I

8 have is for future events such as that, is there any kind of

9 work ongoing or research to try and develop that type of

10 equipment?

11 LTC NALLS: Yes, ma'am, in a lot of respects the

12 Fox vehicle has been modified several times since Desert

13 Shield/Desert Storm. This was a learning experience for us

14 and we've taken steps to make some very significant

15 modifications. One is that we've added a printer -- I'm not

16 sure if this is true of the Marine Corps ones, but it is

17 true of the Army ones -- they've added a printer screen that

18 repeats up for the vehicle commander. During Desert

19 Shield/Desert Storm, the vehicle commander, who sat up

20 front, could not physically look at the MM1 screen and look

21 over the operator's shoulder without crawling through back

22 to the back and having a look at that. That's been

23 improved.

24 The vehicle locating system, VLS, counts wheel

25 revolutions, it doesn't work very well in the desert. We've

1 modified that with a global positioning system, much more

2 accurate, much more reliable system.

3 And procedures have been in place that require

4 people to archive, to save the tapes.

5 DR. LASHOF: You indicated that you are planning

6 to interview all of the personnel that were involved. There

7 were what, 60 Fox vehicles deployed and you intend to

8 interview the crew for all of those.

9 LTC NALLS: Yes, ma'am.

10 DR. LASHOF: Do you have a time table for

11 completing the interviewing of all those individuals?

12 LTC NALLS: I can't give you a firm date today,

13 ma'am. We're taking a look at that. What we've done to

14 date is we've gone back and we have gotten the school

15 academic records from both the German school and the U.S.

16 Army school. We've located -- we have a name and Social

17 Security number of every individual that went through that

18 training. The number is greater than the actual number of

19 people who actually went to Desert Shield/Desert Storm.

20 There should have been about 240-260 people and we've got

21 400 names. So we've had to weed out some of those. We've

22 started the process to locate home phone numbers and

23 addresses for each and every one of them and we've come up

24 with a list of basic questions and our veterans' data

25 outreach people will be making those phone calls.

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1 I can get back to you with an estimate.

2 DR. LASHOF: This is probably not a fair question

3 to you, but it strikes me as I listened to your description

4 of the limitations of the Fox and what we could and couldn't

5 do, that despite the fact that the Iraqis used chemical

6 weapons in Iran, we really weren't prepared to send in

7 equipment that would keep on top of this issue during this

8 war. Is that correct?

9 LTC NALLS: It's my opinion that we did the best

10 we could with what we had, that the people that were put in

11 the position to prepare us for war did the absolute best

12 they could. We got the most up-to-date equipment off of the

13 shelf, we trained our people, they trained in the desert.

14 There are still some things that we could have done better,

15 but -- hindsight being 20/20 -- but given the time

16 constraints they had at the time, I'm not sure that they

17 could have done a whole lot better.

18 MR. VESSER: I would like to address that, just

19 for a minute.

20 DR. LASHOF: Please.

21 MR. VESSER: The war in the desert was a fast-

22 moving war, armored war, a war of movement. And when we

23 went into the war, we had only -- were approaching the

24 threat, we only had in the U.S. inventory the M256 and the

25 M8 alarms. We had no alarm under armor. And that was one

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1 reason that the government at the time approached the German

2 government because we had not purchased equipment that was

3 mounted on armored vehicles so that the crewmen who operated

4 could move as rapidly as armored forces move and have

5 protection and still have some ability to detect. We knew

6 that the vehicle had some limitations because of its design.

7 It was designed, as Mr. Stern will tell you, for warfare in

8 Europe where we expect the Russians to use persistent

9 chemicals rather than the chemicals we actually anticipated

10 might be used against us in the Gulf. But this was an ad hoc

11 measure to try and give our troops the latest in protection.

12 DR. LASHOF: Thank you. Rolando, you had

13 question?

14 MR. RIOS: Yes. In your presentation, you

15 mentioned that there were no chemical mines detected and

16 that that was kind of unusual. What did you mean by that?

17 LTC NALLS: I meant it to be unusual, sir, that

18 surrounding the detections and the alerts in the breaching

19 operations, you have to naturally ask the question what was

20 the source of it. And it was thought by many people that it

21 was a chemical mine. And that verbiage appears in the

22 Marine Corps monograph that was originally suspected of

23 being a chemical mine. However, every single report of a

24 chemical mine that was followed up by explosive ordnance

25 destruction personnel, EOD personnel, every account of a

1 chemical mine that was followed up by the collection people

2 who were in place specifically looking for samples, looking

3 for munitions, looking for things of this kind, not one

4 single chemical mine was found. There are some initial

5 reports suspected to be a chemical mine and then the follow

6 up report, one in particular was found in the city and we

7 located the individual, talked to him about it and he said

8 yes, that was a -- it was a homemade thing that somebody had

9 made, it was obviously designed to look like a chemical

10 mine, it was very crude at that. It was not a chemical

11 mine.

12 MR. RIOS: Did they use chemical mines at any time

13 in the war?

14 LTC NALLS: We have not found evidence of a single

15 one, sir. And after the war, a contracting company was

16 hired by the government of Kuwait to go in and clear 350,000

17 mines, and we've interviewed those people -- did you ever

18 find any mines, not in the heat of battle but in the course

19 of clearing these mine fields -- did you ever find a single

20 chemical mine and the answer was emphatically no. And they

21 were looking for them.

22 MR. RIOS: Have chemical mines ever been used in

23 the Middle East?

24 LTC NALLS: I can't answer that question, sir.

25 MR. VESSER: There was an intelligence report that

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1 Iraq had the capability to make chemical mines. But as LTC

2 Nalls suggested, no chemical mine was found in the theater

3 of operations or in any of the ammunition storage areas that

4 we're aware of that were seized by U.S. forces.

5 MS. GWIN: I have one question. In your

6 description of the Marine breaching case that you're working

7 on, you indicate that the evidence that you're now looking

8 for is primarily based on injuries, you're looking for

9 injured individuals. Does that mean that you will use that

10 kind of information if you find it, to confirm or deny the

11 incident?

12 LTC NALLS: Yes, ma'am. That is a key piece of

13 information. If we could go back to our process slide,

14 which is slide number 8. The issue of confirmation is

15 something that we as a team have wrestled with for sometime

16 trying to determine what would we consider a credible or

17 confirmed chemical incident. And Dr. Rosker's latest

18 statements are that we would use a common sense approach

19 based on the entire -- all the evidence that we have --

20 everything. You consider it not out of context, but in the

21 greater context of everything going on here. And under

22 medical aspects, number 2 there, -- I'm sure the audience is

23 not going to be able to see that, but the types of things

24 that we would consider in our decision in coming to whether

25 or not this is a confirmed incident or not, are deaths and

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1 autopsies. Obviously we've already looked for that and

2 there were no chemical-related deaths and none of the

3 autopsies support that.

4 The next line down is injuries and Purple Hearts.

5 There was one Purple Heart awarded in the entire theater.

6 That's the Fisher incident. But none awarded in the entire

7 Marine Corps. This individual who was supposedly injured

8 had submitted himself through his chain of command for a

9 Purple Heart award after the war. He said look, I had an

10 injury to the back of my hand, I think I deserve a Purple

11 Heart. It was the heat of battle here. There was an

12 investigation convened at the time, we have statements from

13 the medical personnel who examined him, and they had

14 determined that it was not -- did not meet the criteria for

15 a Purple Heart. We have resurfaced that investigation with

16 the idea well, okay, this might not have been substantial

17 enough for a Purple Heart, but was it a chemical injury.

18 And that is a very key part of our investigation, to

19 determine if this was in fact -- no matter how minor it was

20 -- was it due to chemical warfare agents. So that is a key

21 part of the overall scenario.

22 MS. GWIN: And you would use that to verify a

23 detection.

24 LTC NALLS: Oh, absolutely -- absolutely, yes,

25 ma'am.

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1 MAJ CROSS: Are you in contact with that

2 individual now?

3 LTC NALLS: Yes, we are. And if I can -- if I've

4 got the time to explain a minute about how we're going about

5 doing that, it took us a year to find this individual. He

6 got out of the Marine Corps, he moved three or four times,

7 he had an unlisted phone number each time. It took a lot of

8 investigative work to locate the individual once we had a

9 name and Social Security number.

10 We located him and we've located several people

11 who were in his unit and physically saw him and then we

12 hired two civilian experts in chemical warfare agents and

13 their effects, to go and interview these people and

14 interview all the witnesses who specifically saw the

15 injuries. And we've asked as a product from them their

16 assessment -- not Art Nalls' assessment, but their

17 assessment of whether this is a potential chemical warfare

18 agent injury. And we are still in the middle of that

19 process.

20 DR. LASHOF: This question of on what basis --

21 what criteria were used to finally confirm has been an issue

22 this Committee has been asking for a long time, I think. Do

23 you consider that you must have medical evidence in order to

24 confirm? And in the absence, you would -- is that a sine

25 qua non for you?

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1 LTC NALLS: No, ma'am. That certainly adds to it,

2 but that is not an absolute requirement.

3 DR. LASHOF: Okay.

4 MR. TURNER: While you have that slide up,

5 Colonel, is this a methodology that DOD intends to use to

6 evaluate every incident of potential chemical exposure, this

7 detailed UN-based methodology?

8 DR. LASHOF: We use this as a guideline for the

9 types of things that we're looking for and the types of

10 people we're going to seek that information from. Each

11 investigation will be slightly different, but we'll use this

12 as a starting point and we hope to be able to answer these

13 types of questions. For example, if we're interested in

14 determining whether or not there were exposures, there's a

15 question up under -- it's under number 1 A, on the upper

16 left-hand side. It says "Did the unit go to MOPP 4?"

17 MR. TURNER: Sure.

18 LTC NALLS: If you're looking for exposures, that

19 is a very key issue -- did the unit -- was the alarm

20 sounded, did the unit put on their protective gear.

21 MR. TURNER: Yeah, I'm not questioning the

22 individual components.

23 LTC NALLS: Okay. But there are some differences.

24 MR. TURNER: Certainly there'll be differences for

25 each case.

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1 LTC NALLS: Yes.

2 MR. TURNER: Could you give the Committee some --

3 you said it took a year to find this -- track down this one

4 fellow. Can you give the Committee some idea what kind of

5 time frame you think applying this kind of a detailed

6 methodology to an individual case is going to require to

7 crank through to an answer?

8 LTC NALLS: Yes. As I stated earlier, a year ago

9 when we testified, we only had two people working on the

10 chemical warfare investigations. Now we've got 26 and that

11 number is prepared to grow as the need may arise. We very

12 shortly here are going to be coming out with the first

13 series of narratives and issue papers that inform people

14 about the issues -- the general issues on Fox vehicles,

15 M256s and what-not. We're going to publish those also and

16 then the investigating teams will pick up a second round of

17 -- I'll throw more work at them. So we expect by the end of

18 the year, we will get another series of narratives out.

19 MR. TURNER: So by the end of this calendar year.

20 LTC NALLS: Yes, sir.

21 MR. TURNER: Thank you.

22 DR. LASHOF: John.

23 MR. LONGBRAKE: I just had a -- I saw that you put

24 up the conclusions from the PMNBC memo and last year you

25 testified about the Marine Foxes before this Committee and

1 you told the Committee -- you said that we have not been

2 able to substantiate any of them. In fact, every instance

3 where a chemical detection caused the unit to upgrade to a

4 higher MOPP level was determined to be a false alarm. And I

5 just wanted to know if that was still your position.

6 LTC NALLS: I think -- the rest of that paragraph

7 I think was a quote from the 11th Marines Command

8 Chronology, where they had had several instances -- was I

9 quoting the 11th Marines? I don't have the testimony in

10 front of me.

11 But your question was do I still stand behind

12 that?

13 MR. LONGBRAKE: Yes.

14 LTC NALLS: I'm not prepared to draw conclusions

15 on anything until after all of the evidence is in, I think

16 it would be premature to do that. But we haven't found the

17 substantive evidence that would lead me to say right now

18 here today that any of these are confirmed.

19 MAJ KNOX: I just have one more question about the

20 positive detection. You said that it was a positive

21 detection, but it was not -- if I understood you right -- in

22 the library of the 60 chemicals that were programmed?

23 LTC NALLS: Did I use the word "detection?" I

24 think I used the word "alert."

25 MAJ KNOX: Okay.

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1 LTC NALLS: And there's a difference.

2 MAJ KNOX: So even though there was not a database

3 or it wasn't printed, what did the Fox vehicle operator say

4 that the chemicals were?

5 LTC NALLS: It would come up most likely with an

6 "unknown" but I think that that is a more appropriate

7 question to ask Mr. Stern or Mr. Vigus after me. I think I

8 used the word "alert."

9 DR. LASHOF: Well, that's okay.

10 MAJ KNOX: You may have.

11 DR. LASHOF: We will take one more, because I

12 think a number of these issues, as you say, are appropriate

13 for Mr. Stern and Mr. Vigus.

14 MR. KOWALOK: I have a quick question for either

15 GEN Vesser or LTC Nalls. I understand you've made

16 improvements in the protocol for using the Fox vehicle for

17 chemical agent monitoring. Other than changes in

18 procedures, technologically are we any better off than we

19 were in 1990, such that if we needed to deploy to the Gulf

20 tomorrow, would we be better prepared now for detecting

21 chemical agents?

22 MR. VESSER: You have an expert, I suggest you

23 defer that question on the technical side of the vehicle. I

24 would say that we're better prepared in one respect. We

25 have worked closely to look at lessons learned in the Gulf,

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1 and I would say that we're in process of putting in place

2 procedures that will take advantage of the improved training

3 of the Fox vehicle crew members -- and remember that this is

4 a newly improved Fox vehicle that they'll talk to you about

5 on details, but we also will not only create tapes in a way

6 that we didn't create tapes during the Gulf war, but we will

7 keep them as records, so that we will be able to answer

8 questions like this. I spent yesterday morning with the

9 Deputy J7 on the joint staff, who is responsible for

10 doctrine. He understands what we need to incorporate in the

11 next doctrinal publication so that we have records that will

12 at least be as adequate as our technology is about exposure.

13 DR. LASHOF: Thank you very, very much. We

14 appreciate this.

15 We're going to move on and hear from Mr. Vigus and

16 Mr. Stern, who are going to talk further about the technical

17 analysis. I'd like at this point to say that we are running

18 late and I'm taking the prerogative of saying we're just not

19 going to take a break because I think these are important

20 issues and we need to take the time. So if people feel they

21 need a break, they can take it at their own convenience, but

22 we'll keep going straight through until our lunch break.

23 MR. YOUNCE: Is there anyone in here from the VA

24 Hospital? Is there a regional representative here? All

25 this data is real good, but the only thing I'm going to get

78

1 to go home and tell my wife after this today is that I'm

2 going to get a flag from the VA to go across my coffin and

3 $250 for burial.

4 DR. LASHOF: I'm sorry, sir. We've been holding

5 hearings across the country and we've heard from the VA on

6 the medical. There's a great deal more information that is

7 available to you and can be obtained either from the

8 Internet or from our office. There is a Final Interim

9 Report, a Final Report. We'll have other reports. Today's

10 session is to deal with this problem of what were the

11 detections, what were people exposed to. We can't cover

12 every issue at every meeting, and we've tried to progress

13 over the period of time to cover the issues. And the kinds

14 of issues you're asking about we have covered at various

15 time, we will be covering again, but there is a great deal

16 more information. And we'll be glad to have staff get in

17 touch with you and provide a resource to you.

18 Dr. Fran Murphy from the VA Headquarters is here

19 and possibly at the break, you could talk with her.

20 I'll take one more question from the floor, but we

21 really have to keep on schedule. Yes?

22 MR. SIMMONS: Thank you, Madam Chair. My name is

23 LT Simmons, I'm with the 251st Evac Hospital previously. Of

24 course, I'm out now, but I wanted to give you an autographed

25 copy of my second book, I hope it is going to give you some

1 more answers to the Gulf War syndrome.

2 DR. LASHOF: You have testified previously, I

3 understand, and we can have staff be in further touch with

4 you. I couldn't really hear your question, but we'll have

5 to ask staff to meet with you and discuss any additional

6 problems or issues you wish to bring up.

7 So I'd like to thank Mr. Vigus and Stern for

8 joining us now. We're going to talk about some of the

9 technical issues that we need to learn a lot more about.

10 Mr. Stern.

11

12 Fox and M256 detections: DOD technical analyses

13

14 MR. STERN: Thank you. Good morning. My name is

15 Bob Stern. Since this is my first occasion to meet with

16 you, I thought it necessary to give you a quick biography,

17 so that you understand why I'm here.

18 I've been employed at the Chemical and Biological

19 Defense Command for just under 35 years. My degrees are in

20 physical and analytical chemistry and I spent the first 18

21 years of my career at the bench, developing chemical warfare

22 detection equipment and working with live agents. I've

23 progressed through various management and admin functions,

24 and during Desert Shield and Storm, I worked in the office

25 of the Army Project Manager for NBC defense systems. During

1 that period, I was responsible for funding, acquisition and

2 fielding support of several detection systems which were not

3 already fielded, and which were being expedited to the war

4 effort.

5 My purpose today is to teach a simplified class in

6 mass spectrometry, so that you can better understand the

7 operation and the operational procedures of the MM1 mass

8 spectrometer which is in the Fox or the FUCHS. And I am

9 also going to quickly discuss the 256 kit. You will not be

10 able to pass a chemistry class when I am done, but you will

11 have an understanding of the systems that we currently use.

12 Mr. Vigus will then discuss the analysis done at

13 CBDCOM on these seven tapes and if we have both been

14 successful, you will understand how we read and interpret

15 the tapes, based on what you heard from the class.

16 I also have some videotapes of the Fox in

17 operation, and depending on our timing or maybe during

18 lunch, they're available if we want to look at them. Please

19 stop me at any time if there's a question or I need to

20 better explain something.

21 This is the 256 detector ticket. I've brought one

22 here and you guys can take a look at it. It is designed,

23 and the soldiers are trained to use it, as an all-clear kit.

24 Soldiers are to unmask only when a clean 256 test is

25 obtained. It is basically a high school chemistry set on

81

1 the ticket. There are liquid reagents that are in glass

2 ampules, they react with each other and the chemical agent

3 vapor gives a specific color change.

4 Ms. Taylor, for your question, they're broken into

5 categories, as you can see from there and you can see when

6 you handle the ticket. You'll get a blister category, blood

7 category, nerve or leucyte. It will not give the exact

8 agent, it is done by categories. Okay?

9 The one thing that I want to specially point out -

10 - and you can see it fairly good in this viewgraph -- is

11 that the detection of all agents except nerve gives a color

12 change. The ticket starts out white and if there is an

13 agent present, you get a color. In the case of nerve

14 agents, it goes from color to white. So it's opposite. The

15 reason being, the nerve agent test is actually a living

16 animal, it's an enzyme. When the enzyme is alive, it is

17 blue, when it's dead, it has no color. The kit is designed

18 solely to detect dangerous levels of agent vapor so that you

19 can remove your protective mask. It is operated when you

20 are wearing a protective suit, gloves and mask. Only when

21 the complete 20 minutes sampling and analysis is done and no

22 agent detected, are soldiers supposed to remove their masks.

23 Because we're not allowed to operate the ticket in

24 a room like this, due to vapors being given off from the

25 reactions, the one that you all have was done outside and I

82

1 won't say where.

2 (Laughter.)

3 MR. STERN: It was done last night and that's why

4 some of the colors have changed and the liquids have bled on

5 the tickets.

6 If you're interested, I have with me a big data

7 sheet that gets more into the actual chemistry and I didn't

8 think it was necessary for here.

9 The question that I've most recently been asked

10 many times is what happens if you get a partial detection.

11 The problem is I have not been able to get anybody to tell

12 me what they mean by a partial detection. But the obvious

13 answer is you take another sample for another 20 minutes

14 with a fresh ticket, and look at it.

15 During Desert Storm, the soldiers were given a

16 manual which was quickly prepared which had suggestions in

17 it for operating in the desert environment. I brought a

18 bunch of those with me. The thing you want to notice in

19 there is that in the 256 kit area, they're told to save

20 enough liquid to rewet the enzyme ticket a second time

21 during the test. Under those dry conditions, the enzyme

22 liquid would start to dry out, the enzyme would either die

23 or get inactive and the color would start to go back to

24 white. This is why they had to wet it a second time. If

25 procedures weren't followed, this could also give you a

83

1 partial test where the enzyme itself was either dry or dead.

2 Okay, let's do the Fox. Any questions on the 256

3 that we can handle?

4 (No response.)

5 MR. STERN: Okay. First, let me make sure that you

6 understand what the Fox is. This photograph is the XM-93

7 Fox nuclear, biological and chemical reconnaissance vehicle.

8 You've also heard it called the Fox, the TPZ-1 FUCHS, NBC,

9 the Recon system or the Recon truck. It is a six by six

10 wheeled, armored vehicle. That means six wheels and all six

11 are driving wheels. It's standard in the German army, they

12 donated five dozen of them to us during the conflict. They

13 also sent eight to Israel and 11 to the U.K. As you heard,

14 the soldiers were given a short training course in Germany

15 or here, and then they, with the vehicle they trained in,

16 were sent to the war.

17 The liquid chemical warfare agent detection heart

18 of this system is the MM1 mass spectrometer. The class

19 today is to education you on how the MM1 works in the Fox.

20 Let me add, on those referred to as the nuclear, biological

21 and chemical recon vehicle, we do not in the Army have the

22 capability to detect biologicals at this time. The only

23 thing the Fox can do is sample the ground for biologicals,

24 puts it in a sealed bottle and ship it someplace else for

25 analysis. The Army now has just started building a

1 biological detection vehicle system.

2 This is the most detailed graphic you guys are

3 going to see. It shows the major parts of the system and

4 before I'm done, you will hopefully understand the part that

5 each piece plays in the analysis. The box in the center

6 represents the mass spectrometer system, analysis system,

7 computer, and the heart of the sweep.

8 The blue rectangle on the left refers to the inlet

9 system or the mass spectrometer probe that a sample travels

10 through to get inside the system. What passes through that

11 tube is the vapor from a liquid sample, which has been

12 vaporized at the inlet. That tube can also sample outside

13 air, but it is not designed for that purpose.

14 The thing with the circle on the end represents

15 the sample wheels or the way a liquid sample is introduced

16 to the inlet. And the little round odd-shaped animal on the

17 left is the liquid that we want to look for.

18 First, let me design mass spectrometry. In the

19 world today, molecules are combined to make compounds. The

20 compounds are the building blocks of organic chemicals. All

21 chemicals have a molecular weight or mass. The molecules

22 are held together by different strength electrical bonds.

23 When a chemical is exposed to energy, whether it's heat,

24 electrical energy or radioactive energy, the chemical gets

25 excited and starts to change.

85

1 If you hit the chemical with the proper

2 combinations of energy, you can break the chemical into

3 pieces. These molecular pieces, or ions, will always be the

4 same for a specific chemical under the same energy

5 conditions. If you catalog the pieces by their size or

6 their mass and the relative intensity, how high the peak is,

7 then we get a spectrum. Under the same conditions, the mass

8 spectrum of a chemical will always be the same.

9 Therefore, we can build a library of mass

10 spectrums for any chemicals that we can analyze. Although

11 they can be similar, no two chemicals will give the exact

12 same mass spectrum.

13 Mass spec instruments go from very small that they

14 use for analyzing gases in space to large, laboratory room-

15 size university tools that have huge computers and a great

16 deal of power. The MM1 in the Fox weighs about 450 pounds,

17 it runs off 24 volts DC power on the vehicle.

18 When we match the spectrum obtained from a

19 chemical to the exact same spectrum in the mass spec

20 library, it is considered a positive identification. If a

21 mass spec gets more than one chemical sample at a time, the

22 spectrum reflects this. Separation of the sample into its

23 individual chemicals is critical.

24 In the real world, lots of compounds exist on the

25 air and the ground. A typical mass spectrum is cluttered

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1 with lots of ions. Keep that in mind as I get further into

2 the class. Are we okay so far? Sir?

3 MR. RIOS: These devices, as you explained, as

4 designed to measure what you called dangerous levels. In

5 other words, there are levels below which they cannot

6 measure, and there could be levels out there that are in

7 existence that troops could have been exposed to, but that

8 you can't detect with either of these two devices.

9 MR. STERN: Sir, nothing that the Army uses for

10 detection is designed to detect what you're referring to and

11 what I've been taught are chronic levels. All of our

12 systems are designed to warn a soldier when he must take

13 immediate action to protect himself to prevent being unable

14 to fight. The only ticket which is designed for lower

15 levels than that would be the 256 or the M18, which are

16 designed for an all-clear purpose, to tell him okay, it's

17 clean enough to take your mask off.

18 MR. RIOS: But even those are designed to a

19 certain level and not below that level.

20 MR. STERN: That's correct, not for chronic

21 levels, long-term chronic exposures.

22 MS. KIDD-TAYLOR: Actually what we're talking

23 about here is mainly ground contamination versus airborne.

24 MR. STERN: Yes. And I'll get into that and

25 you'll understand how it works for ground liquid

87

1 contamination.

2 MS. KIDD-TAYLOR: Okay.

3 MR. STERN: Okay, let's try the next one.

4 As I have said, a critical part of getting a good

5 mass spectrum is that the system should break apart so that

6 there's only one chemical getting in at a time. For best

7 results, we need to separate the stuff in the air or in our

8 sample into each of its individual chemicals. And then do

9 the analysis on the separate chemicals.

10 The sample inlet tube is designed and filled with

11 coated particles that absorb and desorb organic vapors at

12 different rates. They soak into the particles and then they

13 come back off the particles. This is a principle of gas

14 chromatography, but that's another class.

15 As the sample travels through this tube, it

16 chromatographs, or it separates. Lower molecular weight

17 chemicals travel faster up the tube and heavier chemicals

18 travel slower. If the tube is heated, all the chemicals

19 inside will travel relatively fast and there is less

20 separation. When you cool the tube, the chemicals slow down

21 and you get more separation. The Fox is operated in both a

22 hot and cool mode, and you'll see on the tapes and hear it

23 referred to, as either high or low. And I'll get into that

24 next. Okay so far?

25 (No response.)

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1 MR. STERN: This is probably the most important

2 point I want you to understand. Mass spec works best when

3 the spectrum is obtained from one single pure chemical. It

4 certainly is not possible in the normal environment, and

5 much less the diversified sample you would get on the

6 battlefield.

7 The inlet probe on the Fox is designed to do the

8 best separation of vapors possible under the limitations of

9 time, power and mobility. Separation of vapor takes time.

10 In order to meet mission responsibilities, the system is

11 first operated in wheel high or what we can call a survey

12 mode. And you'll see on the tapes it says "wheel high".

13 The separation tube is heated to 180 degrees Centigrade.

14 Chemicals that are similar in either molecular weight,

15 molecular structure or even the vapor pressure can and will

16 travel up the tube almost together. Consequently, the

17 analyzer will not see one chemical, but several at a time.

18 Ion fragments from more than one chemical can combine and

19 produce an initial false positive test.

20 This leads to combined spectrum peaks. The Fox

21 MM1 uses only four peaks and their relative intensity for an

22 initial alert or warning. This is only the first step in a

23 complete detection verification.

24 For speed and safety, if the ion mass matches and

25 the relative intensities are close for a chemical which we

1 have programmed to alarm, the system will alarm. The high

2 temperature is necessary to get the initial sampling speed

3 which is critical to performing a safe mission. The Army

4 has decided to accept this initial alarm potential in order

5 to get rapid analysis. And this is why the next step is so

6 critical.

7 If an initial alarm occurs, the operator is to

8 stop the vehicle and cool the probe. He switches to the

9 surface low or what I call verification method. The probe

10 is cooled down from 180 to 120 degrees Centigrade. This

11 permits the liquid or the sample to be resampled, heated and

12 its vapors can travel more slowly up the tube. It gives the

13 vapors a chance to separate into the individual chemicals.

14 Since the analyzer is only looking at one chemical

15 at a time, all the peaks will belong to one chemical and an

16 exact match can be made with the library. This match is

17 presented as a spectrum plot and it can be printed for later

18 review. Soldiers are trained to run the spectrum for entry

19 into the on-board database. Printing a copy is an option.

20 For this reason, an alarm verification in the Fox can only

21 be made when both wheel high and surface low operations have

22 been completely executed. The second step must confirm the

23 initial alarm. Unless the surface low, or second step, is

24 completed, an initial warning is not confirmed and not

25 considered a true detection. Total sample and analysis time

1 could take up to 20 minutes.

2 DR. LASHOF: If you want to move through the next

3 one quickly, we do have all of this in front of us.

4 MR. STERN: Okay. Let's run through the rest kind

5 of quick.

6 This is the last chart that I had. Basically what

7 it does, the sample wheel in the red there -- the sample

8 wheel rolls along the ground, contacts the liquid, it's

9 routinely raised up to the bottom of the probe and then the

10 probe, which is hot, vaporizes what's on the wheel.

11 All the soldiers, again, are trained that without

12 the second step or the verification, no initial alarm is

13 confirmed or accepted as a positive alarm. Okay?

14 Mr. Vigus will go into the analysis of the seven

15 tapes. Let me take a quick second and state that we have

16 had the OSD team up to my shop and I strongly urge you,

17 either as a group or individually, if you're in our area, to

18 come spend some time with us in the Edgewood area, touch,

19 feel, see the hardware, learn how all of the detection

20 equipment works and I think it will help you get a better

21 feeling for the chemical detection side of the house.

22 Mark and his people have been there and I think we

23 had a good day.

24 DR. LASHOF: Thank you very much. Did you want to

25 ask Mr. Stern something before Mr. Vigus?

1 MS. KIDD-TAYLOR: Yes, I did. I just wanted to

2 ask one question about the chemicals that can be detected.

3 That is done manually to set the system for a particular

4 chemical out in the field? That's something I don't

5 understand exactly. How do you determine which chemical --

6 MR. STERN: In the Fox?

7 MS. KIDD-TAYLOR: Yes.

8 MR. STERN: In the Fox, in the computer's library

9 or the computer brain, they have entered 60 compounds which

10 everyone tells us are likely to be found. One of those

11 compounds is referred to -- and you'll see it on tapes -- as

12 fats, oils and waxes. Organic chemicals, petroleum and

13 stuff, when you break them apart, give you peaks which are

14 close to some of the peaks that you find from organic

15 chemicals that are chemical agents. So they lumped all

16 those together, and you'll see one of the things in its

17 memory is fats, oils and waxes.

18 If the situation was to arise where they got a

19 detection and they did the second step and the machines says

20 unknown, they can enter that into the memory and it now will

21 have 61 and they can give it a name or do what they want.

22 And if they come across that compound again, the machine

23 will bring number 61 up and say that's what I saw. So they

24 can add to that memory while they're out there.

25 The initial setting is only about eight or ten

92

1 items and the soldiers have the option of taking any of the

2 ones out of that memory of 60 that they want to enter and

3 say the first time, look for these half a dozen especially.

4 And you'll see where due to the belief that we had a lot of

5 mustard over there, some of the agents that they didn't

6 expect were put back into the big memory and mustard agents

7 were brought back up into the initial front memory.

8 MS. KIDD-TAYLOR: So it would be about six or

9 seven at one time then?

10 MR. STERN: Rich.

11 MR. VIGUS: What we have in there, they're called

12 methods. And the primary one is called CWA-wheel high.

13 Chemical warfare agent using the wheels with the high line

14 temperature. If you select that method, you get a set of

15 ten compounds. GA, GB, GD, GX, HD, leucyte, phosgene,

16 hydrogen cyanide, GF and fats, oils and wax. That's the

17 primary set of nine agents and hydrocarbons that would be

18 monitored for at any one time.

19 As Bob said, based on intel or whatever other

20 reasons you might suspect some particular compounds, you can

21 add or delete from that list.

22 MS. KIDD-TAYLOR: Just to follow up with that ten,

23 let's say your equipment detects one that is not identified

24 with that ten, so they go to an unknown and you can bring up

25 several from the memory to see if any of those will match

93

1 from the other 50?

2 MR. VIGUS: The balance of those 60, the other 50

3 that are in there, are other types of obsolete chemical

4 agents, precursors, binary compounds, TNT, a calibration

5 gas, some test chemicals.

6 MR. STERN: In the second step when you do the

7 surface low analysis, the computer looks at all 60 and says

8 I went through the whole library and I couldn't find this

9 thing.

10 MR. VIGUS: And if one of those 60 matches

11 exactly, it will identify it as one of those 60 and tell you

12 what it is. If it's not one of those 60, then it says

13 unknown. Now recognize there's hundreds of thousands of

14 millions of organic compounds that exist. You know, the

15 memory in the machine is only so big and most of those

16 compounds, we don't really care what they are anyway. A

17 major portion of them are covered under hydrocarbons, most

18 organic compounds will break down into that stuff and

19 they're covered under the fats, oils and wax.

20 Many compounds that exist out there, especially

21 industrial chemicals or whatever, are also hazardous,

22 produce ions across the spectrum, might be detected by this

23 machine, giving an initial false alarm in the initial scan

24 to a chemical agent. When they do the spectrum procedure

25 then, it would come up as unknown because it's not one of

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1 the 60 that are in the library.

2 MR. STERN: We had some questions ---

3 DR. LASHOF: Are there other statements, questions

4 for Mr. Vigus before we go ahead?

5 MR. STERN: Do you want to go ahead and do that or

6 should I answer the questions about upgrades? That'll take

7 about two minutes.

8 DR. LASHOF: Go ahead and answer the question.

9 MR. STERN: The question was what have we done

10 since then.

11 The improved Fox which we'll start fielding in

12 about two years, operationally we have changed the soldiers'

13 procedures in there, how they use the mass spec. One of the

14 features we have added is what everybody has referred to as

15 a serial spectrum. Although the computer had the capability,

16 no one ever thought of doing this, but what they now can do

17 is tell the computer, okay, you analyzed and you came up

18 with fats, oils, and waxes. Take all the information you

19 used for fats, oils and waxes and throw that away and now

20 take a look at what's left and see if it matches in your

21 library. It's a serial spectrum. That's a procedure we've

22 added.

23 In the detection arena, we have added the ability

24 to do stand-off detection. We only had 12 detectors at that

25 time, but they now will have an infrared sensor that they

1 can erect from the Fox, from inside, which will look at the

2 sky up to five kilometers, and monitor for an approaching

3 agent cloud.

4 The M8 alarm, which does have false alarm problems

5 when it gets around a lot of exhaust gases, will be replaced

6 with the new ACOTTA (ph) you'll hear about, or the XM22.

7 That has shown to be about a five percent false alarm, as

8 compared to 25 percent on the M8. It also can detect

9 mustard vapor, which the M8 cannot, and VX vapor, which the

10 M8 cannot.

11 So besides redesigning the interior of the

12 vehicle, cutting the crew size down, we have upgraded with

13 new detection capability in the M8 area with the M22. We

14 now have given them the ability to do stand-off detection.

15 That will start fielding in about two years.

16 DR. LASHOF: Mr. Vigus.

17 MR. VIGUS: Thank you.

18 I'm Rich Vigus. At the time, prior to ODS and

19 subsequent to that, I was the engineering team leader on the

20 Fox program working in the PMNBC Defense. Therefore, I

21 guess it fell to me to look at any of these tapes that we

22 received.

23 I would say that with the exception of this latest

24 Marine breaching tape, I have looked at all of them, either

25 done the analysis myself or been in a consultative role,

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1 even though I don't work in that organization any more.

2 I was asked to address the panel and present

3 testimony relative to the analysis of the Fox vehicle MM1

4 tape printouts, specifically the analysis of those seven

5 tape printouts as described in the 17 December '93 memo from

6 the PMNBC Defense Systems to the Army Office of the Deputy

7 Chief of Staff for Operations and Plans.

8 Along with other technical experts, I analyzed

9 these tapes to evaluate the validity of the MMI alarms, as

10 requested by DESOPS. The enclosure to that 17 December '93

11 memo is the complete analysis, including my personal notes.

12 As stated in the memo, it represents a combined evaluation

13 by myself, other Army and industry mass spec experts and

14 representatives from the U.S. Army Chemical School. This

15 evaluation was our best effort, given the limited

16 information that was provided on the tapes, and handwritten

17 notes accompanying the tapes and information provided in

18 follow-up telephone conversations.

19 In my opinion, the reported initial alarms were

20 false, as stated in the memo. I will explain the reasons

21 given in the evaluation. Please refer to your copy of the

22 memorandum, which I believe you have.

23 First of all, let me provide a little chronology

24 of the history of these seven tapes. On November 23, 1993,

25 a fax was sent from the CG 3rd Corps at Fort Hood, which

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1 sent these tapes to a fax number that's at FORSCOM

2 headquarters, Fort McPherson, Georgia. On November 24th,

3 '93, the tapes were forwarded by fax from FORSCOM

4 headquarters to DESOPS. On November 29, '93, the tapes were

5 forwarded by fax from DESOPS to CRDEC, which is Chemical

6 Research Development and Engineering Center, at that time at

7 Edgewood, with a cover memo from COL Dosberker (ph), who was

8 the chief of that office, to our office and also jointly to

9 the chemical school at Fort McClellan, Alabama. That same

10 day, I did the analysis, put my handwritten notes side by

11 side on the tape printout as I went through it line by line

12 or step by step.

13 Between November 29 and the 17th of December,

14 myself and other experts were involved in evaluation of the

15 tape, coordinating our response, getting input from other

16 representatives at the chem school, et cetera.

17 Part of that analysis included, on 6 December '93,

18 a telephone conversation among two other representatives at

19 Edgewood, Mr. David Sickenberger (ph) and Dr. Bill Sarver

20 (ph) with the individual who had provided the tapes. That

21 contact was made through a go-between or a POC. The name of

22 that individual was not revealed to those two people. I

23 myself did not talk to that person.

24 So then on the 17th of December, we finalized the

25 memo, COL Colplaser (ph) who was then the project manager

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1 for NBC Defense, sent it through channels up to DESOPS.

2 On 20 December '93, a memo was sent from DESOPS

3 back to headquarters FORSCOM providing the analysis of the

4 tapes. And that's the last of what I know of what happened

5 to them.

6 DR. LASHOF: Can you short-circuit and give us

7 your conclusions and why you feel the way you do? We're

8 just so short on time and staff can supply us with other

9 memos and the other data.

10 MR. VIGUS: Sure.

11 What it says in the memo basically is, as has been

12 said before, we can't confirm that any of the reported

13 detections were in fact actual chemical agent detections,

14 nor can we deny with 100 percent certainty -- and I think

15 that's the key phrase there, 100 percent certainty --

16 because I don't know that you can ever be that. But from

17 the information that's on the tape and our collective

18 expertise, there are several reasons.

19 All of the detections are at very, very low

20 levels. The machine prints out when it makes an initial

21 detection, it might print out sarin or GB and a number.

22 That number is the amplitude or the height of the peak on a

23 log scale of the response level that it gets. Background

24 levels are typically seen in all of our history with the

25 machine, at the two or two and a half level. The warning

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1 levels and the alarm levels are set for the agents at .6 and

2 .9 log units above that minimum detectable amount, which is

3 a function of the background.

4 All of these levels on these seven tapes, where

5 they present chemical agent detections or initial

6 detections, are very low, they're 1.7, 1.8, 1.9 -- very,

7 very low, below the normal background level. I don't know

8 how that normally would occur unless you had a super-clean

9 environment and a super-clean column or probe.

10 None of the detections were verified or confirmed,

11 using the spectrum procedure done properly. And that may be

12 a function of the particular mission or the scenario or the

13 tactics that were employed for that particular operation, as

14 we've heard in previous testimony.

15 When alarms were generated and spectrums were

16 taken, they were taken directly from the air, not from --

17 without cooling the line down and not from going back to a

18 spot and probing the ground. I can tell that from the tapes

19 because they alarms print out every 30 seconds. If you

20 continue to have samples coming in, you get a printout, an

21 alarm, every 30 seconds. So you'll see places where it

22 prints at 9:24, you get an alarm, at 9:24, you get another

23 alarm, that's the same thing. At 9:25 you get another one.

24 And it would go on and on and on until you reduce the level

25 in the line below the detectable amount.

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1 Where they would do a spectrum immediately

2 thereafter, it prints out also at the time, you know, right

3 there. So it takes about ten minutes normally to go through

4 the procedure, to do the spectrum operation correctly and

5 reduce the line temperature.

6 So you've got the very low levels, not doing the

7 spectrum procedure correctly, not cooling the line down.

8 You're getting a spectrum essentially of the same thing that

9 came in in the initial alarm. We've got a lot of other

10 information that was available in the handwritten notes that

11 accompanied the fax, the location where they were -- not the

12 specific location geographically, but the description. They

13 were close to a road, a lot of vehicle traffic, a lot of

14 exhaust. At one location, they were close to an industrial

15 type area where there could have been a lot of other

16 industrial type chemicals or other kinds of material in the

17 air.

18 The indication on some of these reported alarms

19 shows that the alarm was generated just from the air and not

20 from use of the sampling wheels. That's -- some of them,

21 they are, some of them, they are not. That's indicated on

22 the alarm.

23 The one case where they did take a spectrum and

24 printed out all of the masses, is the one that was referred

25 to before by LTC Nalls, as a spectrum of the air. And in

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1 fact, it is. You have nitrogen, oxygen, water vapor, argon,

2 carbon dioxide and several of the ion masses of the

3 calibration gas that's used to set up the machine, present

4 in that listing of the spectrum.

5 DR. LASHOF: If I can, to try to get at the heart

6 of this, if I understand everything I've heard, and I hope I

7 do. If mustard were in low level in the midst of oil and

8 lots of other chemicals, it would be very difficult for the

9 Fox to be able to separate it out by this method used, is

10 that correct?

11 MR. VIGUS: Depends on the level and what the

12 relative level is.

13 MS. KIDD-TAYLOR: What would be considered low

14 levels?

15 MR. VIGUS: I don't know.

16 MS. KIDD-TAYLOR: .05, .1. I'm just trying to

17 figure out what would be considered not detectable by this.

18 MR. VIGUS: This machine is not designed to detect

19 vapors. Okay?

20 MS. KIDD-TAYLOR: I got that.

21 MR. VIGUS: It's intended to detect liquid on the

22 ground. It would respond with an alarm with about a

23 microgram of liquid transferred to the end of the probe

24 membrane. That's, you know, a millionth of a gram.

25 We have done some tests with testing it as a vapor

1 detector. For example, for nerve agents, the M8 alarm for
2 nerve agents G-type agents, is -- responds at about .2
3 milligrams per cubic meter. The MM1 could detect reliably
4 about 100 milligrams per cubic meter. Okay? So that's --
5 MR. STERN: I think to answer Dr. Lashof, if there
6 was a large quantity of oil, fats and oils and whatever, and
7 a little bit of agent, the machine would have identified the
8 fats and oils and ignored the other stuff. That was the
9 procedures in place at that time under most circumstances.
10 MR. VIGUS: Let me hit that again, it depends. If
11 there was a detectable amount of mustard there in a larger
12 amount of fats, oils and wax or hydrocarbon, as I think
13 there may have been from the burning oilwell fires and the
14 smoke and whatever else is there, diesel exhaust, et cetera.
15 It would provide, as is seen in many cases on these tapes,
16 multiple initial alarms. You get a hit for mustard
17 accompanied by a hit for fats, oils and wax. If there were
18 three or four compounds that you were monitoring for and
19 they were all there, you'd get multiple hits within that
20 same 30 second response. And it would print mustard, fats,
21 oils and wax, or whatever.

22 When you do the spectrum procedure then as a

23 second step to verify it, it would look at each of those and

24 say which one is there in the highest response level --

25 which one gave me the highest response level in the initial

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1 alarm. It would check that spectrum against the spectrum of

2 the 60 compounds in the library and if it was in fact a

3 correct spectrum, it would print that one out, only that

4 one. The one that gave it the highest response level

5 initially.

6 That was a limitation of the training and the

7 operational procedures that we provided to our troops at

8 that time, because we were not aware of the additional

9 capability that the machine has, which has now been

10 incorporated in the training and the procedures, to do that

11 spectrum analysis sequentially on each of the initial

12 alarms. So today, if we had a three agent response, or two

13 agents and fats, oils and wax and they did the series

14 spectrum procedure correctly, you would get a yes or no

15 verification for each of the individual alarms.

16 MR. BROWN: But Mr. Vigus, just to be clear about

17 that, to follow that up; what you're saying is if you had a

18 small amount, but a measurable amount of mustard agent, for

19 example, in the presence of a lot of fats, oils and waxes or

20 oil, you would see it in the alarm mode because it's

21 detecting multiple components --

22 MR. VIGUS: You would see it.

23 MR. BROWN: -- but -- you would see it, you would

24 detect it, the alarm would detect -- would show it. When

25 you go, as is required by military doctrine, by operational

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1 doctrine of this instrument, to the full spectrum mode, it

2 would pick up the largest amount of material there and it

3 would give you an answer of fats, oils and waxes, which

4 might unfortunately be the wrong answer.

5 MR. VIGUS: It could be. But recognize why we set

6 the machine up to do that, we wanted to be absolutely sure

7 that we would not suppress any alarms to chemical agent.

8 MR. BROWN: I understand.

9 MR. VIGUS: We didn't want to miss any false

10 negative, which would be much more detrimental and injurious

11 to the --

12 MR. BROWN: I understand.

13 DR. LASHOF: We understand. We want it to be

14 sensitive, we don't want a false negative; therefore, you

15 get a certain number of false positives.

16 MR. VIGUS: Right.

17 DR. LASHOF: But as I understand further -- now

18 that I've gotten that part clear -- you've made it

19 abundantly clear that you need to do the second step and it

20 has to be done at the low temperature 20 minutes after the

21 first. And we've been told that that was impossible in the

22 battle conditions. Therefore, check me if I'm wrong, there

23 is no way we would have been able to confirm. So when you

24 say you cannot confirm that there was, but you cannot deny

25 100 percent, clearly we never can deny 100 percent.

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1 MR. VIGUS: Well, that depends on --

2 DR. LASHOF: But at what percent can we deny?

3 Have we got a 50 percent chance of saying it wasn't there or

4 we've got no ability to say it wasn't there because we

5 didn't do a second and the Fox just wasn't a good vehicle

6 for this particular kind of battle conditions, and it

7 probably doesn't help us much. Is that an unfair

8 conclusion?

9 MR. VIGUS: I would say that's probably a fair

10 assessment, the last part of what you just said.

11 Remember, as LTC Nalls said, the Fox was designed

12 by the Germans, given to us by the Germans, intended to be

13 used in a European scenario.

14 DR. LASHOF: That's right.

15 MR. VIGUS: Now given that the war happened

16 somewhere else --

17 DR. LASHOF: Right.

18 MR. VIGUS: -- and the particular mission and

19 scenario that the Marines used it for in breaching through

20 the mine field, did not lend itself to the real intended

21 design purpose for the vehicle. They in fact used it

22 primarily as a vapor detector.

23 DR. LASHOF: Right.

24 MR. VIGUS: And you see that on their tapes where

25 --

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1 DR. LASHOF: Yeah.

2 MR. VIGUS: -- although they may have been in the

3 wheel high mode, they may not have been operating the wheels

4 and having them coming up and down. That's indicated by the

5 "A's" on the tape rather than the "C's."

6 I don't want to address the doctrinal or the

7 tactics or the techniques of the war, that's not my bag, so

8 --

9 DR. LASHOF: No, and none of us do. I mean we

10 can't refight a war, we can't redevelop new equipment that

11 wasn't available. We have to deal with the reality. All

12 we're trying to get at, and all of us are trying to get at,

13 is when you say we can't confirm that it was there, it

14 wasn't possible to confirm. So from my understanding of

15 what you're telling me about the technology, we didn't have

16 the technology to confirm whether these false alarms were

17 real or not because we weren't able to do the second level.

18 MR. VIGUS: Yes, we were. Only because of the

19 limitations of the material.

20 DR. LASHOF: Well, that's what I'm saying.

21 MR. VIGUS: In cases where they've done the

22 procedures correctly like this one that's on the tape where

23 it's a spectrum of air, I can tell that that's what that is.

24 DR. LASHOF: Right.

25 MR. VIGUS: And I'm 100 percent confident of that

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1 one.

2 DR. LASHOF: Obviously. But you cannot -- you

3 don't have -- I'm not being critical at this point.

4 MR. VIGUS: I understand.

5 DR. LASHOF: I'm just trying to get at what do we

6 know and what don't we know. And from how I hear this,

7 there is no way we're going to know whether or not there

8 were low levels of chemicals there when these false alarms

9 went off, because our equipment wasn't telling us whether it

10 was there. Is that a wrong conclusion?

11 MR. VIGUS: Without the second step, you cannot

12 confirm.

13 DR. LASHOF: Okay. Therefore, when you say we

14 cannot deny to 100 percent, you can't deny to any percent,

15 can you? I mean, that's my training as a scientist, an

16 epidemiologist.

17 MR. VIGUS: Probably so. I know from working with

18 the instrument and other results that we've obtained, either

19 in the laboratory or in field testing, and use of it, that

20 the machine does false alarm. And we have seen it false

21 alarm in places where --

22 DR. LASHOF: Right. No question, there's no

23 question. They could be false alarms, but we have no way of

24 knowing whether they are or they aren't, from what I hear

25 you saying. Now if that's a wrong conclusion, please, this

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1 is a very important issue to us and to our veterans.

2 MAJ CROSS: And the only way that's going to

3 change is if more tapes appear that maybe the full spectrum

4 shows up on the tapes.

5 MR. VIGUS: And in cases where they have hit the

6 print spectrum button and we get a long list of the complete

7 spectrum, or where we have seen videotape and you can

8 actually see the graphical spectrum on the machine itself,

9 you can look at that and say yeah, that ion was there and

10 that ion was there and that ion was there in the right

11 proportion to each other, and then you can make an informed

12 decision. As I have done in some of the other incidents

13 that have been evaluated.

14 But you're right.

15 DR. LASHOF: I mean, eventually we could be sure

16 of a positive at some time, but there's no way of knowing

17 whether the negative is so.

18 MR. STERN: Some of the events that are going to

19 be discussed this afternoon had sufficient data with the

20 mass spec to work on a confirmation.

21 DR. LASHOF: Right.

22 MR. STERN: The majority of the data we've seen on

23 everything except those events, there just was not enough

24 data to say definitely what we had.

25 DR. LASHOF: Right. And not enough data to say

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1 that it wasn't. That's all we're trying to do, is get a

2 level of understanding of the certainty and uncertainty of

3 both positive and negative.

4 Further questions?

5 MAJ KNOX: Do we know whether or not any of these

6 potential alarms coincided with any other positive

7 detections, say for instance did the checks validate it?

8 MR. STERN: That's what LTC Nalls team is

9 attempting to do, is tie together all the various incidences

10 and see which ones hook together. We never got involved in

11 any of that.

12 DR. LASHOF: Mark and then Joan.

13 MR. BROWN: I want to tap into your expertise on

14 some of the other detectors that were used in the Gulf War,

15 specifically the CAM detectors that were used for chemical

16 detection, the M256, the M18A and the M8 and M9 papers.

17 This is a question for either of you. Are either

18 of you aware of the chemical agent monitor, the CAM, to give

19 false positives for mustard detection based on the presence

20 of other interference?

21 MR. STERN: The CAM can false alarm -- the CAM is

22 also a mass spectrometer, just a different kind. The CAM

23 can false alarm certain greases and lubricants. This is why

24 the Fox teams that carried a CAM with them, are taught that

25 before you start a mission, you survey your vehicle when you

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1 know it's clean of agents and determine if around certain

2 wheel bearings or certain areas of the vehicle, you get a

3 false CAM test. So that after a mission when you do a

4 survey, you know whether, if you get another test in the

5 same area, whether it was due to the initial vehicle grease

6 or not.

7 MR. BROWN: Similarly for the M256 kits, which

8 you've already talked about somewhat, and the M8 and M9

9 paper, which we've heard about at other meetings; can you

10 tell us about what would give any kind of false detections

11 for those detectors?

12 MR. STERN: Okay, the M8 paper, which is the

13 three-color paper -- this is why I would love to have you

14 guys at the shop for a day -- the M8 paper, which has been

15 around since World War II, changes three different colors

16 based on the category of agents it sees. The M8 color will

17 give you a false test to bug spray and to certain organic

18 solvents, but you're not likely to find them in the field.

19 This is not a working CAM, I took the rad source

20 out.

21 The M9 paper, that is a paper which is two inches

22 wide, it's a gray-green color and it's sticky backed. They

23 wrap it around their uniform or they stick it on a box.

24 That detects rain-size drops of liquid agents. It will give

25 you a false alarm again to organic solvents, to motor pool

1 fluids, some of the brake fluids and stuff, and bug spray.

2 You'll see bug spray a lot in this, but bug spray is nothing

3 but a dilute agent.

4 The 256 detector ticket will give you a false

5 alarm again to pesticides. It'll see pesticides and it'll

6 act like a nerve agent. Strong bleach will give you a test

7 similar to mustard and leuocyte. Mustard and leuocyte are

8 the same kinds of compounds. So that's the 256.

9 Any other questions?

10 DR. LASHOF: Joan.

11 MS. PORTER: I have a question about red fuming

12 nitric acid, because it has been featured as a confounder in

13 other possible detections. Can red fuming nitric acid,

14 RFNA, give a false reading for mustard in the Fox vehicle's

15 MM1 analyses?

16 MR. STERN: I will ignore the "can it" and say --

17 in the Fox -- I will ignore the "can it" and tell you what

18 we did in the lab.

19 Red fuming nitric in the laboratory, pure red

20 fuming nitric, gave us a G agent false alarm and then in the

21 second step, the verification was "unknown compound."

22 Under other conditions, with red fuming nitric

23 being handled differently or in a different atmosphere or in

24 a different container or with other contamination, I have no

25 idea what it would respond as.

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1 In a pure condition, pure red fuming nitric, and

2 in clean air, we got a G test. But what it could do in the

3 presence of other things, I have no idea.

4 MS. PORTER: Mr. Vigus.

5 MR. VIGUS: I would say that the answer to that is

6 "possibly." The way that the MM1 mass spec works is it

7 looks at four ions for each of the chemicals that are in the

8 monitor list, the 10 or however many are selected to put in

9 the initial list. So normally they would be looking at 40

10 ions of individual masses. The machine has no way of

11 knowing from what source those ions came, other than the

12 sample that it has ingested, either from liquid being placed

13 on the end of the probe membrane, or from vapor coming in

14 from the air.

15 From mustard, for HD, we used four ions, 109, 111,

16 158 and 160. And they have to be present in the right

17 proportion to each other, in order to produce an initial

18 alarm for mustard.

19 MR. BROWN: If I could just interrupt, it sounds

20 like the answer is something like red fuming nitric acid

21 could at least in theory give a false positive for something

22 else, but when you actually tried that experiment to see

23 what the Fox vehicle would alarm for in the laboratory, you

24 see a G agent.

25 MR. VIGUS: Yes, it did, but recognize that that

1 was a laboratory grade --

2 MR. BROWN: I understand.

3 MR. VIGUS: -- probably very much purer than what

4 was in the tank at the girl's school. Okay?

5 MR. BROWN: But I mean theoretically, something

6 could respond as anything, you don't know, in the real

7 world.

8 MR. VIGUS: Well, red fuming nitric acid, when we

9 did it initially, you know, at a small dose, it produced no

10 alarm, had ion activity across the spectrum. We pressed the

11 spectrum button and asked it to take a spectrum, as the

12 troops are taught, even when they don't get an alarm, when

13 they see a lot of activity all across the spectrum. And it

14 took a spectrum and it produced 1-P at mass 46, which is

15 NO2, 100 percent, and that's all that was there.

16 They did it a second time at three times the

17 amount was put onto the probe head, they got a chemical

18 agent alarm. It was for GF, okay?

19 MR. BROWN: Okay.

20 MR. VIGUS: They then did the spectrum, it came

21 back and it said "unknown." And that's proper for the

22 machine to do that because RFNA is not one of the 60

23 compounds in the library. We looked at the graphical

24 spectrum when that occurred, and that's on the videotape

25 that Bob has. It still had 1-P at mass 46.

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1 Now why did that produce an alarm for GF? I don't

2 know, other than the fact that there may have been very,

3 very low levels of the other ions that are also used for GF

4 present in the sample that was ingested. Recognize where we

5 did that, we did that in the back of a vehicle bay, the door

6 was open to the outside air, it's sucking in this gas coming

7 off the RFNA, you could see it, it was brown, there may have

8 been something else. And even though there may have been

9 only a very small amount of some other compound in the air

10 that was ingested, if those particular ions for GF, which

11 are very similar to ions that could be produced from who

12 knows what, any compound in the air and hydrocarbons, and

13 they were there in the right proportion, you could get an

14 alarm. So yes, RFNA produced an alarm in this case to GF.

15 In the Gulf, it may have produced the alarm to mustard.

16 MR. TURNER: But you don't have any evidence that

17 it produces an alarm for mustard. The one true test you've

18 run produced an alarm for GF. That's the evidence. The

19 rest of what you said is really speculation.

20 MR. VIGUS: Yes, it is.

21 MR. TURNER: Informed speculation, but it's still

22 speculation, isn't it, Mr. Vigus?

23 MR. VIGUS: Yes, it is. But what I would say to

24 you, what I would like to add to that is that in the

25 conditions where that occurred, you have to look at what

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1 else was in that tank, what might have been in the tank, how

2 long it had been there, was it pure RFNA, had it broken

3 down, decomposed, who knows what. You look at that incident

4 and look at all of the evidence that was reported in that

5 fax that said, you know, they shot holes in the tank and a

6 brown gas escaped, et cetera; that is pretty indicative of

7 what it was, because there aren't that many gases that exist

8 in the chemistry world that come off brown.

9 MS. PORTER: Dr. Lashof.

10 DR. LASHOF: Joan. We're going to have to move

11 this along.

12 MS. PORTER: One more follow-up question. Your

13 laboratory analysis, as reflected in your report, indicated

14 that RFNA could not fool any of the detectors for mustard.

15 Is that correct? And when the Fox was confused, it reported

16 out cyclo-sarin.

17 MR. VIGUS: That's correct.

18 MR. STERN: At the initial alarm, and then in the

19 verification mode, it said oops.

20 MS. PORTER: All right.

21 DR. LASHOF: Fine. Andrea.

22 MS. KIDD-TAYLOR: I'm still concerned a little bit

23 about airborne detection levels of any of these chemicals.

24 I still don't hear that any of this equipment that we had

25 could detect low level airborne exposures of specific

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1 chemicals.

2 MR. VIGUS: The equipment that we have to detect

3 chemical agents are designed -- other than the M256 kit --

4 are designed to detect tactical concentrations, when an

5 agent is purposely used in a tactical scenario, offensively

6 against our troops. They're detectors, they're not

7 environmental monitors.

8 MS. KIDD-TAYLOR: Okay, right.

9 MR. VIGUS: They're intended to say it's

10 dangerous, put on your MOPP gear.

11 MAJ KNOX: So there is no equipment to detect low

12 level chemical exposure?

13 MR. VIGUS: This M256 is the only thing, and it's

14 intended to say there's none there, it's safe to unmask,

15 it's all-clear. We do not have, for military mission --

16 MAJ KNOX: Exactly. That was the question.

17 MR. VIGUS: -- things that go out and monitor for

18 whatever level, like you might have in the workplace.

19 MAJ KNOX: Right.

20 DR. LASHOF: It remains an open question, and

21 that's what we're trying to get at.

22 MAJ CROSS: Other than some modifications to the

23 Fox vehicle, we're no further along today in our equipment

24 than we were during the war time?

25 MR. STERN: In terms of sensitivity levels or in

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1 terms of capability? In terms of sensitivity levels, no.

2 There is no attempt to make the equipment more sensitive to

3 chronic levels of agents.

4 MAJ CROSS: Because that's not the mission.

5 MR. STERN: That's not the mission. The equipment

6 is being upgraded in terms of as we get better science, in

7 the air sampling point detectors, we're adding mustard vapor

8 and VX vapor and radically decreasing the false alarm rate.

9 So that's a new detector.

10 We now have the ability to do stand-off detection.

11 Up until that time, we didn't know the agent was there until

12 we were in it. Now we can look up to five kilometers away

13 and see a cloud approaching. That is new technology. We

14 have now fielded the biodetection system. Basically it's a

15 laboratory in a trailer, but we at least have some

16 capability now in the battlefield to do biodetection.

17 Next generation, eight to ten years; yes.

18 MAJ CROSS: So we still have a long way to go yet.

19 MR. STERN: Yes, very much so.

20 DR. LASHOF: I think there are a number of

21 technical things that staff will need to follow up, and some

22 of us, and some of it will come out in the other things

23 we're going to have.

24 I'm going to take one more question from Holly,

25 we're running about 40 minutes behind, we'll then ask the

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1 Marines who were in the Marine breaching thing, to come up

2 and testify, and we'll get into this afternoon some of the

3 other specifics. And some of these issues will come out

4 then.

5 MS. GWIN: I want to ask Dr. Stern for a

6 clarification. When you were talking about the ACOTTA as

7 opposed to the M8, you said you expect to get about a five

8 percent false alarm rate from the ACOTTA, whereas the M8

9 gives you a 25 percent false alarm rate?

10 MR. STERN: This is to identify a list of I think

11 it's about 25. We know these are going to be in the

12 battlefield interferences -- smoke, some things like that.

13 Against that specified list, the M8 gave us a 25 percent,

14 the ACOTTA gives us five percent.

15 MS. GWIN: I want to understand whether that means

16 we should believe that approximately 75 percent of the M8

17 alarms that went off in the Gulf War, should be viewed as

18 possible positive detections rather than all dismissed as

19 false alarms.

20 MR. STERN: No, you cannot apply the same kind of

21 rationale. The thing you have to remember is the soldiers

22 are caught and everybody knows that the M8 alarm has a false

23 alarm rate, especially to exhaust gases. And it's brought

24 out here in their training again. The majority of the time

25 spent in Saudi, as I know it and in Iraq, was in a garrison

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1 environment, where there was a great deal of vehicle

2 traffic.

3 You cannot apply any numbers trying to determine

4 how many M8 alarms were due to either a false alarm from

5 having sensed exhaust gases for known interferences, and at

6 the same time, when they do maintenance on that M8 alarm,

7 one of the operations causes the alarm to go off. In the

8 situations that they were during this conflict, it is easily

9 conceivable that units in the area of the M8 horn heard it

10 ringing, did not know that maintenance was ongoing and would

11 say, hey, I heard an alarm go off. Again, this is data that

12 I don't have, but it's easy to assume it happened that way.

13 When they were servicing the M8, which came quite often

14 because it was sucking a lot of dust, if the horn was

15 tripped, only the soldiers servicing the M8 possibly knew

16 yeah, that horn went off, but that's part of our

17 maintenance.

18 So no, you can never apply numbers to those kinds

19 of things.

20 MS. GWIN: Well, how do you come up with that data

21 then? Is that for specificity or sensitivity?

22 MR. STERN: This is laboratory and field testing

23 that is controlled. We have a list of 25 compounds, around

24 25, that are known battlefield interferences. We make the

25 detectors smell those and we determine whether or not we got

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1 an alarm. Using that kind of controlled testing, the ACOTTA

2 is giving us a five percent hit, whereas the M8, M43A1,

3 gives us 25 percent of those.

4 DR. LASHOF: Okay, I think we'll dig into this at

5 a subsequent time and --

6 MR. VIGUS: Could I ask one question of the

7 Committee?

8 Put up this viewgraph here that represents the

9 locations of the incidences on the seven tapes. And the

10 question was brought up had I done any analysis to try to

11 determine if those locations matched any of the other

12 incidents that we've heard of. And the answer to that is

13 no, but my question to the Committee is have you done that?

14 Is there an overlay that puts up there where some of these

15 other places are? Just for my own benefit.

16 MR. BROWN: That's something we're looking into.

17 DR. LASHOF: If we get it, we'll let you know.

18 MR. BROWN: Obviously that's an important step.

19 DR. LASHOF: All right, I'm going to ask at this

20 point, the Marines -- you know what, I'm not enough of a

21 military person to know -- Chief Warrant Officer -- thank

22 you -- Chief Warrant Officer Robert James, Master Sergeant

23 Michael Bradford, John Laymon, James Kenny and Gunnery

24 Sergeant George Grass to come forward. We're going to take

25 a couple of minutes just to extend the table so there's room

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1 for you. What we're going to do, considering we're now 45

2 minutes behind time, is I'm going to ask all of the Marines

3 to give their testimony. We'll then probably take a break

4 and come back and call you all back up again and do the

5 question and answer after lunch. If that's agreeable to all

6 of you, I'd appreciate that, because these are important

7 issues and we really do want to get at the heart of it.

8 (Brief pause.)

9 MS. GWIN: Okay, I think we can get started. Dr.

10 Lashof will be right back.

11 Chief Warrant Officer James, if you would start

12 off, please.

13

14 Marine breaching and Fox detections during the Gulf War

15

16 CWO JAMES: Thank you, ma'am. Good morning. I'm

17 CWO Bob James, currently the Nuclear, Biological and

18 Chemical Defense Officer for the Marine Corps Forces

19 Atlantic located at Camp Lejeune, North Carolina. Thank you

20 for the opportunity to speak before you today.

21 During the period of mid-August 1990 to March

22 1991, I was assigned to the 1st Marine Expeditionary Forces

23 in Southwest Asia as a Nuclear, Biological and Chemical

24 Defense Officer assigned to the G3 operations section. My

25 routine duties were that as an NBC watch officer in the

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1 combat operations center during the hours from 1800 or 06 in

2 the afternoon until 06 in the morning.

3 I'd like to briefly describe the number of

4 preparations that Marine Forces took prior to the ground

5 war. And with your permission, ma'am, there are a couple of

6 points that I will omit, as you have copies of this.

7 First of all, the Marine Corps ensured that we had

8 sufficient numbers of personnel with the NBC Defense

9 Military Occupational Specialty were on hand and in the

10 units needing their skills. This effort included staffing

11 some units above their normal NBC defense personnel

12 requirements.

13 Secondly, tremendous efforts were made to ensure

14 units had the NBC defense equipment rated for that unit and

15 that it was in a serviceable condition.

16 In addition to the normal protective equipment

17 held within our units, numerous items were forwarded to us,

18 which enhanced our capabilities. These included British

19 protective overgarments, which supplemented our standard

20 garment; the chemical agent alarm; remote sensing chemical

21 agent alarm, which you've heard discussed today. Also,

22 hydrogen sulfide testers, which we provided as a means of

23 detecting the gas which emanated from certain oilfields and

24 can be deadly to unprotected personnel. Large numbers of M17

25 lightweight decontamination systems were made available to

1 enhance our mobile decontamination capability, and finally,

2 as you've heard today, ten Fox vehicles were provided to the

3 Marine Corps.

4 My confidence level in the equipment was good,

5 although I did have concerns as to this equipment's

6 reliability in the heat, dust, pollutants and so forth that

7 we experienced in Southwest Asia. We continually queried

8 central command and/or Marine Corps systems command as to

9 the effectiveness of the equipment under these conditions.

10 And as has been mentioned before, there were certain

11 specific procedures that were supplied to us for operating

12 this equipment in the environment we were in.

13 Generally we were told that the equipment would

14 function as designed. I had very little confidence in the

15 M8 chemical alarm and this was based on my personal

16 experience prior to the Gulf War.

17 We also conducted numerous NBC defense training

18 within SWA. That included individual training, NBC defense

19 team training, contaminated casualty decontamination

20 training and training for medical personnel. All the

21 training described above was then incorporated into what we

22 refer to as mission oriented training. Is where a Marine

23 trains performing his normal mission while operating in the

24 chemical environment. A couple of examples would be an

25 artilleryman firing his howitzer while in full protective

1 clothing, a truck driver hauling equipment in a contaminated
2 environment or a radio operator establishing and maintaining
3 a communications net. What this training produces is a unit
4 trained to function and perform its war-time mission in a
5 chemical environment. The focus here is to treat the use of
6 chemical weapons against us as any other condition on the
7 battlefield, such as heat, cold, darkness, et cetera.
8 The Fox vehicle training has been covered and will
9 be covered in a little bit more detail by the Marines
10 following me.
11 A final word on training. Our Marines took
12 seriously the threat of chemical attack. With the equipment
13 provided, they trained, they believed in the threat and
14 their confidence level was very high when that ground war
15 kicked off.
16 During the latter part of 1990, central command
17 informed the Marine forces that we would receive ten Fox
18 reconnaissance vehicles. The 1st Marine Expeditionary Force
19 began coordinating with headquarters Marine Corps in
20 Washington to identify crews for the vehicles and to provide
21 training for them. The ten vehicles and crews arrived in

22 country in early February.

23 The 1st and 2nd Marine Divisions were provided Fox

24 vehicles for their employment during the ground war. The

25 commanders of each of these divisions further decided how

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1 best to employ the vehicles at their level. Generally

2 speaking, the vehicles went to the task force level, which

3 you may have heard described before.

4 I'd like to briefly describe what the Marines

5 sitting here to my left, who conducted the breaching

6 operation, faced and what guidance was provided to the Fox

7 crews.

8 Mine fields and obstacles were in place by the

9 Iraqis for the purpose of slowing and canalizing our forces,

10 thereby making these forces more susceptible to both direct

11 and indirect fire while attempting to breach. Breaching

12 lanes were established with the intent of getting personnel

13 and equipment through the mine fields and obstacles as

14 quickly and safely as possible.

15 During the breaching operations, the Fox vehicles

16 were usually one of the first vehicles passing through their

17 assigned lane. We directed our Fox vehicles that they were

18 not to halt in the lanes, as this would create a logjam,

19 greatly increasing the vulnerability to attack.

20 Our Marines were trained that if at any time

21 during the breaching operations, indications of a chemical

22 or biological attack occurred, they were to assume the MOPP

23 4 posture, which is the highest level of protection, and

24 continue through the breach. Specific procedures for

25 reducing those MOPP levels and determining the scope of

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1 contamination, would be conducted by unit NBC defense

2 personnel after exiting the breach and when the tactical

3 situation permitted.

4 The Fox mass spectrometer did detect the presence

5 of chemical agents during the breach; however, you've heard

6 today there's been some of the technical analysis that have

7 said that the findings were inconclusive. And Dr. Lashof, I

8 think you hit it off, but as a Marine, I want to say it;

9 these guys did what they were ordered to do and that was to

10 go through the breach and not to stop. So what they did was

11 what they were ordered to do and what the tactical situation

12 on the ground presented itself. They were getting shot at.

13 So I want to make sure that that's perfectly clear -- they

14 did what they were ordered to do.

15 Finally today, I'd like to discuss two specific

16 chemical-related incidents that I recall from the ground war

17 that were reported up the chain of command to central

18 command.

19 The first incident occurred during the initial

20 breaching of the mine fields, when we received information

21 in the 1st Marine Expeditionary Force combat operations

22 center that a vehicle transitting a breach lane in the 2nd

23 Marine Division sector, had struck a land mine and that a

24 Fox vehicle following in trace had detected a chemical

25 agent. I do not recall specifically what type of agent was

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1 reported. Word from the division headquarters was that

2 personnel in the area had assumed a MOPP 4 posture. This

3 information was relayed to central command, NBC defense

4 cell, as being an unconfirmed report.

5 At a later time, I was informed by the division

6 headquarters that tests were conducted by the Fox vehicle on

7 the far side of the breach site and that it was determined

8 that the area now occupied was free from contamination.

9 At a later time after the report of the mine

10 detonating, I recall being told that some sort of detection

11 device was taken to the suspected location, where no

12 contamination was noted. This would have been a number of

13 hours after the initial incident.

14 I do not recall any specific further information

15 on this incident.

16 The second incident, to the best of my memory,

17 occurred the next day. We received reports that an

18 ammunition storage site located outside of Kuwait City, was

19 testing positive for chemical agents. Information from 2nd

20 Marine Division indicated that numerous tests from a variety

21 of detection devices, including two different Fox vehicles,

22 were coming up positive. Again, this was all relayed to the

23 central command NBC defense cell.

24 As this site contained artillery and mortar

25 shells, the Marine Corps explosive ordnance disposal team,

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1 who are trained in the disposal of chemical munitions, was

2 called in. The EOD team's opinion, based on what was

3 relayed from division NBC to me, was that these were in fact

4 chemical munitions. The Fox vehicle crew marked the area as

5 contaminated.

6 Through coordination with central command, a joint

7 captured material exploitation center team was dispatched to

8 the area to collect samples of material, the mass

9 spectrometer tape from the Fox vehicles and any other

10 pertinent information relating to the incident.

11 Within the next 24 hours, I inquired to CENTCOM

12 about any findings and I was told either that I did not have

13 a need to know or that the tests were not yet complete.

14 This was as the ground war was concluding.

15 There were maybe half a dozen incidents when I

16 received fragmented information that may have been

17 indicative of a chemical attack. Usually the information

18 was received or overheard on a radio net by non-NBC defense

19 personnel within the combat operations center. In most of

20 the cases I recall, we heard that unit X had assumed a MOPP

21 4 posture, the highest level of protection. Normally, one

22 would assume that when a unit assumes that particular MOPP

23 level, it is either under a chemical attack or that there is

24 perceived imminent threat of such an attack.

25 On each occasion that information of this type was

1 received, I followed up with communications with the

2 respective NBC officers. All were considered to be false

3 alarms or unsubstantiated. I attribute this possibly to

4 Marines responding to alarms going off on chemical detection

5 equipment or indications of a possible chemical threat, such

6 as a large artillery barrage or seeing mine fields.

7 I was not personally aware of any other chemical

8 warfare related activities. Within days of the end of

9 hostilities, I departed Southwest Asia and returned to my

10 duty station in Quantico, Virginia.

11 This concludes my statement, ma'am, and, as agreed

12 upon, we'll wait until this afternoon for any questions.

13 DR. LASHOF: We'll wait for this afternoon. I

14 think we'll go through all your testimony and we'll confer

15 and come up with as concise a group of questions as we can,

16 to get at the heart of this.

17 Thank you very much. Mr. Bradford.

18 MSG BRADFORD: Good morning. I'm very glad to be

19 here, thank you for inviting me up. I'm MSG Michael S.

20 Bradford, NBC Defense Chief for the 2nd Marine Division at

21 Camp Lejeune, North Carolina. This is my statement:

22 During August of 1990, I arrived in Okinawa, Japan

23 to the 3rd Force Service Support Group as a Gunnery

24 Sergeant, for duty. I was further assigned as the NBC

25 Defense Chief for the 9th Motor Transport Battalion. At the

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1 end of December in 1990, several Marines from the 3rd FSSG

2 and the 1st Marine Air Wing, were assembled to join other

3 Marines at the German's army chemical school in Saunhaufen

4 (ph), Germany, for instruction in the operation of the FUCHS

5 or Fox NBC reconnaissance vehicles.

6 Due to the imminent use of these teams in actual

7 combat situations, the Germans tailored the course of

8 instructions for the Americans by assigning an individual

9 sergeant vehicle commander to each of the American teams to

10 help mentor us through our training. Instruction was also

11 split up to specialize each team member for their assigned

12 task.

13 As vehicle commanders, the Marine staff NCOs and

14 the one senior sergeant that we had as vehicle commanders,

15 divided the crews up to provide the best mix of personnel

16 for each job -- driver, wheel/radio operator and the MM1

17 operator. The training went from the 7th of January '91 to

18 1 February '91.

19 Upon completion of our school, we arrived in Saudi

20 Arabia on 2 February '91. We were met by a team from

21 General Dynamics who checked our vehicles upon arrival. We

22 then checked into the 1 MF headquarters and were further

23 assigned to the 2nd Marine Division. At the Division

24 headquarters, our job was to do daily PMCS, which is

25 preventative maintenance checks and services, confidence

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1 checks and crew drills. We also spent a great deal of time

2 doing capability briefs for anyone and everyone that would

3 listen.

4 Prior to the ground war, we were given our combat

5 assignments and plans were made to get again the General

6 Dynamics technicians forward for another service check for

7 all of our vehicles.

8 My crew was assigned to the 6th Marine Regiment,

9 which ultimately assigned us to the Bravo Company, 1st

10 Battalion, 6th Marines.

11 SSG Cummings' Fox vehicle was designated as the

12 general support vehicle to verify any positive readings that

13 may be found by a forward unit. He remained with the 2nd

14 Marine Division headquarters element. It was up to CWO4

15 Ashley and CWO2 Jones, the Division NBC Defense Officers to

16 dispatch that Fox vehicle to any sites suspected of

17 contamination. It was obvious to us from the very

18 beginning, very early, that it would not be possible for any

19 Fox under fire to stop and compete the entire

20 testing/sampling process necessary to confirm any agent

21 findings.

22 On the morning of the ground war, at approximately

23 0400, we awoke and conducted our PMCSs, background and

24 confidence tests. We were going to run through the breaches

25 with our sampling wheels down to pick up any ground

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1 contamination. As we approached the berm line between Saudi

2 Arabia and Kuwait, there were engineers already there to

3 guide us forward. When we approached what was to be the

4 first breach, called Red Lane 1, we paused as an AMTRAC with

5 a rocket propelled line charge moved forward and blew open a

6 path for us to move through. There was at least one M60

7 mine plow and 3 AAVs -- that's amphibious assault vehicles -

8 - with troops forward of us. And that's at least -- I don't

9 have an exact recollection of exactly what was up there, but

10 that's what was at least there. My best guess is that we

11 were the fifth vehicle through the breach.

12 The oilwell fires were burning on all sides of us

13 and there was a light rain or a heavy mist falling. Our MM1

14 was operating with the wheels and we were noticing

15 interference from the oilwell fires. On a scale of

16 approximately 1 to 10, it was approximately a 2, so we were

17 receiving interference before we got there.

18 We entered the breach lane running our wheels.

19 Approximately halfway through the MM1, alarm -- it did not

20 alert, it did not warn -- it alarmed. I called over the

21 radio, gas, gas, gas and a few minutes later, I followed up

22 with an NBC full report. The MM1 had spiked with at least

23 three chemical agents -- HQ mustard, L-leuocyte and GB-

24 sarin. Upon the alarm, the Marines within the AMTRACs ahead

25 of us went from MOPP 2 to MOPP 4 very quickly.

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1 After the positive reading had been made, the

2 company commander was pressing me for a safe location to

3 allow his Marines to unmask. We went into the air

4 monitoring mode and waited for the levels of the agent to

5 purge out of the MM1. A few minutes later, we passed over a

6 hard ball or paved road and I called that landmark in as a

7 safe place to go to MOPP 2.

8 I have not yet had an opportunity to view the

9 actual tape since we left the desert. LTC Nalls has that

10 tape. We've got a copy of some of the portions of that

11 tape. Reviewing that, it seems that there were 17 minutes

12 worth of readings that we were getting. So between the time

13 of detection and between the time that we found the hard

14 ball road, we were traveling for at least 17 minutes.

15 My logic at the time was that we had used high

16 explosives to set off at least two chemical mines, that we

17 were moving rapidly through the breach and that there was a

18 light rain or a heavy mist in the area and that the MM1 was

19 steadily purging below dangerous levels. That was my logic

20 to determine that the hard ball road would be a safe place

21 to allow the troops to unmask.

22 During the second breach, we were taking artillery

23 and mortar fire. This time, the line charge failed to work,

24 so that AAV was sent a couple of hundred yards to our right

25 to allow it time to be repaired. When they finally did get

1 it fixed, they opened the breach lane from that position as

2 well. Meanwhile, the M60 mine plow attempted to clear out

3 our lane on its own. About 100 yards into the mine field,

4 it became disabled due to the mines. We were ordered

5 forward and followed the command track into the mine field.

6 When the command track attempted to go around the

7 M60 mine plow that hit the two mines, they hit two mines

8 themselves and they lost their left track. We were stuck.

9 I ordered Lance Corporal Laymon and Brantley into MOPP 4 and

10 out of the Fox. The MM1 was showing negative readings, but

11 I wanted them to use the CAM, which is the chemical agent

12 monitor that you heard about, to check out our Fox. They

13 gave me an all-clear to any vapor contamination coming off

14 of our vehicle.

15 However, they did notice that the M9 chemical

16 agent detection paper that we had placed on the outside of

17 our vehicle had turned red in spots. And that had indicated

18 that we had come in contact with some sort of chemical

19 agent.

20 Since both Marines were already on the deck, they

21 assisted the combat engineers in collecting and holding an

22 ever-growing number of EPOWs. I asked for and received

23 several Iraqi masks to be brought over to me to be checked

24 for any residual signs of chemical exposure. We were

25 checking that with a chemical agent monitor. The EPOWs

1 resisted giving up their masks, and only under armed

2 persuasion did they comply. When their masks were returned,

3 they kissed them.

4 Later that night, a sergeant from the AMTRACs

5 reported possible contact with blister agents on the back of

6 his hands. His hands appeared to be red and slightly

7 swollen and he was complaining about them being very itchy.

8 I informed him to keep them clean and to have the corpsman

9 monitor him for signs of blistering.

10 At the first night's pause, we took the Fox around

11 to the AMTRACs that were ahead of us, asking if everybody

12 was all right. We also used the CAM to look for any

13 residual contamination. The Fox was monitoring the air as

14 well.

15 Lance Corporal Brantley had found a shrapnel hole

16 in the side of one of the packs that was strapped to one of

17 the AAVs that had been ahead of us going through the breach.

18 He discovered two bars of G-series nerve agent present

19 inside this pack. Due to it being a very minor reading, we

20 cautioned the Marine not to use his pack that night and we

21 checked again the next morning and we received no readings.

22 I'd like to reinforce the following points:

23 Just a few days prior to entering Red Lane 1,

24 General Dynamics had been forward to service our Fox

25 vehicle. The morning of the ground war, we took background

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1 readings and made our confidence checks in the same type of

2 environment as the detection was made.

3 We were running the wheels prior to entering Red

4 Lane 1. No alarm. As we entered Red Lane 1, still running

5 the wheels, there was no alarm.

6 However, about halfway the breach of Red Lane 1,

7 we alarmed and spiked to the top of the screen with the

8 three chemical agents. This is not a warning, this is not

9 an alert, it was an alarm.

10 During our pause in the second mine field, we

11 noticed the M9 chemical detection paper had red spots,

12 indicating contact with chemicals. The use of the CAM had

13 showed two bars of nerve agent in the pack with the shrapnel

14 hole inside of it.

15 And there's probably many other points that we can

16 talk about later and I'll be more than glad to answer any

17 questions.

18 Thank you.

19 DR. LASHOF: Thank you very much.

20 Mr. Laymon.

21 MR. LAYMON: I'd just like to thank the Committee

22 for inviting me here. This is the first time I've ever

23 heard of a Committee, whenever MSG Bradford called me at

24 home. I exited out of the Marine Corps in March of 1996,

25 Mark Brown contacted me approximately a month ago. I

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1 happened to be on the road at the time on my job.

2 Currently, I'm a network administrator/systems technician

3 for a computer company who does bank processing. I'm a

4 former sergeant in the United States Marine Corps.

5 At this point in time, from December of '90

6 whenever I left Okinawa, Japan, I was a lance corporal, I

7 was bound for Saunhaufen, Germany and then on to Saudi

8 Arabia for the ground war. My vehicle commander was MSG

9 Bradford.

10 We were trained in the German concern in

11 Saunhaufen by the German army. And once again, we had

12 German instructors designated to each crew. They saw that

13 we had what we needed to learn, they were with us every step

14 of the way. Their major point was a lot of learn-it-by-the-

15 book and then go out and practical application. The

16 training in Germany, I believe was adequate for the

17 operations that we had to conduct. As far as confidence, I

18 had confidence in the crew I was with, myself and the

19 equipment we were using.

20 The morning of the ground war, we had been running

21 24-hour on and off watches on MM1, myself and Lance Corporal

22 Brantley, in the back of the vehicle. We'd work

23 approximately eight hours, relieve the other one and change

24 out and get sleep that way. We got up around 0400, made all

25 our operational checks, to include functional tests,

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1 confidence tests, printer checks, automatic testing, came

2 out okay by the evidence of the tapes that were produced on

3 that day.

4 At 0621 as we entered the mine field, we received

5 an alarm on a fat, oil and wax. From 0621 until 0637 on the

6 tapes, we received leuocyte, sarin, GB and HQ mustard. Of

7 these, none of them were detected from the surface, they

8 were all detected in an air mode, automatically. We were

9 running in CWA wheel high, we were using the wheels. The

10 reason I know and can remember we were using the wheels at

11 0626 on the tapes, a spectrum was run on a fat, oil and wax.

12 That fat, oil and wax was detected from the surface. The

13 only way it could have been detected from the surface while

14 moving is if you were using the wheels. And the way I know

15 it was detected from the surface, just prior to the readout

16 numbers, you will see an S, that S means it was detected

17 from the surface.

18 Once again, the weather conditions was light rain

19 to mist. It was very difficult from my position to see

20 anything outside the vehicle. From the operator number one

21 seat, there's no exact window to look out of. There's a one

22 inch crack between the machines that you can look through

23 out the portside window and there's also another window back

24 over your right-hand shoulder. I relied solely on the

25 internal communications, what Lance Corporal Brantley was

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1 saying to at the time Gunnery Sergeant Bradford. I could

2 understand that mortar rounds were coming and chasing us and

3 they were trying to zone in and hit us. They had no means

4 of getting to us because we kept our vehicle constantly on

5 the move, either forward or backwards.

6 It has been reiterated time and time again that a

7 spectrum was not taken. The spectrum key was hit, yet by

8 the book, a spectrum, graphic spectrum, cannot be printed

9 out, the graphic itself. What actually would print out I

10 cannot remember and tell you that at this time.

11 We did take background readings. Those background

12 readings appear on the screen in what is known as a picket

13 fence and you have that. If you have readings constantly

14 below that picket fence, there should be no readings at all,

15 is what we were taught. There was no detection prior to the

16 mine field breaching incident.

17 Once again, fat, oil and wax was the first alarm

18 on the tapes and it did not do an automatic print until

19 0626. From 0626 until 0637, it did three automatic 30-

20 second prints. Of those prints, HQ mustard did a 30 second

21 print twice within a matter of two minutes, at 0635 and

22 0637, it automatically printed for HQ mustard twice.

23 As far as the tapes go, once these tapes were --

24 we got to our resting place, our rally point, if you will, I

25 took the tapes, reviewed them with my vehicle commander, MSG

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1 Bradford. We went over them, that was the last I saw of

2 them. They went to his hands, they went up the chain of

3 command. I have no idea what happened to them or to the

4 yellow canary at the time, yellow canary being the message

5 slip, which we sent out for a gas call.

6 Myself and Lance Corporal Brantley personally

7 dismounted the vehicle, checked AMTRACs for signs of

8 chemical agents, Lance Corporal Brantley was operating the

9 CAM at the time, I was there as a buddy team. If someone

10 goes down, somebody is there to pick them up.

11 As far as contacts for the committees, the first

12 contact I had with any type of committee was from SGT

13 Timothy Anderson, a Marine stationed at Camp Lejeune, a good

14 friend of mine. He gave me a 1-800 number for the

15 Presidential Advisory Committee. I called it, I got a

16 message and I said I'll call back. I went on the road,

17 never had time to actually get back.

18 I came off the road one weekend and I went to my

19 parents house and Mark Brown had contacted my parents. I

20 called him back and talked to him very briefly, just to find

21 out who he was and what he wanted, didn't really explain

22 anything about what had happened. I called MSG Bradford.

23 He said well, I gave him your name. I said okay, well, if

24 you gave him my name, then I'll talk to him. I finally

25 talked to him, we discussed a lot on the phone about what

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1 has happened but that is the only contact with any committee

2 as far as the Persian Gulf War is concerned that I have had

3 with, is with the Presidential Advisory Committee. The DOD

4 committees never contacted me. I was on active duty until

5 March of '96, I was never interviewed by any of them.

6 We discussed briefly with other Fox crews about

7 chemical agent detections. The one that stands out in my

8 mind was with the corporal that was in the rear who was

9 checking NBC gear and received so-called leuocyte burns upon

10 his arm. That's now SSG Santos.

11 At this time, I cannot remember any positive other

12 readings. Like I said, I was a lance corporal at the time.

13 I was in the back of a vehicle, I was not working the

14 radios, I had internal communication.

15 I'll be willing to answer any questions at the

16 end.

17 DR. LASHOF: Thank you very much. Mr. Kenny.

18 SSG KENNY: I just want to thank the committee for

19 allowing me to testify. My name is SSG James E. Kenny, III,

20 I'm with 4th Supply Battalion, 4th FSSG in Newport News,

21 Virginia. I am the NBC Chief for that unit. And my

22 statement goes:

23 Upon my arrival to Southwest Asia, I was assigned

24 to an NBC Fox recon vehicle as the MM1 operator, mobile mass

25 spectrometer. My vehicle commander was at that time SSG

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1 Grass, the driver was Corporal Mason and the probe operator

2 or wheels operator was SGT James. Our team was assigned to

3 1st Marine Division, Task Force Ripper.

4 My mission on the Fox team was to calibrate,

5 actually to run spectrum with a known calibration gas, and

6 run the MM1 and report to SSG Grass any chemical findings

7 that might occur. After reading Gunnery Sergeant Grass'

8 statement and testimony, I concur with the chemical findings

9 that he has indicated in his statement and testimony.

10 Specific incidents include (1) small traces of nerve agent

11 detected during the eight lane survey of the mine field

12 breach. As I remember, it took 30 to 45 minutes to run

13 through and survey all eight lanes. During this time, small

14 readings of nerve agent did appear on the computer screen,

15 but the MM1 was not able to get a spectrum of this.

16 (2) In the evening hours at Al Jabar air field an

17 S mustard agent was detected. The Fox vehicle was stopped

18 for the night and we were running the MM1 in air mode.

19 Since the vehicle was off, the MM1 was running on its own

20 battery supply. The vehicle had to be started to charge the

21 MM1's battery every so often. It was very dark, so dark,

22 you couldn't see your hand in front of your face. SSG Grass

23 and Corporal Mason were on the outside top of the vehicle

24 while SSG James and myself were inside the vehicle

25 monitoring. All of a sudden, the MM1's alarm went off for S

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1 mustard and I yelled up to SSG Grass and Corporal Mason to

2 get into the Fox because I had just gotten the alarm.

3 SSG Grass then yelled gas, gas, gas over the net

4 to warn the other Marines to put on the field protective

5 masks. After a few minutes, the readings went away and the

6 command told us everyone to unmask, it was all clear. A

7 couple of spectrums were printed out during this and I gave

8 the printouts to SSG Grass.

9 The third incident, ammunition storage bunkers

10 just outside Kuwait City. An S mustard, HT mustard and a

11 benzene bromide agent were detected. The bunkers were dug

12 into the ground and most of them had some type of cover over

13 them. Many of the crates inside the bunkers had markings on

14 the outside that indicated they were from Jordan and

15 Holland. After surveying a number of the bunker areas

16 without any abnormal readings, the Fox was stopped and all

17 four of us got out to take a break and look a little closer

18 at some of the bunkers. The MM1 was still in operation.

19 While we were outside the vehicle, the MM1's alarm went off.

20 It scared all of us back into the vehicle. Once inside,

21 there were a number of readings on the MM1's computer

22 screen. They were S mustard, HT mustard and a benzene

23 bromide agent. Again, a couple of spectrums were run and

24 the printouts were given to SSG Grass.

25 There's a couple of other categories I had down at

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1 the bottom. As far as for the fat, oil and wax readings,

2 the MM1 has a category set up to read common gases that

3 might be found on the battlefield, such as diesel fuel,

4 gasoline, smoke, et cetera. This section of the computer

5 screen usually had some type of reading in it at all times,

6 as would be expected.

7 As far as the oil fire readings, I programmed the

8 MM1 to detect the burning oil fires. I set up a field

9 called unknown-1 for this purpose and there was always some

10 type of reading in this field also.

11 Other Fox operators' detections, after the war was

12 over and all the Foxes were back with 2nd Marine Division,

13 other Fox operators told me they had gotten nerve agent

14 readings during the war. I was told there had been a liquid

15 sample taken and it was going to be sent to a laboratory for

16 further processing.

17 And the final thing is a chemical agent injury. I

18 only saw one chemical agent injury during the war and that

19 was with a Corporal Santos, as Laymon has said. He was one

20 of the other Fox operators. I saw the injury after the war

21 was over and we were back in Camp 15. When he first showed

22 it to me, it was a very small area that seemed to be

23 affected, but then it started spreading up his arm. He went

24 to medical to have it checked out and they supposedly took a

25 sample and confirmed that he indeed had come in contact with

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1 a blister type agent.

2 That's all I have.

3 DR. LASHOF: Thank you very much. And Gunnery

4 Sergeant Grass, I thank you very much for joining us for the

5 questioning. You have testified before and we all remember

6 very clearly the excellent testimony you gave and it is in

7 our books, so that we've had a chance to review it.

8 We have about 15 minutes that we can begin and

9 take some questions before our lunch break. We'll break at

10 12:40 and then resume at 1:50 and continue questioning then.

11 But we might as well use this little bit of time to start

12 the questioning.

13 Any questions from the Committee at this point?

14 MAJ CROSS: The first thing I want to ask a

15 question about, Mr. Laymon, when you were explaining some of

16 your observations of the readings and the tape, I could see

17 Mr. Vigus in the background shaking his head that that was

18 incorrect. So it appears to me that right off the bat,

19 there's some disagreement in maybe the training that you had

20 and maybe the things that he's worked on and there's a

21 little bit of concern on my part about that.

22 MR. LAYMON: I'm not a mass spectrometer expert, I

23 just know what the Germans taught me as far as the printouts

24 at the time.

25 MAJ CROSS: Right.

1 MR. LAYMON: They said this is this and this is

2 that. You know, I have the book here that the Germans

3 issued to us and it was translated from German into English.

4 I don't know what year or anything like that. I know it was

5 at least six and a half years ago though.

6 Mr. Vigus is the technical expert on the situation

7 and I can just tell you my account of what happened, what I

8 was doing at the time.

9 MAJ CROSS: Right. Okay.

10 DR. LASHOF: Marguerite.

11 MAJ KNOX: SGT Bradford, what day was that, the

12 first day of the ground war, was that ten days into Desert

13 Storm?

14 MSG BRADFORD: Ma'am, I had lost all sense of

15 time, but they tell me it was the 24th of February, early in

16 the morning.

17 DR. LASHOF: Andrea.

18 MS. KIDD-TAYLOR: You mentioned that there was

19 only one individual who actually had the blister agent

20 indications, the symptoms that he had on his hand or the

21 sign on his hand and arm. I guess my other question would

22 be to any of you, do you know of any of the others who

23 served with you or yourselves, whether any of you have

24 developed any of the symptoms of Gulf War veterans'

25 illnesses that we've heard so much about over the last year

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1 and a half?

2 MSG BRADFORD: I've got intestinal problems

3 myself, ma'am and when I went to the doctor in Hawaii, I was

4 told that there was no Gulf War illness, there was no

5 validity to that, and I've got some joint pain now and I

6 don't know -- I can't attribute that to the Gulf or just --

7 MS. KIDD-TAYLOR: Did you have it before or --

8 MSG BRADFORD: No, ma'am. It wasn't since after

9 the Gulf. Obviously, knees, football and rugby and

10 skydiving took my knees for that. But other than that, the

11 rest of the joints are kind of questionable.

12 MS. KIDD-TAYLOR: Anyone else?

13 CWO JAMES: I've had problems with sleeplessness

14 at night, not being able -- actually waking up wide awake.

15 I go back to sleep, I can't say that I'm fatigued, it

16 doesn't cause me any problems during the day, but that

17 definitely is something that started after the Gulf War.

18 MR. LAYMON: Prior to exiting the Marine Corps, I

19 was diagnosed with bad knees obviously. I have a package in

20 with the VA that's been pending since 30 days after I got

21 out of the Marine Corps. So April of '96, I'm still waiting

22 for the VA to evaluate me on my knees. My mother made an

23 observation that I have memory loss. As far as that, you

24 know, I think I'm in good health.

25 MR. KENNY: I've had some memory loss and frequent

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1 headaches. Whether that's attributed to the war or not, I'm

2 not sure. Besides that, I feel basically fit.

3 GNYSGT GRASS: I've had severe headaches when I

4 first came back and memory loss and the headaches have gone

5 away and my memory loss, that depends on what my wife is

6 trying to get me to do.

7 (Laughter.)

8 GNYSGT GRASS: I have a rash on my back side that

9 won't go away, it's the kind of rash that the more you

10 scratch it -- I can dig myself bloody and the harder I dig

11 it, the better it feels. I've tried different creams,

12 medical gave me different creams for it and it doesn't go

13 away. In warm weather, it seems to get worse and more

14 itchy. It doesn't spread, it just stays in that one

15 specific area.

16 MR. LONGBRAKE: MSG Bradford, are you confident

17 that your Fox detected chemicals in the breaching lanes?

18 MSG BRADFORD: Yes, sir, I am. From the training

19 that we had and the conversations that I had with an Army

20 Fox instructor at Fort McClellan, Alabama, SSG Williams, who

21 said that the German manufacturer actually sent an

22 individual on over there, an engineer, over to the Fox

23 school, and if we got a detection and got an alarm, there

24 was something out there. And I am extremely confident that

25 there is something out there in the form of chemical agents.

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1 MR. LONGBRAKE: What do you believe was the source

2 of those detections?

3 MSG BRADFORD: At the time, I was confident that

4 there was at least two chemical mines in the mine field.

5 Logically speaking, you would not mix all three of those

6 chemicals into one mine. I was confident and I still am

7 confident that there was at least two chemical mines that

8 were out there. And if there were no chemical mines in the

9 mine field, the only other source that I could think of

10 would be artillery or mortars that came in on the breach,

11 which I did not personally witness, but I could hear over

12 the radio that we were taking artillery and mortars during

13 the first breach. And if that wasn't the case -- and this

14 is all speculation now -- if that was not the case, then

15 obviously we just happened to roll over the top of chemical

16 agents that had been spilled and had been sitting out there

17 for six months. But I would discount that highly due to the

18 fact that we spiked across the spectrum, and even a

19 persistent agent sitting out there in those conditions for

20 six months would dissipate greatly. In training, we were

21 training with very low levels in Germany in order to

22 separate those out and get a spectrum. We got very high

23 levels that peaked across our screen. So I am extremely

24 confident that there were chemicals in that breach lane.

25 Does that answer your question, sir?

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1 MR. LONGBRAKE: Yes, thank you.

2 MR. RIOS: Let me ask, I guess it's kind of a

3 repetitive question, but since you all were there, you

4 witnessed what you went through and you saw the machinery,

5 the equipment that you had, is there any doubt in your mind

6 that there were chemical agents in the atmosphere there --

7 is there any doubt, Mr. James?

8 CWO JAMES: Sir, again, I had no direct knowledge

9 or did I see any instrument --

10 MR. RIOS: No, I'm not asking if you saw any, but

11 from -- just from being there.

12 CWO JAMES: Well, certainly, sir, the pollutants

13 in the air were everywhere. Even before the war started,

14 back within Saudi Arabia at the port Nal Jabal, the plants

15 that are there, the refineries, the industrial pollutants

16 are just everywhere. That's not something that started with

17 the war.

18 I guess the point that I would make, sir, is that

19 there is absolutely, positively no doubt in my mind that

20 when these Marines say that they saw a spike or they saw

21 some bars on the CAM or they saw a color change on a 256

22 kit, that they saw that.

23 MS. KIDD-TAYLOR: But it couldn't be very

24 specific.

25 CWO JAMES: But I will say then that I would -- as

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1 a Marine that's been around a few years and who has a

2 responsibility to train Marines and equip them, then I want

3 someone to tell me that okay, maybe they did see that, but

4 we had a problem with our training or the equipment we

5 provided, and we need to correct that then. I would argue

6 the point well, okay, maybe there was an equipment problem

7 or a training problem. I'll be open to that, I'll be open

8 to discussing that. But the fact that they saw what they

9 just described and other personnel have appeared and said

10 those things, they saw what they saw. And if it wasn't a

11 chemical, then again that leads me to believe we've got a

12 training problem or we've got an equipment problem and let's

13 fix it.

14 MR. RIOS: What about you, Mr. Bradford, there's

15 no doubt in your mind that there were chemicals in the

16 atmosphere?

17 MSG BRADFORD: That is affirmative, sir. I'm

18 confident that there were chemicals out there.

19 MR. RIOS: How about you, Mr. Laymon.

20 MR. LAYMON: I'm very confident that there were

21 chemicals out there. I was looking at the actual screen

22 whenever they hit.

23 MR. RIOS: Mr. Kenny.

24 MR. KENNY: Yes, sir, I'm confident that there

25 were.

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1 MAJ CROSS: MSG Bradford, I know you made a

2 specific distinction about an alarm versus an alert versus a

3 warning. And I'm sitting here and I'm confident that you

4 had enough confidence in the vehicle and the equipment and

5 you had enough confidence in your training that when that

6 alarm went off, you knew what that was.

7 MSG BRADFORD: Yes, sir.

8 MAJ CROSS: There's no doubt in my mind that you

9 knew what that was. And, you know, I agree with you 100

10 percent. I myself was there, I was right behind you. You

11 know, I heard the alarms and I saw the reports. It was

12 there.

13 So, you know, we heard some testimony earlier on

14 about the vehicle and its capabilities. I was impressed

15 with what I saw the vehicle do and, you know, if it gave an

16 alarm, I'd say it was a positive alarm -- more likely a

17 positive than a false reading. And that's just my gut

18 feeling too.

19 DR. LASHOF: Have any of you been interviewed by

20 any of the people at DOD during the investigation of these

21 alarms and could you tell us what those interviews were like

22 and what testimony you've given to the investigators? Chief

23 James.

24 CWO JAMES: Just recently, within the last couple

25 of months, I have been interviewed by LTC Nalls and his

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1 group and also a group that the Miter Corporation is heading

2 -- I say heading up, they're actually doing the interviews

3 and that's more related to the intelligence gathering or

4 information that we had during the Gulf War. So I have

5 talked to those two groups, never in a forum such as this,

6 but I have provided statements to them.

7 DR. LASHOF: Your impression of their reaction to

8 your identification of chemicals in the Gulf.

9 CWO JAMES: Reaction by the other committees? I

10 will be very honest with you, ma'am, what I have seen is

11 what I considered to be an uncoordinated effort, not sharing

12 of information between the groups. It seems like to me

13 there would be a central collection point for this, a data

14 bank, and the fact that we appear before -- I have talked to

15 numerous groups and it seems each time it's new information,

16 it's providing this information over again, is disturbing to

17 me. Again, it seems like there ought to be a central data

18 bank and there should be a coordinated effort and I can't

19 say that I see that.

20 DR. LASHOF: MSG Bradford.

21 MSG BRADFORD: Yes, ma'am, I concur with CWO James

22 as far as the feeling that the United States is kind of

23 uncoordinated in its attempts to come down. Coming off of

24 the plane from the very first moments, actually even before

25 that, even before I left the desert, we were claiming that

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1 there were chemicals in the Gulf and we were told that they

2 didn't exist, they weren't there. And I was also advised

3 before I left the Gulf that I maybe want to rethink my

4 position because if I go blabbing to the press or blabbing

5 to anybody else that there were chemical there, that I might

6 be held liable for something, and my line of reasoning was

7 well, if it never existed, how can I be held liable for it.

8 So I continued to tell the truth as I saw it.

9 Every class that I have given in NBC Defense,

10 every opportunity, every forum that I've had, I've made sure

11 that my Marines knew from the most senior officer to the

12 most junior private, that there were chemicals that were out

13 there. And I've reinforced that over and over again.

14 As far as the people that I've talked to, GNYSGT

15 Grass was nice enough to go ahead and give my name during

16 testimony in front of Congress. LTC Nalls went ahead and

17 contacted me, I had a long conversation with him on the

18 phone and then subsequently on the phone at several times;

19 he came down to Camp Lejeune, I gave a statement there. I'm

20 still reviewing that statement, going to make a few changes,

21 give it back to him. The Miter Corporation came on in and

22 was sitting there that afternoon with CWO James and they

23 asked specific questions and we gave statements for that.

24 I'm willing to do as much as I possibly can, to

25 let anybody that needs to know what's going on, and I want

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1 to be very cooperative, but it should be a concern for a lot

2 of people that are in this business to get a coordinated

3 effort, in my opinion.

4 DR. LASHOF: Thank you very much.

5 MSG BRADFORD: Yes, ma'am.

6 DR. LASHOF: Mr. Laymon.

7 MR. LAYMON: No, ma'am, the only committee I've

8 been contacted by is yours and that was through Mark Brown

9 thanks to MSG Bradford releasing my name and number.

10 DR. LASHOF: Okay. Mr. Kenny.

11 MR. KENNY: No, ma'am, the only interview I've had

12 is with Mr. Turner and this committee.

13 DR. LASHOF: Okay, and we have your answer on the

14 record on that too.

15 I think we'll take our lunch break now and we will

16 resume at 1:50 and we'll resume with the questioning of you

17 if you all can come back, we'd appreciate it. Thank you

18 very much, we stand adjourned.

19 (Whereupon, a luncheon recess was taken at

20 12:40 p.m., the meeting to resume at 1:50 p.m.)

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1 AFTERNOON SESSION

2 DR. LASHOF: We're ready to resume the meeting.

3 Thank you very much.

4 We'll open up the questions. First, Marguerite,

5 do you want to start?

6 MAJ KNOX: I just had a question, how many of you

7 have undergone the comprehensive clinical evaluation exam

8 that DOD provides as active duty soldiers or since you've

9 gotten out of active duty status?

10 MR. LAYMON: I didn't know there was such a thing.

11 MAJ KNOX: You never heard of such a thing?

12 VOICE: Would you repeat the question, please?

13 MAJ KNOX: I'm sorry, the question was I was

14 curious as to how many had undergone the comprehensive

15 clinical evaluation, the exam that DOD gives for Gulf War

16 veterans.

17 GNYSGT GRASS: I've been through it.

18 MAJ KNOX: You've been through it?

19 GNYSGT GRASS: Yes, ma'am.

20 DR. LASHOF: Mr. Kenny.

21 MR. KENNY: No.

22 DR. LASHOF: No. Did you know about it?

23 MR. KENNY: No, ma'am.

24 DR. LASHOF: You've never received any notice or

25 gotten any letter from DOD or read anything about it?

1 MR. KENNY: No, ma'am.

2 DR. LASHOF: And you're still in the service?

3 MR. KENNY: Reserves, yes, ma'am.

4 DR. LASHOF: You're in the Reserves and you've not

5 been notified. Mr. Laymon.

6 MR. LAYMON: No, I haven't, I had no idea about

7 it, I've received no notification of it.

8 DR. LASHOF: Interesting.

9 MAJ KNOX: SSG Bradford.

10 MSG BRADFORD: No, ma'am, I've understood that as

11 an active member, that they'd send us to a regional center

12 and I heard that it was in Norfolk, Virginia, and that they

13 would do a screening at your unit and then get you sent up

14 there. But I've never been notified, that's just been

15 general information I guess.

16 CWO JAMES: I am aware of it, have not

17 participated in the program.

18 MAJ KNOX: Can you tell us why you didn't

19 participate?

20 CWO JAMES: Personal reasons, I just personally

21 did not desire to participate in it.

22 MR. LONGBRAKE: I just have another question for

23 MSG Bradford. I understand that we provided you with the

24 Chemical and Biological Defense Command analysis of the

25 tapes from your Fox vehicle, and I was just wondering if

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1 that analysis has altered your feelings about the chemical

2 detections of your Fox?

3 MSG BRADFORD: Yes, sir, you did provide me with

4 that analysis. I read the analysis, I talked to everybody

5 that I could think of as far as associated with the MM1, to

6 make sure that I wasn't missing something in what I had

7 done, and what we were supposed to have done. Second and

8 third guessed myself a lot. And until I called SSG

9 Williams, who was a United States Army instructor for the

10 MM1 down at the Fox reconnaissance school in Fort McClellan,

11 Alabama and talked to him for about half an hour, did I come

12 away with the fact that we did everything correct, and that

13 if it alarmed, then there was something out there and that

14 if that something -- because the MM1, it says right there at

15 the beginning of the manual for it, it is sensitive and it

16 is designed to give you its readings. It was engineered to

17 be a chemical defense piece of gear and it's been used and

18 tested by the Germans and since the Gulf, the United States

19 Army has used it extensively, and I talked to the operator

20 down there, SSG Williams, who is the actual instructor and

21 an operator. And everything that we did, to the best of my

22 ability, he's confident, from what I explained to him, and

23 I'm confident that what we did got us a reading.

24 Does that answer your question, sir?

25 MR. LONGBRAKE: Yes, sir, thank you very much.

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1 DR. LASHOF: Let me follow that up a little bit.

2 You've heard the testimony given by Mr. Vigus and Mr. Stern

3 about the need to do a second confirmatory, more specific to

4 break it out because you can get so many of these chemicals

5 giving the same first spike. And it's clear that that's not

6 possible to do under the conditions that you were working

7 under and I want to make it very clear that no one is being

8 critical that you didn't do that, that was impossible to do.

9 But in view of that testimony, do you still feel

10 and could you tell me on what grounds you still feel that

11 what you did see, even without that confirmatory, was

12 indicative of mustard or sarin.

13 MSG BRADFORD: Yes, ma'am. In Germany, we would

14 go and they would have a couple of different test agents for

15 us and we were going to have to discover what it is that the

16 test agent was, and one drill that I remember very

17 distinctly, we were out on a fairly large field and we were

18 doing big figure 8's and we came across a positive reading,

19 we stopped, we backed up and we were making our indications

20 on just a very small, slight reading. The German

21 instructors came up, were kind of confused at what we were

22 doing and we showed them, verified that we're tracking down

23 a small reading here and we're coming up with positive

24 indications now, and we're able to break it up and go into

25 the low mode and take all the steps that are necessary to

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1 make this positive identification. And they said ah, what

2 you've found here is an old test spot because the German

3 Army was using this area before. And they didn't expect us

4 to find it.

5 So we're well familiar with the need to lower the

6 temperature, to make those separations, to be able to

7 identify a chemical agent, especially when there's a small

8 amount. That is real critical, because you will get the

9 unknowns, you'll get something that'll just be a quick

10 blurp, in and out. If it's an unknown or if it's a quick

11 blurp in and out, you should still consider that there may

12 be a chemical agent in the area, you should be erring on the

13 possibility that it's there, rather than saying that it's

14 not, in our position.

15 What we had was we had a mass amount from the

16 ground, from the wheels, that spiked our readings, that was

17 for a very long period of time still giving us readings into

18 the MM1. We were, like Mr. Laymon here told us, 17 minutes

19 worth of the readings still coming through the MM1. This

20 increased my confidence considerably that it was there. To

21 stop and do a validation process, you know, and it wasn't a

22 small amount, it wasn't a trace amount. We spiked. I don't

23 know how to explain. He explained it to me that, you know,

24 how the screen is white with a little bit of black, and I

25 said yeah, well the screen is black with a little bit of

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1 white, because I grilled him. I could not see the screen

2 myself, so he had to paint pictures for me as far as what

3 was there. And I was asking him for as many details as he

4 could give me, and asking to the point of his frustration,

5 where he kept, you know repeating the process over and over

6 again as far as what it was he was seeing.

7 So I'm extremely confident that it was there.

8 DR. LASHOF: GNYSGT Grass, in your experience, how

9 do you respond to the same question?

10 GNYSGT GRASS: Well, jus as MSG Bradford said, when

11 we were going through the breaches, we didn't have a lot of

12 time to slow down. We detected the low levels of nerve

13 agent going through there and once we were sitting up at

14 Jabar air field, we were sitting in place and then when the

15 alarm went off for the sulfur mustard, before I called gas

16 over there -- before I alerted the entire unit that was

17 around us, I made sure we ran the proper background checks,

18 made sure it wasn't some of the oil fires, because it was

19 pretty dark at that time from the oil fires, and then once

20 we made sure that it wasn't a false reading, according to

21 the procedures we were taught, that's when I yelled gas. Up

22 there by Kuwait City at the -- I'll refer to it as the

23 chemical weapons storage area because I'm sure that's what

24 it was, we had plenty of time, we were not on the move, it

25 was a clear, sunny -- it was a clear day. There were no

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1 vehicles around us to throw anything off. Sitting right in

2 front of us were 155 rounds. We backed the Fox back up to

3 the bunker, lowered the probe down into there, as SSG Kenny

4 was doing the procedures on the mass spectrometer and it

5 would have run as many spectrums as we needed right there.

6 I'm fully confident that all the procedures that

7 we did and everything that we had seen were exactly what we

8 had to do. Unfortunately, the tickets aren't around that we

9 had, as proof.

10 DR. LASHOF: Thank you.

11 MAJ KNOX: If DOD is not tactically concerned

12 about low level chemical exposure, were you trained that

13 once you saw a certain level of chemical, that you were

14 supposed to notify everyone, or just when you identified it

15 at all, that you were supposed to call gas, gas, gas? I

16 mean, because these are low level exposures, is that

17 correct?

18 GNYSGT GRASS: Yes, ma'am. If I'm not even just

19 on the Fox vehicle and I'm out there testing, because

20 procedures for us -- I know it's been briefed earlier that

21 the M256 kit is only used for unmasking procedures, that's

22 not the way that we teach it. We teach it, you can go out

23 there as a survey team using the M256 kits to verify that

24 there may be chemicals out there. So once you have a

25 verification that there is some kind of chemical out there,

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1 you're going to notify someone, whether it's low levels or

2 high levels, to put somebody into the proper protective

3 posture.

4 MAJ KNOX: Are you frightened by low level

5 chemical exposure?

6 GNYSGT GRASS: It's a -- whether it's low level or

7 high level, it's just a matter of time before -- it just

8 gives different effects onto the body, it just takes longer.

9 Whether I have a liquid blister agent on my hand or a vapor

10 in the air, in the long run, it's going to cause the same

11 kind of problems, just one does it a little faster than the

12 other.

13 MAJ KNOX: Are you taught that in NBC school?

14 GNYSGT GRASS: Yes, ma'am.

15 MR. TURNER: GNYSGT Grass, you talked about three

16 incidents, the one where you had traces going through the

17 breach, on that one there were no spectrums taken, is that

18 correct?

19 GNYSGT GRASS: Yes, sir, that's correct.

20 MR. TURNER: Then the second one is at Al Jabar,

21 and there were printed out tickets, actual printed spectrum

22 tapes printed out from that, is that correct?

23 GNYSGT GRASS: Yes, sir, that's correct.

24 MR. TURNER: And what happened to those?

25 GNYSGT GRASS: Well once -- I kept those and once

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1 we determined there was no more vapor in the area, that's

2 when I had gotten out of the vehicle and we tried to report

3 it up through Division and when it was determined, talking

4 to CWO Beidenbender talking into Division, that they passed

5 it off as either vapors from the oil fires or vapors from

6 the tanks and the tracks that were around us, and I held

7 onto that ticket right there until I gave all the tickets,

8 either from the three printouts from the ammo dump and from

9 Jabar air field, that's when I gave them all back to CWO

10 Pattrell in the chain of command.

11 MR. TURNER: And today, can those tickets be

12 found, does anybody know where they are?

13 GNYSGT GRASS: No, sir.

14 MR. TURNER: What do you do, what is your current

15 assignment with the Marine Corps, GNYSGT Grass?

16 GNYSGT GRASS: I'm the non-commissioned officer in

17 charge of the decontamination element of the Marine Corps

18 chemical, biological incident response force.

19 MR. TURNER: But what does that force do?

20 GNYSGT GRASS: Basically what we do is we're a

21 chemical, biological unit that helps prevent the acts of

22 terrorism using chemical or biological -- helps prevent the

23 effects of chemical or biological terrorists.

24 MR. TURNER: Do you consider it an honor to have

25 been chosen for that unit?

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1 GNYSGT GRASS: Yes, sir, absolutely.

2 MR. TURNER: MSG Bradford, what are your current

3 duties, you teach NBC, don't you?

4 MSG BRADFORD: Yes, sir, I'm the -- actually as

5 the Division NBC chief, I oversee my immediate platoon, I do

6 inspections, both on the logistics side and on the

7 operations and tactical side for all of our 22 battalions

8 that have to go through a yearly inspection cycle.

9 MR. TURNER: Is that at Camp Lejeune?

10 MSG BRADFORD: That's at Camp Lejeune, yes, sir.

11 And prior to that, when I came aboard at the Division, I was

12 a TAD, temporary additional duty to the CBIRF also, and

13 helped get them started and stood up until about August, in

14 which case I came back and assumed my duties as the Division

15 NBC chief.

16 MR. TURNER: Thank you.

17 DR. LASHOF: Mark.

18 MR. BROWN: Thank you. I have a question for both

19 operators, both you operators and both you Fox vehicle

20 commanders; and that is to do with this idea that the oil

21 well smoke somehow could have caused interference with your

22 ability to detect chemical agents. And it seems to me as I

23 hear that account of that possibility, it seems to me that

24 the implication is that when you started those breaching

25 operations, when the ground war started, and you had some

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1 detections, that you'd never seen oil well smoke before.

2 And my question is, isn't it the case that you were familiar

3 with oil well -- from the times that you first had access to

4 your Fox vehicles in the Gulf War theater prior to the

5 breaching, you had -- you were operating the equipment and

6 saw backgrounds and you were familiar with what this oil

7 well background smoke looked like, it was a constant, it

8 wasn't something that suddenly started at the beginning of

9 the ground war. Am I correct in that?

10 MSG BRADFORD: That is correct, sir. Specifically

11 at 0400 when we got up and we took the background reading

12 and at that time, it was probably more intense as far as the

13 smoke where we were than at any other given time, but we

14 still took a background reading. As we were going through

15 the berm line, and we were waiting for the line charge to be

16 set off and we were sitting in between the oil well fires,

17 obviously then that was more intense than where we had been

18 at 0400. But as we're sitting there, we are getting a

19 reading, we are watching the background. But it is not

20 spiking, it is not alarming, it is not giving us a warning

21 or an alert of any kind.

22 MR. BROWN: And how about you, Mr. Laymon and Mr.

23 Kenny, you were both operators, were you familiar with these

24 oil, fats and wax backgrounds that was due to the oil well

25 smoke, is that something that you had seen before the start

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1 of the ground war and the breaching operations?

2 MR. LAYMON: Yes, sir.

3 MR. KENNY: Yes.

4 MR. BROWN: And then you said at this point at

5 0400 just before the breaching operations were going to

6 begin, just before the ground war was going to open up, that

7 you were at a relatively high level of oil well smoke that

8 was particularly dense right then, and yet did you see --

9 when it was particularly dense, did you see alarms at that

10 point for things other than fats, oils and waxes? In other

11 words, in a situation where you had high levels of oil well

12 smoke, did you see chemical agents at that time?

13 MR. LAYMON: No.

14 MR. KENNY: No, sir.

15 MR. LAYMON: None that would go over the picket

16 fence of the background reading.

17 DR. LASHOF: Just one final question and I think

18 we have to wrap this up because we are going to talk about

19 some of the other detections. In our what we thought was

20 our final report before the President decided to continue

21 this Committee, we looked at the question of what scientific

22 evidence we had that low level exposure to chemical weapons

23 in the absence of any acute symptoms, whether there was any

24 scientific evidence that that caused disease, and in our

25 report, we said that our review of all the scientific

1 evidence would indicate that in the absence of acute

2 symptoms, we did not see evidence that illness could be

3 caused by low level exposure to chemicals.

4 We did, however, recommend that further research

5 be done on this because we felt the research wasn't

6 completely adequate and especially in combination.

7 But in your response to Marguerite Knox' question,

8 you kind of indicated that you had been taught that low

9 level could cause disease. I would like to ask each of you

10 whether any of you have been taught that and whether you

11 have any reference or scientific data that you could supply

12 to the Committee concerning any evidence that low level

13 actually causes chronic disease.

14 CWO JAMES: Ma'am, I can tell you that prior to

15 the Gulf War and during it, as has been alluded to earlier,

16 our equipment is generally set to operate at a threshold

17 level and I'm sure there's a definition for that, but it

18 does not mean that it detects the presence of any amount of

19 a chemical agent. It is set at a certain level which

20 someone has determined where you would start to see the

21 onset of fairly immediate symptoms. In the case of a nerve

22 agent, the first one you would see would be pinpointed

23 pupils. So our instruments warrant if you saw a reading, it

24 wasn't the lowest absolute level out there, it was somewhere

25 above that. And we knew that, we knew that even if it

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1 wasn't picking up something, that there was a possibility

2 that there were levels of chemical agent there.

3 Did we discuss that much back during the Gulf War

4 or immediately prior to that? No, ma'am, we did not. That

5 has been a lesson learned from the Gulf War, that now as we

6 develop equipment and procedures and training for medical

7 people, for instance, we need to rethink what detection

8 really means and at what level do we start wanting alarms to

9 go off or bars to show up on a CAM, something like that.

10 So I cannot tell you that I was aware of any long-

11 term exposure hazards from chemicals. We certainly -- I say

12 we -- I, in the training that I've received and the

13 knowledge that I had, I certainly did not believe that a

14 chemical could travel as far as we now think they might and

15 cause problems, health problems. We thought that with the

16 dissipation, the wind breaking up, the rain, all these other

17 environmental factors, I never believed that a cloud or an

18 agent could travel that far. So I personally did not

19 understand the effects of low level hazards from chemicals

20 during the Gulf War.

21 I understand now that there is a hazard. I can't

22 necessarily define or say how we would detect that or how we

23 would know or how it would manifest itself in someone there,

24 that a Marine on the battlefield could pick up.

25 DR. LASHOF: Anybody else want to comment on that?

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1 MSG BRADFORD: Yes, ma'am. As an instructor, one

2 of the things that I make clear to the Marines that I

3 instruct is militarily significant quantities. In other

4 words, they're still able to complete their military duties

5 at the time. Battlefield concentrations, those are the

6 terms that I use to get them to understand that NBC is a

7 survivable event and that if they take their precautions and

8 even if they are exposed, that they still have a chance for

9 survival. We don't talk about quality of life, we don't

10 talk about long-term effects 20 years after they're retired.

11 But that's kind of understood that because these are

12 fuelative in nature for the most part, they're not good.

13 And we do get their attention with the threat of how nasty

14 they are. Sometimes we've overdone that; instead of

15 instilling confidence, we started out by instilling fear.

16 But in our minds, with the detection equipment that we've

17 had, if there is any indication that it's out there, as the

18 Chief Warrant Officer explained, that's getting to the point

19 where an individual is going to be harmed, harmed quickly,

20 and will not be able to be militarily significant, he won't

21 be able to accomplish his duties.

22 DR. LASHOF: Any other different point of view?

23 GNYSGT GRASS: Even the live nerve agent chamber

24 at Fort McClellan, Alabama that we would go through, there's

25 a small amount of nerve agent put onto the tank or the

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1 weapon and the M8 alarm is in there going off and the

2 instructors that are inside this chamber are very concerned

3 if any of us happen to get exposed by that little eyedropper

4 worth of nerve agent. So the significance of whether it's

5 low level or vast amounts of chemicals, even down there at

6 the controlled environment inside this nerve agent chamber,

7 they're very concerned about exposure to small amounts of

8 chemicals down there.

9 DR. LASHOF: Andrea.

10 MS. KIDD-TAYLOR: Just as a follow-up to that

11 question. In testing or putting out a low level dosage of

12 chemical, has it been determined that there is any

13 relationship or health effect that is developed where the

14 person has just been inside the chamber or tested inside the

15 chamber? Or maybe I misunderstood what you said.

16 GNYSGT GRASS: I wouldn't be able to answer that

17 question because I don't work at the chamber down there. But

18 I'm sure that they do studies because they do that

19 cholinesterase check in case anybody has been exposed down

20 there. They take your blood prior to going in, so they've

21 got a baseline and then for the remainder of the school, at

22 least three times after you've come out, they're checking to

23 ensure that your blood matches the baseline.

24 DR. LASHOF: Well, I think we'll wrap it up with

25 that. There's a great deal more to be learned about this

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1 whole issue and we will be continuing to explore it. And I

2 want to again thank you all very much for coming, and your

3 testimony is very helpful.

4 Now, LTC Dee Dodson Morris.

5

6 Ongoing DOD investigations: Kuwaiti Girls School, Mustard

7 Exposure, Camp Monterey and Czech/French Detections

8

9 MR. VESSER: Good afternoon, Dr. Lashof, I'm glad

10 to continue to appear before you for the Department of

11 Defense, to talk about the progress several of our case

12 narratives have taken.

13 With me again is MS Anne Rathwell Davis and her

14 Deputy, LTC Dee Morris, U.S. Army, who will talk in detail

15 about the progress of these case narratives.

16 I would like, after she completes her discussion,

17 to summarize what these cases seem to me to suggest.

18 DR. LASHOF: We appreciate that, thank you very

19 much.

20 LTC MORRIS: Dr. Lashof, members of the Committee

21 and members of the staff, I'm LTC Dee Dodson Morris, Deputy

22 Director of the Investigations and Analysis Directorate,

23 Office of the Special Assistant for Gulf War Illnesses. I

24 will be continuing the case discussion begun by LTC Nalls

25 this morning.

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1 Next slide please. You saw this slide in LTC

2 Nalls' briefing this morning. There's a couple of points

3 that I'd like to highlight before continuing. In the first

4 bullet, we talk about the chemical investigation process.

5 Based on a lot of the information that you've heard from a

6 variety of people testifying before me, it should be clear

7 to you that a single type of evidence is not necessarily

8 determinative and it is because of the fact that we have a

9 comprehensive process and we go out and investigate in

10 addition to looking at scientific evidence, that we are able

11 to move forward with our investigations to determine what

12 may have caused the illnesses that our veterans suffer from.

13 This morning we heard about how the Fox system and

14 the M256 kit work and what they can and cannot do. The four

15 cases that I will now discuss highlight some of these

16 points.

17 Next slide please. Go to slide 5 please.

18 If I had my druthers, the evidence that we had

19 with Camp Monterey would be the type of evidence that I'd

20 like to see for every case. This particular case was a

21 reported positive detection of the nerve agent sarin by a

22 Fox reconnaissance vehicle after the Gulf War. The

23 information that we used to initiate this case was provided

24 to us last fall by an attorney who was representing an

25 individual who worked for General Dynamics and was

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1 responsible for maintenance of the MM1 mass spectrometer on

2 the Fox vehicles.

3 Next slide please. In the course of our

4 investigation, we interviewed the attorney's client, we

5 interviewed the commander of Camp Monterey and we located

6 and interviewed one of the operators of the Fox detection

7 system involved. We also conducted a background information

8 check through our databases and did in fact in the course of

9 this investigation compile a complete list of the soldiers

10 who were involved. And while we have not interviewed all of

11 those people to date, we are continuing to talk to some of

12 them in the event that we get additional information which

13 could change the story as we currently view it.

14 We've also looked at videos and photos of Camp

15 Monterey which were submitted by veterans, to try and create

16 a better picture in our minds as to what occurred.

17 In addition, because the attorney's letter

18 included tapes from two Fox vehicles, we submitted those

19 tapes for scientific analysis, an independent scientific

20 analysis from three separate organizations. The first

21 organization was the Chemical and Biological Defense

22 Command, the second was Brooker Analytical Systems, Inc.,

23 which is the manufacturer of the MM1, and as a final sort of

24 gold standard test, we submitted the tapes for review to the

25 National Institute of Standards and Technology.

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1 Next slide please. As you saw in LTC Nalls'

2 initial discussions on his cases, we had had some

3 unproductive leads. In this particular case, during a

4 literature search, we found nothing talking about Camp

5 Monterey, this investigation literally popped out last fall.

6 And while we say none in the column for evidence being

7 sought, when we publish our narrative on this case, it will

8 be published as an interim report. And therefore, if anyone

9 comes up as a result of reading that report and reviewing

10 it, with additional information that we have so far been

11 unable to locate, we will subsequently incorporate it in any

12 follow-up reports.

13 The evidence that we have to date in the Camp

14 Monterey case indicates that the chemical agent detected was

15 the riot control agent CS or tear gas, and not sarin. This

16 is supported by the interviewee reports of the events, which

17 are consistent with each other and are consistent with the

18 analysis of the tapes. And therefore, our case narrative at

19 this point in time is in final coordination and should be

20 published soon.

21 Going back to Camp Monterey briefly, it's

22 important to note here that while we were investigating

23 this, as a chemical officer, I was somewhat troubled by the

24 fact that we would have an alert for CS and then the actual

25 -- excuse me, we'd have an alert for sarin and that the

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1 actual agent detected would be CS. And so when we went to

2 Brooker as the manufacturer of the MM1, we specifically

3 asked them to explain to us in detail how that occurred, and

4 their explanation was clear and it turns out that this

5 particular phenomenon is expected and it's based on the four

6 peak analysis. It turns out that when you're looking for

7 sarin, they look for one peak twice in the four peak

8 analysis and it's only one point off of one of the peaks

9 that they look at for CS, and based on the sensitivity

10 parameters of the MM1, this is something that should be

11 expected.

12 Moving on to the mustard exposure. This

13 particular case involves a single incident of exposure to a

14 soldier assigned to the 3rd Armored Division immediately

15 following the war. This incident was documented during the

16 Gulf War and the predecessor organization to ours opened the

17 case due to the fact that it was mentioned in the Riegel

18 report. We have interviewed the exposed soldier, we've also

19 reviewed contemporaneous reports such as those provided by

20 COL Dunne, and log entries that traced the reporting of this

21 particular incident up the chain of command. Dr. Dunne also

22 provided testimony to this Committee and we've also found

23 other references to it in our database record searches.

24 Shortly after this event, CBDCOM was asked to

25 review an MM1 tape and they also were given the privilege of

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1 reviewing a video that documented the detection.

2 Next slide please. This is not quite so clean as

3 Camp Monterey. There have been a number of query results

4 which didn't provide us with any additional substantive

5 information and upon later testing of the clothing and the

6 soldier's urine, there were no indications of mustard. I'd

7 like to note that the initial alert on the vest was positive

8 and that there were two Foxes involved, and that according

9 to the video, one of the alerts was in fact verified with a

10 spectrum.

11 We're not currently seeking any additional

12 information but my comment concerning that particular fact

13 at Camp Monterey holds true here. If additional information

14 comes to light upon the publication of our narrative, we

15 will of course consider it and include it in an update.

16 And although the physical evidence in this

17 particular case is not conclusive, the overall evidence

18 suggests that this was in fact a mustard exposure, and a

19 thorough and immediate reaction to exposure at the time of

20 the incident such as this was an isolated incident. There

21 were no other people that were with this individual soldier,

22 who have demonstrated any injuries.

23 Case narrative for this particular case is in

24 review and will be published soon.

25 The third case is the Kuwaiti Girls School. This

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1 is again a reported positive detection by a Fox

2 reconnaissance vehicle after the Gulf War. In this

3 particular case, the reported detection is of the blister

4 agent mustard. This particular case has had an extensive

5 life. The commander of the chemical company at the incident

6 used it as an example when recommending a curriculum change

7 to the infantry school in early 1994. This particular

8 statement that he made to his chain of command was

9 subsequently reviewed by the Defense Science Board and the

10 Riegel Committee, and as a result, when the Persian Gulf

11 illnesses investigation team was formed, we did open a case

12 on this due to the fact that it had been mentioned in the

13 Riegel report.

14 Next slide please. We've done extensive

15 interviews on this particular case. We've talked to the

16 unit commander and his first sergeant. There were earlier

17 interviews done of the chief of security systems-Kuwait and

18 the regimental and task force chemical officers, and we have

19 done a number of back and forth phone calls with the experts

20 at Edgewood who have initially reviewed the tapes when they

21 were first sent from the Gulf.

22 The unit commander has made a prepared statement,

23 the task force chemical officer also has an extensive

24 statement and we have two memoranda from the British

25 chemical establishment at Porton Down, detailing the

1 analysis on the container contents, which they had

2 performed. We also have the initial report prepared by the

3 21st EOD squadron group, British Royal Engineers, which is

4 dated two days after they first went and looked at the tank,

5 and we've looked at all the various answers to questions

6 that were provided to both the Riegel Committee and other

7 organizations looking at this.

8 Because the particular information was based on

9 whether or not the Fox could or could not be fooled, we did

10 make a specific request that red fuming nitric acid be

11 tested in a number of detection items, to include the MM1,

12 the CAM, the 256 kit, the M18 kit and the various papers, M8

13 and M9.

14 As was mentioned earlier today, the MM1 did in

15 fact, after it was more or less saturated with red fuming

16 nitric acid at lab grade, initially alert for cyclo-sarin,

17 not mustard, which is subsequently cleared during a spectral

18 analysis.

19 We have had no unproductive investigative leads on

20 this case thus far. Everything has served to provide a

21 little bit more, and hopefully we'll be able to pick this

22 particular one apart. There are extensive pieces of

23 evidence that we're still continuing to look for. We are in

24 the process of identifying the remaining Fox operators and

25 do intend to interview every one of them. We would like to

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1 see the specific analysis results from Porton Down, and I

2 understand that your Committee has in fact formally

3 requested that those analysis results be provided. We will

4 be asking for the current status of the British soldier

5 involved. We're still looking for copies of the Fox tapes,

6 we have not yet found copies of those tapes at Edgewood, but

7 there are a number of other places where they might have

8 been sent, and so we're continuing to follow that lead. And

9 we're also looking at disposition of similar containers by

10 Corps of Engineers contractors.

11 One of the things that we recently discovered was

12 that the Corps of Engineers did in fact come across -- or

13 contractors for the Corps of Engineers did in fact come

14 across a number of similar types of containers and there is

15 some documentation that these containers in Kuwait City

16 contained red fuming nitric acid. So we're looking at the

17 records of the contractor to see what they may have found.

18 In this particular case, we believe it is too soon

19 to make a determination one way or the other. The facts are

20 conflicting, the stories that we have from people who were

21 physically present at the incident do in fact conflict on

22 major points. We, therefore, believe that the contents of

23 the container is still not -- it's still too soon to tell

24 what the contents were, but unfortunately there may be some

25 conclusive evidence, such as the tapes which may no longer

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1 exist.

2 This case is still under active investigation and

3 it's one of the ones that I'm personally investigating.

4 Next slide please. The final case that you asked

5 me to discuss is the Czech/French detections.

6 This case involves seven incidents which were

7 reported by either Czechoslovak or French forces during the

8 first week of the air war, concerning the possible detection

9 of nerve or blister agents at very low concentrations.

10 The reports of these particular detections were

11 reported in CENTCOM logs and other places. There was an

12 investigation initiated on this by the Defense Science Board

13 study and again, the Persian Gulf illnesses investigation

14 team opened a case due to the fact that this had been

15 mentioned in the Riegel report.

16 We've had four people call us on this particular

17 case, on the incident reporting line, and we have

18 interviewed all of those. And we have been developing

19 sources from other source -- from other logs and reports

20 such as the ARCENT chemical officers, other special forces

21 personnel and anyone else with knowledge of the incident.

22 There has been extensive reporting on this case

23 previous to our investigation and we have reviewed the

24 existing summaries and reports and testimony by the Defense

25 Science Board, MAJ GEN Blanck, Senator Riegel's committee,

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1 Senator Shelby's committee, the Defense Intelligence Agency

2 and the Central Intelligence Agency. In addition to that,

3 we have conducted an extensive search of our databases.

4 One of the problems that we experience in all of

5 our investigations is that some interviewees cannot recall

6 details and many people report things that they have only

7 heard from other sources. This is also the case here. We

8 have also found that the Defense Attache Office in Paris has

9 repeatedly tried to contact the French government on this

10 subject and the French government is extremely reluctant to

11 provide us with any additional information reference their

12 equipment and their methodologies, and this has been the

13 case since 1994.

14 We are continuing to seek substantiating

15 information on the Czech and French reports from first-hand

16 witnesses or contemporaneous logs and records. And also

17 identification of specific units and/or individuals who

18 might have been in the immediate vicinity of the reported

19 detections at the times of those detections.

20 Next slide please. We have found no evidence that

21 would cause us to change our initial assessment, which was

22 published last summer, that the Czech detections of 19 and

23 24 January are credible; however, have not been verified.

24 And the credibility of the remaining detections still cannot

25 be assessed.

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1 The case narrative for this case is in review and

2 will be published soon, and hopefully this is one

3 opportunity where we will get the chance to solicit some

4 folks who may know something, to come to us and give us some

5 additional information that will help us further explain it.

6 As a number of other briefers have indicated

7 today, the investigations and analysis of chemical and

8 biological warfare agents and specifically LTC Nalls'

9 division, is organized and is making significant progress on

10 the investigation of a number of cases. We have a number of

11 related investigation efforts underway and these efforts

12 support our current and future analysis. Our specific

13 chemical warfare agent cases are progressing as scheduled

14 and narratives should be published in accordance with the

15 schedule that we have previously provided you.

16 I'm ready to take your questions.

17 MS. NISHIMI: I think GEN Vesser had --

18 MR. VESSER: First, let me say that I'm glad we've

19 had the opportunity to hear the testimony of the Marine Fox

20 vehicle crew members who were operating during the mine

21 field breaching incident. We are especially glad to hear

22 the testimony of the new witness whom the Committee located,

23 SGT Laymon.

24 I would ask everyone to remember that these

25 Marines were in a hazardous situation and did what was

1 expected of them. Remember that their actions took in a

2 high intensity war of short duration. To reduce risk to

3 themselves and others from enemy fire, they were ordered to

4 keep moving through the mine field breach. The threat to

5 their lives from enemy action, the potential for deliberate

6 use of chemical weapons against U.S. forces for the first

7 time since the first world war, and possibly the first ever

8 modern use of VW, put them under stress.

9 In the circumstances, they performed as trained

10 and got the best information the Fox vehicle could give

11 them, given its limitations that you have heard about this

12 morning. I would say to you that they called it as they saw

13 it to protect their fellow Marines, for which the nation

14 owes them a debt of gratitude. But we have to look at all

15 the evidence and some of the evidence is on the tape. We

16 think that the tape has some anomalies with some of the

17 testimony that you heard this morning, and we'll be working

18 with those witnesses to resolve those anomalies.

19 These Fox vehicles incorporated the latest state

20 of the art detection equipment into a lightly armored

21 vehicle. You have heard how it was not designed to perform

22 some of the tasks it was assigned. And that in the mode it

23 was most often employed, it was programmed to alert with

24 some paucity false positives to make certain our troops had

25 early warning. We in the Department of Defense are glad

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1 that it was programmed to give those alerts, so that troops

2 like those in the mine field breach got into MOPP 4. There

3 were no chemical casualties other than the one that's still

4 under investigation and the one that LTC Morris has

5 referenced, that we know of.

6 The exact determination of substance detected

7 required further sampling and time-consuming testing for

8 validation. Despite these difficulties, the challenge to

9 absorb training quickly by the crews whom you heard,

10 maintenance and logistic problems, getting parts for the

11 vehicles, and the innovation on the part of commanders to

12 develop ad hoc tactics for the vehicles' employment, it

13 offered our commanders a capability for detecting chemical

14 agents not otherwise available in fast moving offensive

15 operations, and it offered protection to those who had to do

16 it.

17 The less threatening situations which LTC Morris

18 described in which the Fox vehicle could be employed to its

19 full capability, has left us with some records that have

20 contributed to more conclusive assessments. We hope would

21 have -- we would hope we could find more of these records.

22 Their relevance in assessment underscore the need for

23 emphasis on creating records and keeping them.

24 You also heard some of the Marine witnesses say or

25 suggest that our effort in Defense is uncoordinated. We

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1 have the investigations handled by Dr. Rosker's office and

2 three other investigations going on. Dr. Rosker has

3 responsibility and authority to integrate within the

4 Department of Defense the results of those efforts. It's

5 clear that even though we make effort so that we only have

6 to talk to a witness once in Dr. Rosker's office, that other

7 elements are looking for different kinds of information.

8 The Special Assistant for Intelligence who had also been in

9 touch with CWO James, for instance, has a different

10 portfolio than our office has. In addition, of course, we

11 have the DODIG looking for the missing CENTCOM chemical logs

12 and the Army investigation which parallels our own. So it's

13 not surprising that many witnesses are contacted by more

14 than one investigator.

15 Thank you.

16 DR. LASHOF: Thank you. I also appreciate the

17 fact that you have heard the Marines testimony and it was

18 very clear that we are both going to be following up on a

19 number of the issues that were raised by their testimony and

20 by the other testimony this morning. So we're not putting

21 this to bed today, that's for sure, and we will be working

22 with you on these issues.

23 I think you still could coordinate the questioning

24 of them. At least you shouldn't have to bring them back too

25 many times and get everybody together.

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1 But let us move on to questions that we might have

2 for LTC Morris. Are there any questions? Joan.

3 MS. PORTER: I have a question concerning the

4 Kuwaiti Girls School case. If it were red fuming nitric

5 acid, which is used in large quantities in rocket

6 propellant, in the tank at the girls school, why would there

7 be only one such tank which was a relatively small tank?

8 LTC MORRIS: That's one of the mysteries that

9 we're trying to address. One of the things that we have

10 found through our contacts with the Corps of Engineers is

11 that this was not the only tank that was found in Kuwait

12 City in a single emplacement. And so we're not really sure

13 why it was there by itself. Other photographs that we have

14 seen of fueling and defueling efforts do indicate that they

15 would be used in a larger number, although the tank size is

16 the same. But again, that's an anomaly we're trying to

17 figure out.

18 MS. PORTER: Isn't it true that tanks of this size

19 and description were used for the storage of other chemical

20 agents?

21 LTC MORRIS: We have indications that similar

22 tanks were used for the storage of mustard. We have not

23 been able to make a determination as to whether these are

24 identical and I think, as I recall, when we chatted about

25 this during a staff visit, we have a similar case in the

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1 United States where we use ton containers for the storage of

2 chemical agents, and those same ton containers can be found

3 at large pools for the storage of chlorine. And really the

4 only difference between the two is the size of the valve.

5 And so again, we're looking into what were the

6 possibilities here and we'll weigh all the evidence

7 accordingly.

8 MS. PORTER: So they did have multiple uses, some

9 of which were chemical warfare agents?

10 LTC MORRIS: We believe so, yes, ma'am.

11 MS. PORTER: DOD had laboratory tests for RFNA on

12 a variety of different detectors and our understanding from

13 the discussion this morning and previous discussions with

14 your staff and others was that none of these detectors

15 registered mustard in the presence of RFNA. In fact, in the

16 presence of RFNA, the Fox would register cyclo-sarin, or

17 unknown. Is that correct?

18 LTC MORRIS: Yes, ma'am, that is. And just to

19 reiterate, the tests were done with lab grade red fuming

20 nitric acid and we really have no idea what was in the tank

21 right now, so we're not really sure what concentration it

22 might have been.

23 MS. PORTER: But that would lead you to believe

24 then that RFNA wouldn't have registered mustard on the Fox.

25 LTC MORRIS: That's one piece of evidence that

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1 we're considering, yes, ma'am.

2 MS. PORTER: A question on the Czech detections.

3 Last spring the Department of Defense testified before this

4 Committee that no U.S. personnel were in the vicinity of the

5 Czech/French detections on those dates indicated, except

6 those U.S. Fox operators who were called in to try to verify

7 what the Czechs and the French had supposedly detected. Is

8 that still your belief or has that belief been modified

9 somewhat?

10 LTC MORRIS: We have some indications that there

11 were special forces individuals who were operating singly or

12 in small groups with some of these organizations, and we're

13 continuing to follow those.

14 MS. PORTER: You're actively pursuing trying to

15 locate those persons?

16 LTC MORRIS: Yes, ma'am, I think if you go back to

17 my slide, that was one of the people we're looking for.

18 MS. NISHIMI: I have a question about the mustard

19 case, I think it's slide 12. In response to the committee's

20 letter report issued at the end of April, DOD pled, quote,

21 not guilty, with respect to our reporting that the standard

22 for possible exposure was, quote, best evidence, or quote,

23 preponderance of evidence, which we -- I'm looking for the

24 one that says evidence to date indicates -- I'm sorry.

25 Anyway, which was drawn directly from the Salt Lake City

1 transcript. And I was under the impression now that DOD's

2 position was that, quote, common sense would be used and --

3 okay, I see you've changed slides, that was my question.

4 The slide that you had delivered to the Committee still

5 retained the preponderance of evidence standard.

6 LTC MORRIS: Right.

7 MS. NISHIMI: And I wanted you to clarify that.

8 LTC MORRIS: In this particular case, the physical

9 evidence here is not conclusive; however, largely because of

10 the fact that there was a contemporaneous diagnosis of this

11 soldier's injuries by a qualified physician, we felt that

12 the prudent thing, and that the evidence indicated, that a

13 reasonable person could believe that this individual was in

14 fact injured by mustard and that is our position.

15 MS. NISHIMI: And so just to clarify. DOD's

16 threshold, if you will, is this common sense threshold, not

17 the preponderance of evidence threshold, is that correct?

18 LTC MORRIS: Yes, ma'am. It will be different for

19 every case.

20 DR. LASHOF: Well, how you assess evidence will be

21 different for every case, not the standard you're going to

22 hold to. Either you hold to a certain level or another

23 level and then you decide whether your evidence is up to

24 that level. What we were getting at was not requiring it be

25 at the highest level to prove beyond a reasonable doubt, as

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1 one might in a murder case, but talk to what's reasonable

2 evidence, which is what we're looking for, I think. And I

3 gather you concur now that reasonable evidence is what we

4 ought to look for.

5 Any other questions?

6 (No response.)

7 DR. LASHOF: If not, thank you very much, we

8 appreciate your attention and help and we will keep working

9 on these.

10 I'm going to ask the Committee to look at Tab E

11 and we're changing the agenda slightly, and we'll hold off,

12 if we can, from Mr. Parker, who was going to testify about

13 the risk communication plan. But I would like us to address

14 ourselves to the Committee's report dealing with these cases

15 first and then we'll go to Brigadier General Parker for the

16 risk communication plan.

17 So if you will look at your Tab E in our books, we

18 have some Committee staff review of findings and

19 recommendations that were in our report and then some

20 findings and a discussion about the Kuwait Girls School case

21 and the Camp Monterey chemical weapon detection. Do you

22 want to lead that discussion, Robyn?

23 MS. NISHIMI: Yes. The first memo just extracts

24 the findings and recommendations from the Committee's Final

25 Report concerning CBW investigation. And as you recall, we

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1 said in the finding that in the face of credible evidence of

2 the presence or release of chemical warfare agents, low

3 level exposure of U.S. personnel at the affected site must

4 be presumed, while efforts to develop more precise measures

5 of exposure continue.

6 DR. LASHOF: Incidentally, that was on page 46,

7 not 26, so you may want to correct your notes.

8 MS. NISHIMI: And the recommendation as follow-up

9 to that finding was that in determining the extent of

10 possible chemical warfare agent exposure, the Committee

11 identifies the following as essential principles; and that

12 is that troops within the presumptive exposure area should

13 be notified and encouraged to enroll in the CCEP or the VA

14 registry program.

15 As you further recall from the Committee's final

16 report, we looked at three site-specific chemical agent

17 exposures, the mustard exposure for SGT Fisher, the nerve

18 and mustard agent detections by the Czech forces and then

19 the nerve agent releases at Khamisiyah.

20 Based on the staff's work for two other cases, we

21 are prepared to make a recommendation to the Committee

22 regarding the Sabajea (ph) Girls School at Kuwait and the

23 Camp Monterey incident. And I'd like to walk the Committee

24 through that.

25 As you've heard from LTC Morris' testimony, there

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1 was an incident at the Kuwaiti Girls School and British and

2 U.S. troops were present there. The basic question for the

3 Committee to consider is were those detections credible and

4 then if they are judged credible, what is the Committee's

5 recommendation as follow-up to that.

6 As to the question about whether the detections

7 are credible, the staff position is that the evidence from

8 the best detectors available in the field indicates mustard

9 was present at the site of the tank. Evidence from

10 subsequent laboratory analyses in the U.K. as well as re-

11 analysis of the Fox spectrum tapes at Edgewood, suggest the

12 substance in the tank might have been RFNA, the red fuming

13 nitric acid. To date, the detailed Porton Down report has

14 not been made available either to DOD or to the Committee,

15 although it has been requested, and the Fox tapes no longer

16 exist.

17 The staff recognizes that detectors are fallible

18 and they can be confused by hydrocarbons and other

19 environmental contaminants. However, the number of positive

20 detections registered by different methodologies at the time

21 of the incidents is weighty, in our opinion. Moreover,

22 recent laboratory tests conducted at Edgewood at the request

23 of DOD, showed that none of these detectors registered a

24 false positive for mustard when exposed to RFNA. So lacking

25 100 percent certainty and there will never be 100 percent

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1 certainty, staff recommends that the Committee find the

2 detections at the Sabajea Kuwaiti Girls School as credible.

3 The prudent course of action following this

4 finding is as follows: Staff further suggests that the

5 Committee recommend, which is consistent with our final

6 report, that DOD undertake a targeted notification of these

7 individuals, of the possible, though remote, health risks

8 and the availability of the medical and diagnostic treatment

9 programs established by the government. That would be the

10 CCEP obviously and the VA registry programs.

11 Staff also recommends the Committee make the

12 finding that it is highly unlikely the public health risk

13 extends beyond the limited group of people.

14 DOD's investigations in these matters, as you have

15 just heard, are ongoing and so further information could be

16 forthcoming, and if a firm conclusion is reached, then the

17 same troops could be renotified as to that position. But it

18 is the staff's finding at this point that the Committee

19 conclude that these detections are credible and that the

20 notification proceed, because exposure should be presumed.

21 With respect to the Camp Monterey -- you want to

22 do them separately? Okay.

23 DR. LASHOF: Please. Let's ask if there are any

24 questions the Committee has in regard to this. One

25 question, how many people are involved again in the Kuwaiti

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1 exposure?

2 MS. NISHIMI: The precise number is not entirely

3 clear at this point, there is one British soldier who

4 suffered an injury that's consistent with exposure to a

5 blister agent and then there were individuals operating two

6 Fox vehicles and perhaps some MPs who were in the area. It

7 is a confined population, but the recommendation following

8 the Committee's final report would be that these individuals

9 be notified of a presumed mustard exposure.

10 DR. LASHOF: Has there been any effort up to date

11 to talk to any of the other individuals besides the British

12 soldier?

13 MS. NISHIMI: Joan.

14 MS. PORTER: I'm sorry? Your question again?

15 DR. LASHOF: Has there been effort to talk to the

16 other individuals at the Kuwaiti School?

17 MS. PORTER: Our staff has not talked to them.

18 Members of the OSAGWI staff have spoken with individuals

19 present. We are trying to obtain information from the

20 British quite actively, but that has not been forthcoming.

21 DR. LASHOF: Any other questions for staff?

22 MS. KIDD-TAYLOR: I think it's getting clear from

23 you, Robyn, we really don't know an exact number of persons

24 exposed at the girls school?

25 MS. PORTER: I would imagine that it's less than

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1 20 persons, not accounting for any personnel who might have

2 disposed of the tank, which we don't know at this point.

3 MS. NISHIMI: Is there any objection by the

4 Committee to moving forward with this recommendation?

5 (No response.)

6 DR. LASHOF: Hearing no objection, I think the

7 Committee will accept this, subject to further change in our

8 final report if those others get more information.

9 MS. NISHIMI: One of the reasons obviously at this

10 point to move forward is the Committee has been charged with

11 looking at several site-specific exposures, and we believe

12 it prudent to start knocking some of these off the table as

13 the staff becomes comfortable with the current level of

14 analysis.

15 And again, we view that the two things, RFNA, not

16 as a confounding factor for mustard, and the range and

17 breadth of the different methodologies in place at the time

18 of the detection, -- there were different methodologies,

19 same result. We view that as weighty evidence.

20 In the case of the Camp Monterey chemical

21 incident, as has been described by LTC Morris and as is

22 described briefly in the memo, a few Army soldiers became

23 ill while moving wooden crates containing some substance at

24 a building at Camp Monterey in Kuwait.

25 There were Fox vehicle detections and subsequent

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1 analyses by the National Institute of Standards and

2 Technology. The conclusion was that the mass spec reading

3 was CS and that CS could cause a false detection for GB.

4 That was DOD's conclusion, that was also the assessment --

5 NIST concurred with that assessment. Staff concurs with

6 both of these assessments and so we recommend the Committee

7 find detections credible for CS, but not for GB.

8 The follow-up question is what is the prudent

9 course of action in the near term. And therefore, similar to

10 and consistent with the Committee's final report, staff

11 suggests the Committee recommend that DOD undertake targeted

12 notification again of these individuals of any possible

13 long-term health risks associated with acute illness

14 following exposure to CS. And again, inform them of the

15 available medical and diagnostic programs currently underway

16 by the government.

17 DR. LASHOF: Any questions of the Committee? This

18 is one where we have complete agreement with us and DOD for

19 a change.

20 MS. KIDD-TAYLOR: I just had one about because of

21 the false positive for GB, a false alarm for GB. Can

22 someone describe why that is, why was it a false alarm for

23 GB in this case for Camp Monterey?

24 MR. BROWN: Yeah. We looked at this analysis, it

25 was provided initially by the manufacturer of the mass

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1 spectrometer, Brooker, and also some of our staff have some

2 experience in mass spectrometry and reviewed this argument

3 and basically it's a case of mistaken identity that's

4 specific for the example of tear gas and GB. The problem is

5 the way CS behaves in this mass spectrometer, it just

6 happens by bad luck coincidence to give the right ions as it

7 fragments in the right ratio to fool the machine into

8 thinking there may be GB present. It's a specific example

9 for that particular combination of chemicals.

10 MS. KIDD-TAYLOR: They appear to be the same.

11 MR. BROWN: In the alarm mode, they look the same.

12 They can be differentiated in the full spectrum mode.

13 MS. KIDD-TAYLOR: Okay.

14 DR. LASHOF: And they were differentiated in the

15 full spectrum mode.

16 MR. BROWN: And indeed they were.

17 MS. KIDD-TAYLOR: Okay.

18 DR. LASHOF: Is there any reservation about this

19 recommendation?

20 (No response.)

21 DR. LASHOF: If not, we'll indicate to staff that

22 that's accepted and this is one we can put to bed for the

23 moment.

24 MS. NISHIMI: For the moment.

25 DR. LASHOF: For the moment. Everything is for

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1 the moment.

2 Okay, anything else on this issue before I ask

3 Brigadier General Parker to come forward and brief us on the

4 risk communication plan?

5 (No response.)

6 DR. LASHOF: If not, then I call Brigadier General

7 Parker for the Persian Gulf Coordinating Board. And what

8 we'll be doing over our next few meetings is going through

9 reviews of the implementation of the recommendations made in

10 our final report, and you are blessed with one of the first

11 ones to tell us what you're doing about risk communication.

12 And listening to testimony this morning and previous

13 mornings, we need a lot of work on risk communication.

14

15 Implementation of Final Report recommendations: Risk

16 communication

17

18 BRIGADIER GENERAL PARKER: Well, Dr. Lashof, thank

19 you very much for inviting me here today to talk about this

20 vital subject, and you've taken the words out of my mouth

21 with your opening remarks there.

22 I would like to read my testimony, then make a

23 comment about our symposium on risk communication and then

24 ask for questions from the Committee.

25 I am Brigadier General John S. Parker, United

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1 States Army Assistant Surgeon General for Health Services,

2 Financial Policy and Readiness Operations. I am speaking to

3 you today for the Clinical Working Group which I chair, of

4 the Persian Gulf Veterans' Coordinating Board. The Board

5 oversees an interagency effort by the Department of Defense,

6 Health and Human Services and Veterans' Affairs to meet the

7 needs of our Gulf War veterans.

8 One of the clearest lessons of the Gulf War for us

9 has been the importance of effective risk communication for

10 those who have served in the Gulf and for those who will

11 serve in future conflicts and other deployments. Risk

12 communication involves providing accurate, timely and

13 credible information to service members about the potential

14 health problems they may face during their service, in

15 combat and in peace, about the protection that we provide

16 against those hazards and during and after a specific

17 operation has taken place, about what really happened that

18 might pose an immediate or future health risk.

19 Like other critical elements affecting the welfare

20 of service members, we recognize that risk communication is

21 a responsibility for commanders at all levels. In their

22 combined response to the President concerning the

23 Presidential Advisory Committee's final report, the

24 Secretaries of the three departments agreed with the

25 Committee's call for improved risk communication.

1 Planning is well underway to make the required

2 improvements. By the time I present this testimony to you

3 today, the Clinical Working Group will have sponsored a day

4 long meeting, on the 29th of April, involving risk

5 communication experts of the three departments. The meeting

6 will incorporate the experience of recognized risk

7 communication authorities from the United States Public

8 Health Service, who deal with similar issues for the

9 residents of areas in the United States designated as toxic

10 waste sites under the Superfund program.

11 A follow-up meeting on the 30th of April will

12 involve risk communication planning and training for the

13 staff of the Secretary of Defense's special assistant for

14 Gulf War Illnesses, joined by Department of Veterans'

15 Administration representatives.

16 We have a coordinated, well-resourced effort

17 underway to meet the commitment of the three departments to

18 improve risk communication and outreach. DOD's special

19 assistant for Gulf War illnesses conducts round tables with

20 veterans' organizations as well as holding town hall

21 meetings with concerned veterans in nine cities over the

22 coming two months, and operates a toll-free incident

23 reporting line which provides debriefings by a trained

24 investigator.

25 The Secretary of the Department of Veterans'

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1 Affairs has conducted public forums in six cities to explain

2 the Department of Veterans' Administration's programs for

3 Gulf War veterans and to receive input on the Department of

4 Veterans' Administration's undiagnosed illness compensation

5 program.

6 The Department of Veterans' Administration has an

7 ongoing program of quarterly Persian Gulf newsletters,

8 public service announcements and community-based outreach.

9 The Department for Veterans' Affairs will soon conduct its

10 third continuing education conference on health consequences

11 of Persian Gulf War service for 400 health care

12 professionals from the Department of Veterans'

13 Administration, the Department of Defense and the Department

14 of Health and Human Services. And is about to publish a

15 printed continuing medical education self-study program

16 directed at these efforts. These efforts will ensure that

17 health care providers will have the most up to date

18 information for communication with their Gulf War veteran

19 patients.

20 The Clinical Working Group supports efforts to

21 assure consistency of information provided to Gulf War

22 veterans on the three government World Wide Web sites

23 operated by the Presidential Advisory Committee, the

24 Department of Defense and the Department of Veterans'

25 Affairs. The working group now has in final development

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1 fact sheets in plain language on the following topics:

2 Depleted uranium, illnesses of Persian Gulf veterans,

3 Persian Gulf Veterans' Coordinating Board, medical care

4 programs, case series registries, overall research approach

5 and pyridostigmine bromide. Fact sheets on an additional

6 ten topics are in current preparation.

7 The Department of Defense has already used the

8 Gulf War experience to improve risk communication, as shown

9 by its work in Operations Joint Endeavor and Joint Guard in

10 Bosnia. And it is planning for protection of our total

11 force with immunization against a biological warfare threat

12 agent, anthrax.

13 Service members deployed to Bosnia have been

14 individually briefed and provided with specific pamphlets on

15 the infectious environmental health threats of that region

16 and on the appropriate medical countermeasures. Disease

17 surveillance in theater and post-deployment medical

18 screening of returning service members is coupled with clear

19 information on how to access help for any potential problem.

20 A robust team of environmental and occupational

21 health experts has done extensive air, soil and other

22 monitoring with full communication of its results.

23 Communication to all service members of information about

24 infectious disease threats, about the hazards related to

25 land mines and about occupational hazards related to normal

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1 maintenance activities under field conditions has been a

2 strong element of the health effort.

3 The provision of a new vaccine against tick-borne

4 encephalitis to personnel at risk for this disease involved

5 a major risk communication effort. The vaccine was provided

6 with full explanation of its U.S. Food and Drug

7 Administration investigational status and well-documented

8 informed consent by those who volunteered to receive it.

9 And with complete follow-up tracking of vaccine recipients

10 for efficacy and safety.

11 The Department of Defense has continued concern

12 about the potential threat posed by anthrax, a biological

13 warfare agent that can cause fatal illness after inhalation

14 exposure. In considering how best to protect all service

15 members against this threat, the Department of Defense is

16 evaluating the potential to immunize all service members

17 using a currently approved and licensed vaccine.

18 A central element of such a policy would be a

19 robust risk communication program to fully inform recipients

20 of the medical and operational need for vaccination as well

21 as all potential risks involved with this vaccine and how to

22 get effective help for any potential problem.

23 To conclude, the ongoing efforts of the Clinical

24 Working Group of the leadership of the three departments and

25 of their cooperating agencies will become the foundation of

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1 the Persian Gulf Veterans Coordinating Board's risk

2 communication plan. The plan will coordinate individual

3 departmental efforts, supplement and enhance efforts,

4 initiate new and improved risk communication vehicles and

5 ensure that the government's risk communication plan is

6 comprehensive, coherent and effective.

7 We recognize the enormous challenges involved in

8 communicating with veterans on this dynamic and

9 controversial issue. We are committed to meeting those

10 challenges.

11 And before I ask for questions, I would say that

12 we invited recognized authorities from the Public Health

13 Service, Dr. Max Lum, Tim Tinker from the Health and Human

14 Services to be with us on the 29th of April. They did for

15 our group of people what it usually would take about five or

16 six days of a seminar to do. They picked the nuggets and

17 they drilled us and what we gleaned from that whole

18 experience was one that made us all take one step backwards

19 and say we've approached this a little bit like doctors

20 making decisions up front about what veterans need to hear,

21 not what they want to know about and what they need to hear.

22 We have understood that risk communication means an exchange

23 of information and that means answering questions truthfully

24 and honestly and answering the questions that they want

25 answered.

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1 In that vein, I am also a member of the Veterans'

2 Administration Persian Gulf Expert Scientific Community. I

3 tried before this meeting to contact Dr. Bingham to ask for

4 her assistance, but it is my intent the next time I talk

5 with Dr. Bingham, is to ask her to sponsor a veterans'

6 advisory panel so that we can have a forum for veterans and

7 list the things that they want to hear about, how they want

8 to hear about it and how they want that communicated.

9 Ladies and gentlemen, that concludes my remarks.

10 DR. LASHOF: Thank you very much. Let me start

11 with a question about the paper you have in preparation, the

12 fact sheets that you have in preparation. Those are being

13 drawn up obviously and the question is whether they're being

14 drawn up by the doctors who think they know what the

15 veterans want to hear and whether you have a plan to have --

16 to test those, field test them on a group of veterans before

17 you print them all up and distribute them.

18 BRIGADIER GENERAL PARKER: Dr. Lashof, if we

19 hadn't had that seminar, we would have fallen into that

20 trap, we would have printed them and sent them out. I will

21 mention that at that seminar, we did invite the veterans'

22 service organizations representatives to be there. We did

23 have one member show. We will pretest the fact sheets, make

24 sure that they're readable and that they're understandable

25 and do they even want to hear about this. So we will do

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1 that.

2 I really think the group of people that are on

3 this Committee are dedicated and we will try to follow some

4 of the science of risk communication.

5 DR. LASHOF: Good, thank you.

6 MS. KIDD-TAYLOR: That was my question exactly,

7 you know, it's different for a scientist to write a fact

8 sheet versus a lay person to understand it.

9 DR. LASHOF: Yes.

10 MR. CASSELS: You gave us a list of six or seven

11 fact sheets that were under development, plus the statement

12 that ten additional topics are in preparation. Are the low

13 level effects or potential low level effects, health effects

14 of low level exposure to chemical agents a subject of one of

15 those fact sheets?

16 BRIGADIER GENERAL PARKER: Sir, I suspect it is,

17 but it's very difficult to talk about something that you

18 don't know anything about. And I find myself personally in

19 the same position as the Committee, trying to find out what

20 that answer is. There is a paucity of information about

21 long-term health effects for very low levels of exposure to

22 chemicals. Now yes, there is some literature that has come

23 out of Edgewood and past studies, but whether that could be

24 extrapolated into this situation, I don't know. We are

25 looking at that and certainly that is a topic that the

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1 veterans are going to tell us that they want to hear about.

2 MR. CASSELS: There's no question about that. Has

3 the Clinical Working Group had an opportunity yet to review

4 the recent Institute of Medicine report on the adequacy of

5 the CCEP program to evaluate possible health effects of low

6 level chemical exposures? And if so, what response do you

7 have or does the Working Group have to that report?

8 BRIGADIER GENERAL PARKER: Well, we have had an

9 opportunity to read it. I -- we're very concerned about low

10 level exposures of chemicals. There are some research

11 projects underway. I wish there were more, they seem to be

12 very small in number. We are -- we have not initiated it

13 yet, but we do have a protocol that's probably one of the

14 first epidemiological protocols to look at engineers that

15 were related to the Khamisiyah incident against a cohort

16 group of engineers within Persian Gulf, deployed to Persian

17 Gulf but not at Khamisiyah, and compare that to a cohort

18 group that never left CONUS, of engineers.

19 I am optimistic that if the Institute of Medicine

20 tackles this epidemiological research, we may get some

21 information that we did not have.

22 MR. CASSELS: Basically they concluded that when

23 properly carried out, the CCEP protocol was adequate to

24 evaluate the potential health effects. How confident are

25 you that the protocols in the hands of all of the many

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1 people involved are in fact being competently carried out

2 and in accordance with the protocol's requirements?

3 BRIGADIER GENERAL PARKER: Sir, in my previous

4 assignment before I came to Washington, I was the commander

5 at one of our medical centers in Denver. And we saw quite a

6 few of the veterans and I was confident in the protocol of

7 the CCEP that our physicians and health care providers that

8 were seeing the veterans would be able to elicit signs and

9 symptoms of an exposure of a known level. I am not

10 confident, because we do not know exactly what to look for

11 for extremely low levels of exposure. And that's what I'm

12 hoping some of these research projects on extremely low

13 levels of exposure to chemicals and combinations of

14 chemicals will teach us in the medical community what

15 further to look for. I don't think we have ignored looking

16 for everything. If you're an investigator and you don't

17 know what to look for, it's hard to find it -- it really is.

18 MR. CASSELS: Thank you, General.

19 DR. LASHOF: Other questions for General Parker?

20 MS. NISHIMI: General Parker, the action plan that

21 the Department has prepared in response to the final report

22 mentioned a risk communication plan would be submitted to us

23 by May 1, and I was wondering what your revised time line

24 for that is now.

25 BRIGADIER GENERAL PARKER: I would like an

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1 opportunity to have that Veterans' Advisory Panel organized,

2 be able to meet with those veterans and then work with the

3 information that I get from that and have our Committee work

4 with that and my closest estimate for a report on a plan

5 would be the fall of this year.

6 DR. LASHOF: Any other questions. Marguerite?

7 MAJ KNOX: I just have one and I don't know that

8 it really deals with Gulf War illnesses, but you've got here

9 listed that you still had continued concern about the

10 potential threat posed by anthrax. Can you give us some

11 idea of what that is?

12 BRIGADIER GENERAL PARKER: Well, as you all know,

13 the Secretary of Defense, the national command authorities,

14 have a responsibility to make sure that the soldiers,

15 sailors, marines and airmen that are deployed to fight our

16 nation's wars and protect the national security of the

17 United States, are protected. They are dealing with an

18 issue that we know that there is a threat not only to

19 chemical agents, but we know there are threats to biological

20 agents and our Secretary is dealing with this right now.

21 There has been a lot of work. The Secretary has not made a

22 decision about this immunization program, but he will

23 wrestle with this with his experts as to whether the forces

24 need to be protected against the potential threat of

25 inhalation of anthrax spores.

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1 MAJ KNOX: For those veterans who served in the

2 Gulf War who already have received anthrax, will they be

3 omitted that opportunity?

4 BRIGADIER GENERAL PARKER: Well, the folks that

5 have received the vaccine, those who know about it and those

6 of us who know who they are, I think titers will be a very

7 important indicators as to whether they are immunized

8 efficaciously or whether they -- if they're still in the

9 force and still subject to being deployed, should be

10 revaccinated.

11 DR. LASHOF: Thank you.

12 MS. NISHIMI: One other question. On the fact

13 sheets that you have under development again, the action

14 plan, it indicated a May 1 look for the Committee. Do you

15 have some time frame for when those might be made available

16 to us as well?

17 BRIGADIER GENERAL PARKER: If I were not going to

18 pretest them, I'd give them to you today. They are in final

19 draft. The Committee will use its contacts with the

20 veterans' service organizations that we have through the

21 Coordinating Group and through the expert scientific

22 committee to get those pretested and I expect that within 30

23 to 45 days we will be able to distribute those.

24 MS. NISHIMI: Thank you.

25 MAJ CROSS: General Parker, the nine hometown

1 meetings that are going on, is that coming out of your

2 office?

3 BRIGADIER GENERAL PARKER: No, sir, Mr. Rosker

4 himself is conducting those meetings.

5 MAJ CROSS: All right. Is there a schedule

6 published? I myself have not seen one. And I think we took

7 a little criticism this morning about this meeting may not

8 having been well publicized either, so obviously I just want

9 to ask the question, is that being publicized and how is it

10 being publicized.

11 BRIGADIER GENERAL PARKER: Well, General Vesser is

12 here. The publication of the town hall meetings, sir, how

13 is that done?

14 MR. VESSER: The publication of the town hall

15 meetings is done by the veterans's associations, VFOs

16 organized it, the VFW and the American Legion who was

17 sponsoring it on location. In addition, we had an effort to

18 get the word out ourselves through a local press campaign

19 using any medical facilities that were in the area. It was

20 all publicized.

21 CPT KILROY: If I could add something. I'm in

22 public affairs and actually I was involved in that, we had a

23 news release that went out over the Internet but also to the

24 folks who generally covered the Pentagon, and prior to any

25 of the DOD town hall meetings, I would call up the local

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1 media in that area and explain that there was going to be a

2 town hall meeting and fax a release.

3 DR. LASHOF: Have you actually held some of those

4 already now?

5 CPT KILROY: Yes, ma'am.

6 DR. LASHOF: What has the attendance been like,

7 how many people have shown up for the town hall meeting?

8 MR. VESSER: The town hall meetings have run from

9 roughly 70 to about 140, of whom half or more were Gulf War

10 veterans.

11 DR. LASHOF: Do you think that's a good response

12 or not? What were you hoping to get in these areas?

13 MR. VESSER: I think it's very difficult to

14 project what we're going to get. We're hearing primarily

15 from Gulf War veterans who have problems. We have people

16 who are coming from various places, several of the veterans

17 have traveled some distance to get there and we were glad to

18 hear from them.

19 DR. LASHOF: Thank you very much, we appreciate

20 this and we'll look forward to reading those after they're

21 field tested and getting the complete plan when you're ready

22 with it.

23 BRIGADIER GENERAL PARKER: Thank you, Dr. Lashof.

24 DR. LASHOF: Okay, that finishes the testimony and

25 we'll now turn to the other work of the Committee, our plans

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1 and final committee discussion. Robyn, I'll turn it back to

2 you for that.

3 MS. NISHIMI: The Committee's next event will be

4 on June 24th in Memphis, which I believe you all have been

5 informed about. As with this meeting, we will continue the

6 Committee's work in reviewing and examining various CBW

7 incidents. We will examine some aspect of the

8 implementation plan and then there will be, as there was at

9 this meeting, additional staff work and staff attention to

10 some of the incidents. We will revisit these incidents

11 should that be merited. We plan to march through the next

12 six months in this fashion and complete our work. Right

13 now, I would say we are still on target. As the memo

14 indicated, we'll reassess that as the time goes on.

15 DR. LASHOF: Hopefully four months, not six

16 months.

17 MS. NISHIMI: Oh, I'm sorry.

18 DR. LASHOF: This is May already. We would hope

19 that September will be our final meeting and we'll be

20 prepared in September to give this letter report, which I

21 guess is what it is now. It can't be a final report because

22 we gave a final report in January.

23 MS. NISHIMI: It'll be a special report.

24 DR. LASHOF: A special report. We have to figure

25 out how to term these reports since -- a special addendum or

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1 something report.

2 Questions the Committee has for any of the staff

3 at this point? Any areas the Committee wants the staff to

4 pay special attention to in following this up?

5 (No response.)

6 DR. LASHOF: Well, I would just say I think it's

7 clear from the testimony we've had this morning concerning

8 the detections, that we still have a lot of issues to be

9 resolved around on what basis one can decide something is

10 negative or not. It's easier to almost decide that when

11 it's positive is when we decide that we have enough evidence

12 to say there wasn't something. And further discussion by

13 staff with DOD on the Fox probably is still in order.

14 MS. KIDD-TAYLOR: Just for clarity from this

15 morning, I would like to make sure that there will be

16 additional detection regarding airborne levels of exposure

17 so that the equipment will actually be able to do that in

18 the future, or whether that is actually ongoing sort of

19 research to develop that type of equipment that could be

20 used in the field.

21 DR. LASHOF: Staff have any questions for the

22 Committee that they would like clarification on before they

23 proceed?

24 (No response.)

25 DR. LASHOF: If not, I thank you staff, thank you

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1 Committee, thank all of you for coming. Our meeting stands

2 adjourned.

3 (Whereupon, the meeting was adjourned at 3:34

4 p.m.)

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